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Advances in Path-Planning Algorithms for Agricultural Robots

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Keywords: agricultural robot | algorithm | path planning

ABSTRACT

With the rapid advancement of intelligent technologies, the application of robots in agriculture has expanded significantly. Path planning, a critical technology for the autonomous navigation of agricultural robots, has emerged as a key research direction. This paper classifies path-planning algorithms into four categories: traditional classical algorithms, modern intelligent bionic algorithms, sampling-based planning algorithms, and machine learning algorithms. It systematically examines the concepts and characteristics of each algorithm type, evaluates their suitability across various agricultural environments, compares their convergence speeds and computational efficiencies, and discusses potential improvement strategies. The analysis reveals that traditional classical algorithms offer high precision and stability in structured farmland environments but lack dynamic adaptability. Modern intelligent bionic algorithms enhance path robustness in complex terrains through group collaboration and global optimization mechanisms, yet they face challenges with slow convergence and parameter sensitivity; sampling-based planning algorithms excel in obstacle avoidance within unstructured, dynamic scenarios, but the quality of the generated paths depends heavily on the sampling strategy; machine learning algorithms enable environment-adaptive decision-making through data-driven approaches, though they require substantial labeled data and significant computing resources. Further comparisons suggest that path-planning algorithms' future development trend will involve integrating multiple algorithms' strengths and leveraging advanced technologies such as artificial intelligence, cloud computing, and edge computing to improve adaptability, real-time performance, and intelligent decision-making capabilities in complex agricultural environments. This paper provides theoretical support and practical guidance for research on path planning for agricultural robots and offers new insights for accelerating the development of modern agriculture.

1 | Introduction

Automation and intelligent technologies have been deeply integrated into modern agricultural development (Padhiary et al. 2024). As a significant application of artificial intelligence, robots have achieved remarkable advancements over their 60-year evolution. Agricultural robots, as critical equipment in modern agriculture, are widely utilized in planting, animal husbandry, and agricultural product marketing. Their operational efficiency and quality directly influence the benefits of agricultural production. Path planning, one of the core technologies for agricultural robots, plays a pivotal role in enhancing operational precision in complex farmland environments, reducing energy consumption, and improving intelligence levels. In the context of agricultural robot applications, improving the performance and efficiency of path planning during operations is a critical issue

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that demands immediate attention. By optimizing existing path-planning algorithms and integrating intelligent optimization technologies such as machine learning and deep learning, robots can dynamically adjust their paths based on historical data and real-time environmental information, which is fundamental to addressing these challenges. The future development trend of agricultural robot path-planning algorithms will emphasize enhancing algorithm intelligence, improving dynamic planning capabilities, and integrating multiple algorithms for collaborative operations. Mature agricultural robot path-planning systems incorporate advanced technologies such as artificial intelligence, cloud computing (Antonopoulos and Gillam 2010), edge computing (Shi et al. 2016), real-time positioning systems, and high-precision sensors to achieve precise perception and positioning of farmland environments. These systems have been successfully deployed in agricultural production, promoting modernization and agricultural intelligence.

Agricultural robot path planning involves determining the optimal or feasible path from the starting point to the destination for the robot in the farmland environment and defining its movement trajectory. Path-planning algorithms (Tang et al. 2024) are mathematical methods that, based on real-time environmental information such as terrain and obstacle distribution, calculate and analyze to derive the optimal path to enhance operational efficiency and accuracy. This encompasses global path planning to optimize overall travel routes, local path planning to respond to real-time environmental changes, and different scenarios for static and dynamic environments to ensure efficient and safe navigation through complex agricultural settings.

This paper systematically reviews the research progress of agricultural robot path-planning algorithms, clarifies their technical characteristics and application boundaries, and provides theoretical support for subsequent research and practice. Specifically, it includes (1) scientific classification of existing algorithms, covering traditional classical algorithms, modern intelligent bionic algorithms, sampling-based planning algorithms, and machine learning algorithms (as shown in Figure 1), revealing the core principles and innovation directions of each type of algorithm; (2) comparative analysis of the feasibility, applicability, and limitations of different algorithms based on the requirements of typical agricultural scenarios; (3) summarizing algorithm fusion and improvement strategies, proposing future trends such as multimodal perception-driven, lightweight computing architecture, and human–machine collaborative optimization, providing new ideas for the intelligent upgrade of agricultural robot path-planning technology. Through systematic summarization and induction, this paper aims to promote the deep adaptation of path-planning algorithms in agricultural environments and facilitate the transformation of agricultural robots from single-operation tools to autonomous decision-making systems.

2 | Requirement Analysis of Path-Planning Algorithm for Agricultural Robots

In field operations, agricultural robots must navigate complex terrains such as gullies, mounds, and diverse soil types. Path-planning algorithms must prioritize the shortest path and ensure stable robot movement to prevent it from getting stuck in

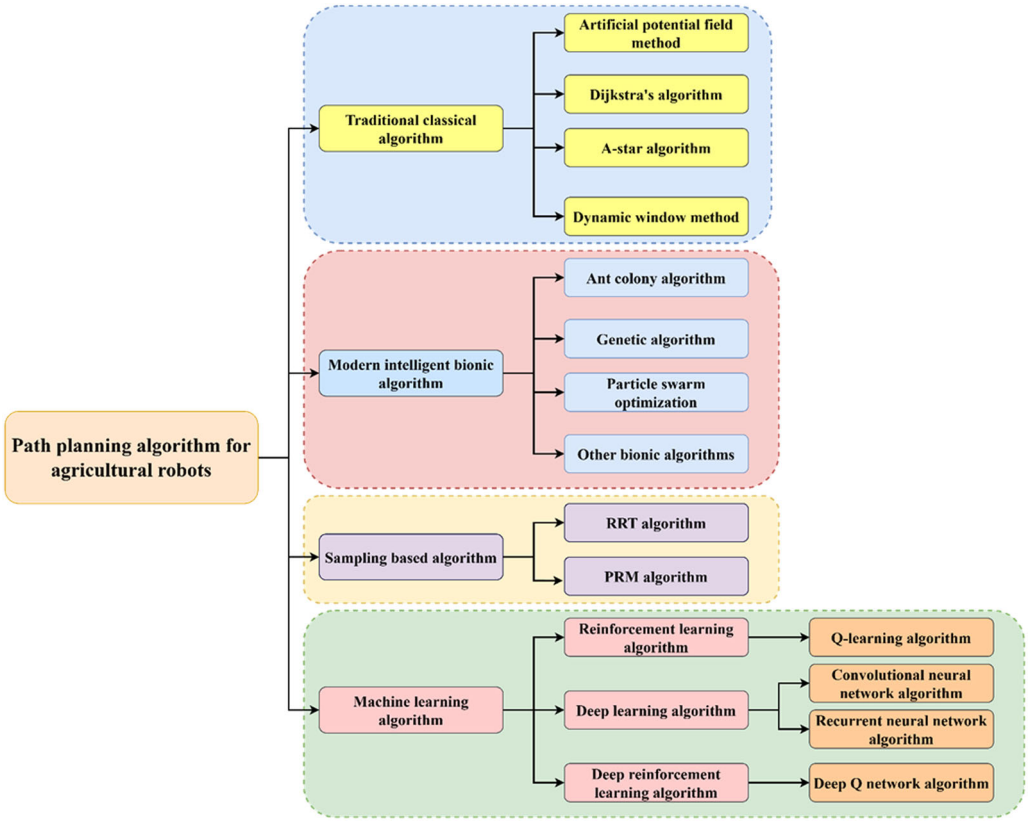


FIGURE 1 | The structural diagram of the path-planning algorithm for agricultural robots.

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mud or damaging crops. For instance, during seeding operations, the algorithm should plan a path that ensures comprehensive coverage while avoiding redundant seeding; in weeding operations, it must precisely guide the robot to areas with dense weeds while circumventing crops. These path-planning strategies significantly enhance operational efficiency, reduce resource waste, and ensure high-quality and efficient field operations.

During irrigation operations, precise irrigation volume and area management are critical for optimizing water resource utilization. The path-planning algorithm integrates soil moisture data, crop growth requirements, and irrigation equipment capabilities to determine the optimal irrigation path. This involves setting the starting point, sequence, and irrigation rate, ensuring each area receives an appropriate amount of water while minimizing water waste and preventing soil damage from over-irrigation. Advanced path-planning algorithms enable more accurate and efficient irrigation practices.

In facility agriculture, such as greenhouses or plant factories, the controlled environment presents unique challenges due to limited space and dense crop distribution. Agricultural robots performing monitoring, fertilizing, and pollinating tasks require precise path planning to avoid crop collisions. The algorithm plans the optimal travel route based on the facility layout and crop growth conditions, including obstacle avoidance, optimized turning radii, and reduced idle time. Additionally, real-time data adjustments allow the algorithm to adapt to environmental changes within the facility, ensuring efficient task completion within confined spaces.

During harvesting operations, agricultural robots must identify crop maturity and accurately plan the optimal harvesting path. By analyzing crop distribution, maturity levels, and the robot's operational capabilities, the path-planning algorithm designs a route that provides comprehensive coverage, minimizes omissions, and avoids repeated harvesting. It also considers crop vulnerability and the robot's grasping accuracy to minimize damage during harvesting. Real-time data-driven adjustments enable the algorithm to dynamically modify the harvesting plan in response to sudden weather changes or abnormal crop growth conditions. Through intelligent path planning, harvesting operations can be conducted more precisely and efficiently, enhancing agricultural production efficiency.

3 | Traditional Classical Algorithms

Traditional classic algorithms refer to well-established path-planning algorithms that have been widely adopted in agricultural robotics and proven effective over time. Specifically, these include the artificial potential field (APF) method grounded in physical abstraction (Khatib 1986), the Dijkstra algorithm for discrete search-based optimization (D. Fan and Shi 2010), the A* algorithm for heuristic-based pathfinding (X. Liu and Gong 2011), and the dynamic window approach (DWA) for real-time motion planning (Fox et al. 1997). These traditional algorithms highlight path-planning techniques' diversity and practical applicability in agricultural robotics.

3.1 | APF Method

The APF method, introduced by Khatib in 1986, is a virtual force-based approach that constructs an APF to simulate natural gravitational and repulsive forces. This method utilizes a virtual force field to guide the movement of robots. When a robot approaches an obstacle, it experiences a repulsive force from the obstacle region, prompting it to adjust its trajectory to avoid collision automatically. Meanwhile, the target operation point is designated as the gravitational center, generating a strong, attractive force that pulls the robot toward the target. Through the synergistic effect of these attractive and repulsive forces, agricultural robots can autonomously plan paths that effectively avoid obstacles while ensuring direct navigation to the target in complex and dynamic farmland environments, as illustrated in Figure 2.

As depicted in Figure 2, assume that the agricultural robot's initial position is at $X = (x, y)$. X represents the position vector of the agricultural robot. $X_t = (x_t, y_t)$ represents the target position vector. K_{att} is the constant of the gravitational potential field and the gravitational field between the target position and the agricultural robot:

$$U_{att} = \frac{1}{2} K_{att} \cdot \rho^2(X, X_t), \tag{1}$$

$\rho(X, X_t)$where $\rho(X, X_t)$ represents the relative distance from the agricultural robot to the target position.

$$\rho(X, X_t) = \sqrt{(x - x_t)^2 + (y - y_t)^2}. \tag{2}$$

The negative gradient direction of the gravitational potential field serves as the gravitational direction exerted on the agricultural robot:

$$F_{att}(X) = -\nabla U_{att} = -K_{att}|X - X_t|. \tag{3}$$

As the distance between agricultural robots and obstacles has varying degrees of influence on the planned path, the repulsive field influence radius ρ_0 centered on the obstacle is set. The distance ρ between the agricultural robot and the obstacle.

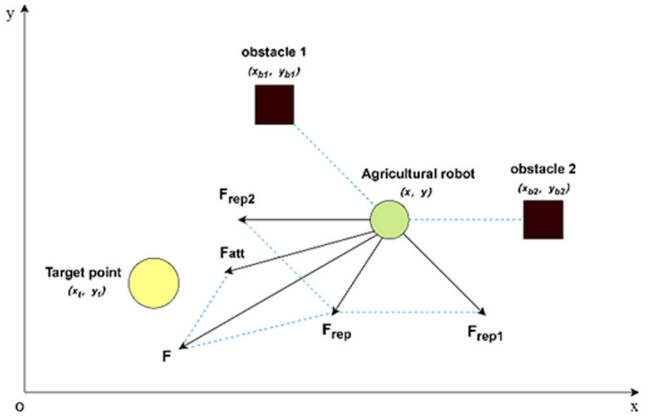


FIGURE 2 | Schematic diagram of artificial potential field.

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When $\rho > \rho_0$ occurs, the agricultural robot is no longer subject to the influence of the repulsive field, and the repulsive force is zero at this moment.

When $\rho \leq \rho_0$ occurs, the distance ρ between the obstacle and the agricultural robot is linearly related to the repulsive field U_{rep}, from which the repulsive field can be derived:

$$U_{\text{rep}(i)}(X) = \begin{cases} 0, & \rho > \rho_0, \\ \frac{1}{2} K_{\text{rep}} \left[\frac{\rho_0 - \rho_{(i)}(X)}{\rho_{(i)}(X)\rho_0} \right]^2, & \rho \leqslant \rho_0, \end{cases} \tag{4}$$

where $\rho_{(i)}(X)$ is the Euclidean distance between the robot and the obstacle, $(x_{\text{b}(i)}, y_{\text{b}(i)})$ is the position vector of the obstacle, and K_{rep} is the constant of the repulsive potential field. Among them,

$$\text{specifically, } \rho_{(i)}(X) = \sqrt{(x - x_{\text{b}(i)})^2 + (y - y_{\text{b}(i)})^2}.$$

The direction of the negative gradient of the repulsive potential field is the direction of the repulsive force exerted on the agricultural robot, namely,

$$\begin{aligned} F_{\text{rep}(i)}(X) &= -\nabla U_{\text{rep}(i)}(X) \\ &= \begin{cases} 0, & \rho > \rho_0, \\ K_{\text{rep}} \left[\dfrac{\rho_0 - \rho_{(i)}(X)}{\rho_{(i)}(X)^3 \rho_0} \right] \dfrac{\partial \rho_{(i)}(X)}{\partial X}, & \rho \leqslant \rho_0. \end{cases} \end{aligned} \tag{5}$$

The resultant force exerted on the robot can be derived by taking the vector sum of the gravitational force it undergoes and all the repulsive forces.

$$F(X) = F_{\text{att}}(X) + \sum_{i}^{n} F_{\text{rep}(i)}(X), \tag{6}$$

where n is the number of obstacles.

APF demonstrates significant advantages in the path planning of agricultural robots. This algorithm is characterized by strong real-time performance and high computational efficiency. Constructing a virtual force field that combines target attraction with obstacle repulsion can rapidly generate smooth paths while fulfilling the dynamic obstacle-avoidance requirements in farmland environments. Unlike other methods, this approach does not rely on complex map modeling but instead dynamically adjusts paths based on real-time sensor data. When machinery such as tractors and seeders are operational in agricultural scenarios, APF leverages a multimodal perception system to generate gravitational and repulsive fields, enabling effective dynamic obstacle avoidance for agricultural machinery, as detailed in Tables 1 and 2.

Although APF has been extensively employed in local dynamic planning for agricultural robots, it suffers from certain limitations, such as the tendency to become trapped in local optima and the inability to reach target positions, as illustrated in Figures 3–5.

Currently, to address the tendency of APF to fall into local optima, primary improvement strategies involve introducing virtual target points or obstacles. To address the dynamic obstacle-avoidance challenge for apple-picking robots in unstructured orchard environments, Wei and Fengyi (2013) proposed an enhanced APF. Using kinematic modeling to analyze the interaction between the robotic arm and obstacles, they introduced dynamic virtual guide points to optimize the target search path, considering the random distribution of obstacles within the tree canopy. Long (2020), based on APF theory, developed a velocity potential field model that uses an attractive velocity field to guide the robotic arm toward the target and a repulsive velocity field to avoid obstacles. Yingkun (2018) proposed a differentiated obstacle threat assessment system, optimized the repulsive potential field function, and incorporated a strategy of virtual target points in the local extremum neighborhood to resolve this issue. Miao et al. (2021) addressed the issue of path overlap in multirobot cooperative agricultural scenarios by incorporating an interrobot repulsive potential field into the traditional repulsive model, dynamically generating guiding points, and optimizing the distribution of target points using a coverage evaluation function. Xie et al. (2019) extended the two-dimensional (2D) APF theory to three dimensions by analyzing the spatial distribution of obstacles in an orchard. They designed a smooth and effective three-dimensional (3D) motion trajectory for obstacle avoidance.

Research has focused on modifying potential field functions to address issues such as target unreachability. C. Zhang, Wang, et al. (2023) focused on the complex constraints in path planning for orchard mowing robots. They addressed this challenging issue by introducing an elliptical repulsive potential field as a boundary constraint and integrating a variable polynomial function with a distance-related weighting factor. Azzabi and Nouri (2019) proposed an enhanced gravitational potential field model by incorporating the minimum distance parameter between obstacles and the target point into the traditional function, thereby strengthening the gravitational attraction to the target. Pack and Kee (Park and Kee 2021) constructed a potential energy field based on the reciprocal distance between the agricultural robot and the target point and the distance between the robot and obstacles. S. Feng et al. (2021) proposed integrating the safety distance parameter into the potential field function, redefining the repulsive force domain of obstacles, and incorporating dynamic velocity constraints into the repulsive force calculation to enhance the safety and robustness of path planning. Z. Zhang, Dai, et al. (2023) addressed the path deviation and oscillation failure resulting from the abrupt force change in the traditional APF innovatively by employing normalization and high-order exponential functions to adjust the force of the potential field precisely.

In the future agricultural applications of APF, improvements can be achieved from two key aspects. First, the dynamic potential field should be deeply coupled with multimodal environmental perception to overcome the limitations of traditional static APF modeling. This involves dynamically updating potential field parameters based on real-time sensor data, integrating preplanning strategies informed by crop growth models, and introducing probabilistic potential fields to enhance robustness in occluded areas. Second, the deep integration of APF with intelligent algorithms should be promoted.

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TABLE 1 | Some application scenarios and influences of artificial potential field (APF).

Actual scenarios	Concrete impacts
Weeding robot (Y. Chen, Tao, et al. 2024)	APF, when integrated with visual recognition technology, enables the generation of weeding paths by utilizing repulsive fields to avoid crops and attractive fields to guide the robot toward areas with dense weed populations.
Fruit and vegetable picking robot (Zhuang et al. 2023)	During automated harvesting navigation, the positions of mature fruits are identified via visual recognition to generate an attraction field. Simultaneously, repulsion fields are established for plant stems, support structures, and other obstacles to avoid collision.
DJI MG-1P drone (F. Li, Xiukai, et al. 2023)	The path generated by the APF method is highly adaptable to the wheat field environment. The gravitational field guides the harvester to follow the contour lines, while the repulsive field ensures obstacle avoidance with centimeter-level precision in edge bypassing.
The tractors developed by CNH Industrial Group (J. Zhang, Kang, et al. 2024)	In the vineyard environment, the APF algorithm enabled the equipment to circumvent trellis supports and irrigation infrastructure precisely, thereby substantially enhancing operational efficiency.
The unmanned tractor Dongfanghong of China National Tractor Corporation (jianguo 2020)	During the testing phase in the Heilongjiang Agricultural Reclamation Area, obstacle avoidance was achieved by incorporating the A* algorithm into global path planning based on the APF method. This approach successfully reduced the path overlap rate by more than 15%.
In the orchard, agricultural plant protection drones (Hao et al. 2023)	Agricultural plant protection UAVs construct a three-dimensional potential field model of farmland using onboard LiDAR and RGB-D cameras. Static obstacles, such as utility poles and trees, are represented as repulsive fields within this model, enabling the UAVs to accomplish precise plant protection tasks efficiently.

Abbreviations: LiDAR, light detection and ranging; UAV, uncrewed aerial vehicle.

TABLE 2 | Challenges and solutions of artificial potential field in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① High humidity can cause fogging on the sensor lens, thereby reducing the accuracy of obstacle recognition.	① By integrating LiDAR and ultrasonic sensors, humidity-induced noise is mitigated through Bayesian filtering.
② The high-temperature and high-humidity conditions in greenhouses accelerate the degradation of electronic components.	② Humidity sensor data are utilized to adjust the intensity of the repulsive field dynamically.
③ Dense branches of fruit trees obstruct the sensor's line of sight, leading to potential missed detections of obstacles.	③ A three-dimensional environmental model using surround LiDAR to identify occluded regions is constructed.
④ The mechanical arm is susceptible to collisions with branches when maneuvering in confined spaces.	④ Virtual repulsive points are introduced in areas with dense branches to enhance local obstacle-avoidance performance.
⑤ Path conflicts may occur when multiple uncrewed aerial vehicles converge at the edges of fields.	⑤ Unmanned aerial vehicles synchronize local potential field information via the 5G network to construct a global potential field.

Abbreviation: LiDAR, light detection and ranging.

Specifically, machine learning techniques can be leveraged to dynamically and precisely adjust parameters and models, while bionic algorithms, such as those simulating biological group behaviors, can be incorporated to strengthen global search capabilities and optimize path planning.

3.2 | Dijkstra Algorithm

Dijkstra's algorithm is a classic greedy algorithm used to solve the single-source shortest path problem in weighted graphs with nonnegative edge weights. Proposed by Dutch computer scientist Edsger Wybe Dijkstra in 1959, this algorithm operates by starting from a source node and progressively expanding outward. At each step, it selects the unvisited node with the smallest tentative distance from the source and updates the shortest path estimates for its neighbors. This iterative process continues until the destination node is reached or all reachable nodes have been visited. Dijkstra's algorithm ensures that the shortest path to each node is found systematically.

Let $G = (V, E)$, where V is the set of nodes and E is the set of edges. The edge weight function $w(u, v) \geq 0$ represents the weight (nonnegative) of edge $s \in V$. The starting point is s, and the goal is to calculate the shortest path from $d[v]$ to all other nodes. The distance array $d[v]$ stores the estimated shortest

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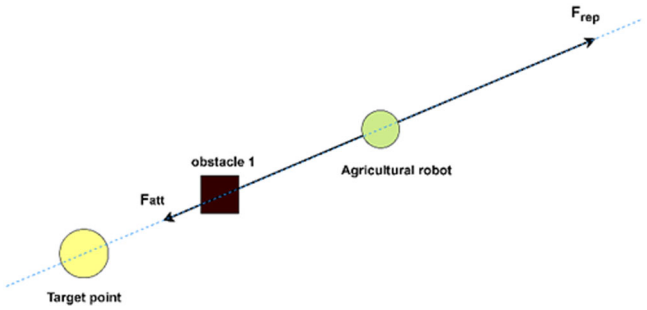


FIGURE 3 | Schematic diagram of the local optimal problem with a single obstacle.

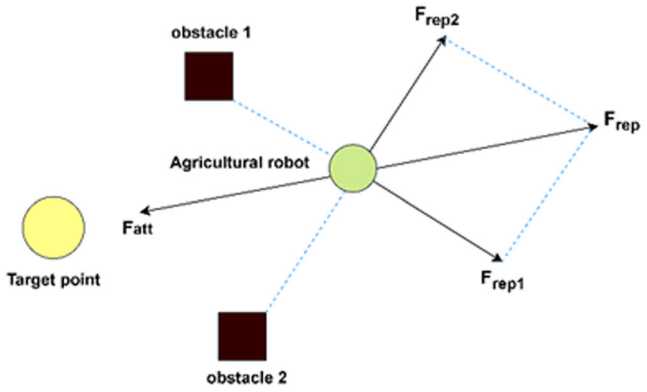


FIGURE 4 | Schematic diagram of the local optimal problem with multiple obstacles.

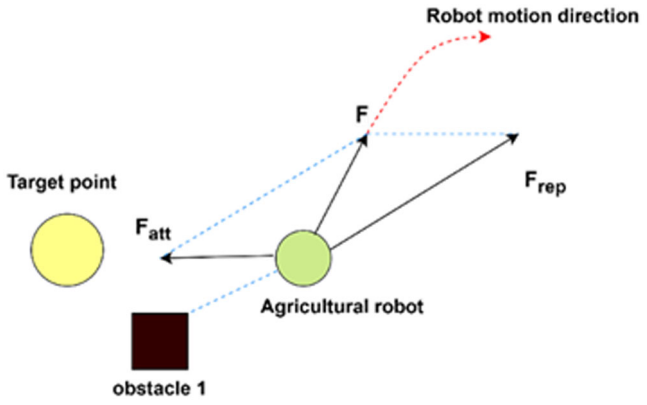


FIGURE 5 | Schematic diagram of the problem of unreachable target.

distance from s to node v. The predecessor array $prev[v]$ records the predecessor node of node v on the shortest path, which is used for backtracking the path. The set $S \subseteq V$ represents the set of nodes whose shortest paths have been determined.

Initialization:

$$d[v] = \begin{cases} 0 & \text{if } v = s, \\ \infty & \text{otherwise,} \end{cases} \quad S = \varnothing. \tag{7}$$

When $S \neq V$ holds, the first step: Select an unvisited node and choose $u \in V \backslash S$ such that $d[u]$ is minimized.

The second step: Mark it as visited and incorporate u into the set S.

The third step: For all neighbors $v \in Adj(u)$ of u, if $d[v] > d[u] + w(u, v)$, then update $d[v] \leftarrow d[u] + w(u, v)$: and record the predecessor $prev[v] = u$ of the path.

The fourth step: When all nodes have been added to S (i.e., $S = V$), at this point, $d[v]$ represents the shortest path length from s to v.

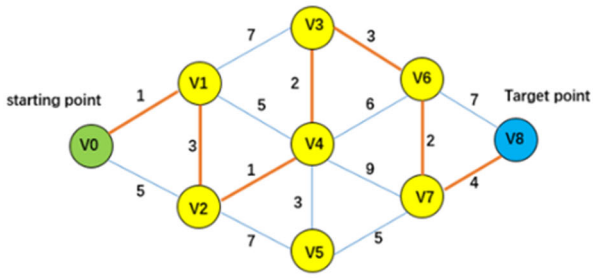
Taking the weighted roadmap (a) as an exemplar, (b) represents the calculation process of generating the optimal path from V8 to V0 in the graph. Meanwhile, the calculation principle of Dijkstra's algorithm is depicted in Figure 6.

In the path planning of agricultural robots, the Dijkstra algorithm models the agricultural environment as a graph structure, where the robot's initial position serves as the starting node, the farmland or operational points are represented as target nodes and feasible paths between them are treated as edges. The weights of these edges are determined based on the presence and distribution of obstacles. By iteratively exploring and updating the shortest paths from the starting node to all other nodes, the algorithm ensures that the agricultural robot operates efficiently and stably along an optimal collision-free trajectory, as demonstrated in Tables 3 and 4.

The Dijkstra algorithm, widely adopted in agricultural robot path planning, leverages its global optimality and reliability by traversing nodes in a breadth-first manner to identify the shortest path in static environments. Its straightforward logic and ease of implementation render it suitable for small-scale precise operations, enabling the planning of optimal obstacle-avoiding paths for traversing multiple target areas in intelligent agricultural machinery navigation. However, the algorithm exhibits certain limitations. As the scale of farmland increases or node density rises, search efficiency declines significantly, hindering its ability to meet real-time requirements. Moreover, its applicability is restricted to static, known environments, rendering it incapable of handling temporary obstacles. Additionally, due to insufficient consideration of terrain complexity, the algorithm may generate infeasible paths that fail to align with practical operational constraints.

Many scholars have proposed improvements in response to the limitations of the Dijkstra algorithm. The primary enhancements focus on efficiency optimization and dynamic processing capabilities. First, a bidirectional search strategy is adopted, which expands the search tree simultaneously from both the starting point and the destination, thereby reducing the number of traversed nodes. Second, a priority queue mechanism is introduced to optimize the order of node visits, effectively lowering the time complexity. Third, incremental search techniques are integrated, enabling recalculations of paths only in changed regions to efficiently address the dynamic insertion and removal of temporary obstacles. Alshammrei et al. (2022) enhanced the Dijkstra algorithm for agricultural robot path planning by precomputing the optimal path offline. During real-time operation, ultrasonic sensors monitored the passage conditions between nodes, and upon detecting obstacles, the

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(a)

Node	D (Distance)	P (Predecessor)
V0	0	0
V1	1	0
V2	4	1
V3	7	4
V4	5	2
V5	8	4
V6	10	3
V7	12	6
V8	16 (optimal)	7

(b)

FIGURE 6 | Schematic diagram of Dijkstra's algorithm. D-distance: Represents the current estimated shortest distance from the source node to each of the other nodes. P-predecessor: It is used to record the predecessor node of each node in the shortest path. *Note:* The optimal path in the figure is V8 → V7 → V6 → V3 → V4 → V2 → V1 → V0. The optimal path steps are 16.

TABLE 3 | Some application scenarios and influences of Dijkstra's algorithms.

Actual scenarios	Concrete impacts
Automatic navigation and picking in orchards (Y. Chen 2024)	On the basis of the locations of fruit trees, a grid map is constructed to calculate the shortest path and minimize the mechanical arm's travel distance. This approach ensures high picking efficiency and the integrity of the fruits during harvesting.
Dynamic feeding path for pastures (Xiaoyan et al. 2019)	In complex pasture environments, generate fixed inspection routes and mark key monitoring points by combining node weights.
Optimization of large-scale farmland harvesting (Gorter 2019)	The farmland is partitioned into grid nodes, and an optimal shortest-path algorithm is employed to plan a full-coverage route that avoids redundant harvesting. Node weights are dynamically adjusted based on suitability criteria, prioritizing harvesting appropriate areas and improving resource utilization.
John Deere's JD5-854-1 series tractors are used for farm work (Dou et al. 2024)	The farmland map is modeled as a weighted graph, and Dijkstra's algorithm is utilized to compute the shortest path from the starting point to the destination. Priority is given to selecting flat and obstacle-free routes, while GPS data are integrated for real-time adjustments to minimize redundant coverage.
DJI Agras T40 agricultural drone sprays pesticides (Girijalaxmi et al. 2023)	The Dijkstra algorithm determines the shortest paths from the take-off point to each spraying zone while dynamically adjusting the routes based on real-time LiDAR obstacle-avoidance data.
Netafim pressure compensated drip irrigation pipes from Israel (Kim et al. 2023)	This system represents the irrigation pipe network as a graph structure, where nodes correspond to water outlet valves, and edge weights are defined by pipe length and terrain elevation differences. The Dijkstra algorithm determines the optimal path from the central pumping station to the field valves.

Abbreviations: GPS, global positioning system; LiDAR, light detection and ranging.

system dynamically updated the directed graph to recalculate the path. W. Ji et al. (2014) proposed a two-stage approach combining an improved Dijkstra algorithm for initial path generation with adaptive ant colony optimization (ACO) to ensure safety in real-time obstacle avoidance for apple-picking robots in dynamic and unstructured environments. H. Liu et al. (2025) introduced the Integrated Harris Hawk Optimizer algorithm, which combines the Dijkstra algorithm with the Harris Hawk Optimizer, to address the challenges of corn harvesting in hilly terrains. They developed a collaborative model incorporating quantum-inspired strategies, new operators, and updated mechanisms to optimize population distribution. Jia (2024) optimized path planning in wireless sensor networks for tea plantations by integrating data dissimilarity and node distance metrics into edge weight quantification, enabling efficient path searches based on start and end points while addressing complex terrain issues. F. Li, Xiukai, et al. (2023) proposed an enhanced bidirectional Dijkstra algorithm framework for transplanting machine path planning. This framework integrated a secondary optimization strategy for node prioritization and combined A* algorithm-based obstacle avoidance with image rendering for visual simulation.

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TABLE 4 | Challenges and solutions of Dijkstra's algorithm in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① In pastures, the dynamic movement of livestock herds and temporary farm equipment can obstruct preset paths.	① Integrate real-time LiDAR-based dynamic obstacle detection to trigger local Dijkstra replanning.
② Worker activities in orchards result in real-time changes to planned paths.	② Establish dynamic node weight thresholds to update paths exclusively for critical obstacles.
③ Changes in node connectivity in muddy farmlands introduce significant delays for traditional path-planning algorithms.	③ Implement hierarchical Dijkstra, initially planning the region's main pathways and refining local paths.
④ The rugged terrain in mountainous orchards poses substantial challenges for node discretization.	④ Pre-generate a sparse node graph using terrain elevation data to reduce computational complexity.
⑤ Path intersections during the collaborative operation of multiple seeders can lead to deadlocks.	⑤ Assign independent subgraphs to each robot and introduce time window constraints to coordinate their passage sequences.

Abbreviation: LiDAR, light detection and ranging.

The future development of Dijkstra's algorithm will primarily focus on two key areas. First, due to the high computational complexity and limited real-time performance of traditional algorithms, which are not well-suited for dynamic agricultural scenarios, it is essential to accelerate the integration of advanced algorithms and promote intelligent upgrades. This can be achieved by incorporating heuristic algorithms or machine learning techniques, leveraging deep learning for obstacle prediction, and collaboratively generating optimized obstacle-avoidance paths while maintaining a balance between weight factors and computational efficiency. Second, to address its high computational load, heterogeneous computing architectures such as cloud computing, edge computing, and field-programmable gate array (FPGA)–based hardware acceleration can be utilized to enhance performance. Additionally, challenges such as communication latency, energy consumption, and insufficient network coverage in remote farmlands must be resolved, with an emphasis on developing lightweight embedded solutions.

3.3 | A* Algorithm

The A* (A-Star) algorithm is a widely adopted heuristic search method in the path planning of agricultural robots, particularly suited for static environments. Introduced by Peter Hart, Nils Nilsson, and Bertram Raphael in 1968, it aims to find the optimal path from a start point to a destination in complex environments. The A* algorithm combines the strengths of best-first search and Dijkstra's algorithm by incorporating a heuristic function, as shown in Formula (1). During the search process, it maintains a priority queue of nodes to be expanded. It evaluates each node using an evaluation function that balances the cost from the start node and the estimated cost to the goal. The node with the lowest evaluation value is selected for expansion until the target node is reached. The A* algorithm typically performs an eight-directional movement search (denoted as A1 → A8), as illustrated in Figure 7.

The traditional cost function of the A* algorithm is shown in Equation (8):

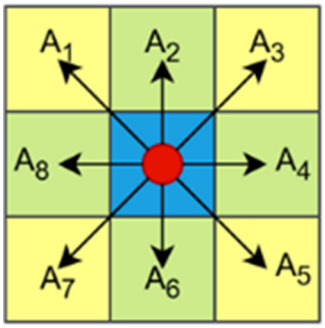


FIGURE 7 | Schematic diagram of eight-direction motion search.

$$f(n) = g(n) + h(n). \tag{8}$$

In the A* algorithm, $f(n)$ denotes the total estimated cost from the starting node through node n to the target node. Specifically, $g(n)$ represents the actual cost from the starting node to node n, while $h(n)$ represents the heuristic estimate of the cost from node n to the target node. The heuristic function $h(n)$ plays a crucial role in influencing both the performance and search efficiency of the A* algorithm. Commonly used heuristic functions include Manhattan distance, Euclidean distance, and Chebyshev distance, as illustrated in Figure 8.

In applying the A* algorithm for path planning in agricultural environments, farmland and orchards are first discretized into a grid map, where each grid cell represents a node. The A* algorithm is then employed to evaluate the estimated cost from the current position to the target operation point, prioritizing paths more likely to approach the target while avoiding obstacles in the extensive agricultural environment, as illustrated in Figure 9.

When combined with heuristic information for path planning in agricultural robotics, the A* algorithm effectively reduces the search space while ensuring optimal path generation. It is particularly suited to farmland environments with limited dynamic obstacles. By leveraging an estimation function to guide the search direction, the A* algorithm exhibits high

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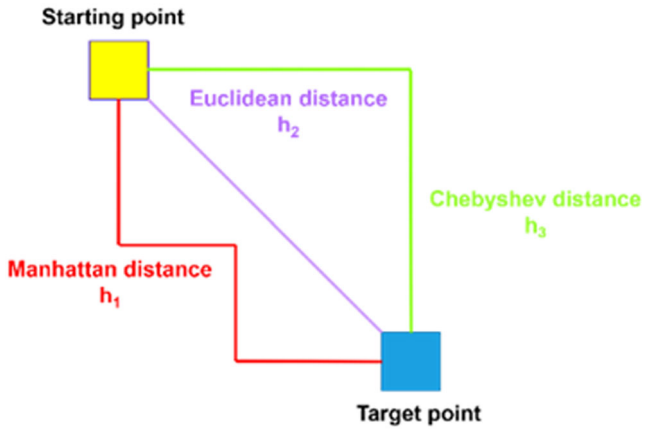


FIGURE 8 | Schematic diagrams of three kinds of distances.

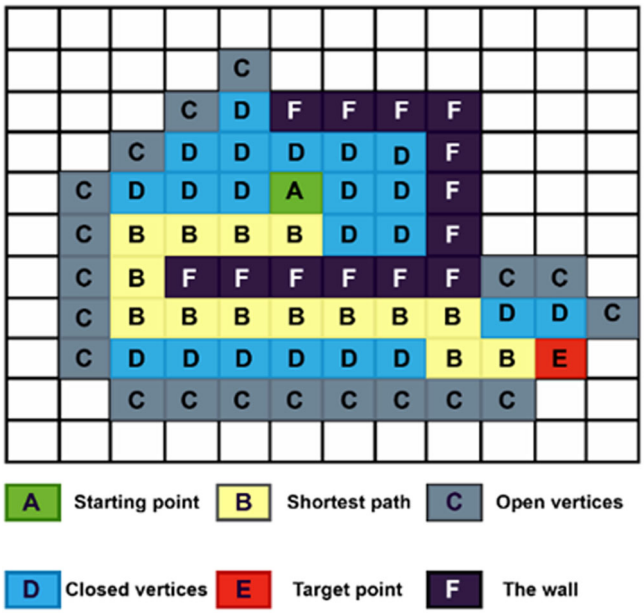


FIGURE 9 | Schematic diagram of A* algorithm.

efficiency in static global path planning, especially for long-distance operational tasks, as demonstrated in Tables 5 and 6.

The primary limitation of the A* algorithm in agricultural robot path planning is its heavy reliance on the heuristic function. Inaccuracies in farmland environment modeling can lead to suboptimal paths, deviating from the optimal solution. The A* algorithm requires a pre-built, highly accurate global map, which challenges adapting to real-time changes in dynamic farmland environments. Additionally, as environmental complexity increases, so does the algorithm's computational complexity, potentially causing communication delays and resource contention during multirobot collaborative operations. While the paths generated by the A* algorithm are often geometrically optimal, they do not fully account for the efficiency requirements of agricultural operations.

The A* algorithm can be enhanced in four key aspects in the agricultural domain. First, the heuristic function can be improved to increase planning efficiency. Second, hierarchical and bidirectional search strategies can be utilized to reduce computational complexity. Third, incremental updates can reinforce dynamic adaptability to handle environmental changes effectively. Fourth, multiobjective optimization can be integrated to satisfy diverse requirements in complex scenarios. M. Wang, Xu, Zhang, et al. (2024) proposed a navigation method that integrates the improved A-Star algorithm with SVR to meet the requirements for autonomous navigation in orchards. The enhanced A-Star algorithm is utilized to optimize path planning, thereby significantly improving navigation accuracy and path smoothness in complex orchard environments. Chatzisavvas, Sanida, et al. (2024) proposed an intelligent agricultural navigation optimization approach by enhancing the A* algorithm, enabling uncrewed ground vehicles to avoid obstacles effectively. They innovatively combined the Euclidean distance function with Catmull-Rom spline interpolation, ensuring agricultural robots' accurate and efficient operation. Zhao et al. (2023) addressed the path-planning challenge for agricultural robots by utilizing sensors to construct real-time maps, refining the heuristic function of the conventional A* algorithm based on obstacle density, extending the search neighborhood, eliminating redundant nodes, and incorporating DWA for dynamic obstacle avoidance. J. Li, Zhang, et al. (2023) addressed the dynamic obstacle-avoidance requirements of agricultural drones in farmland by improving the RJA-star algorithm, thereby significantly enhancing the efficiency of path planning for drones in complex environments. D. Zhang, Chen, et al. (2024) addressed the limitations of the traditional A* algorithm in automated guided vehicle (AGV) path planning by dynamically adjusting weight parameters through the cost function to minimize unnecessary turns. Additionally, they integrated APF to construct a composite heuristic function and incorporated an obstacle penalty term to enhance safety at turning points. This improved algorithm significantly reduces the number of turning points, ensures a safe distance from obstacles, and optimizes the turning trajectories of AGVs.

Owing to the efficiency and flexibility of the A* algorithm, it has been extended and improved by numerous scholars, leading to the development of new variants. For instance, the D* algorithm (Stentz 1994), proposed by Anthony Stentz in 1994, is a dynamic variant of A*, designed to address the shortest path problem for robots in dynamic environments. Unlike A*, which searches forward from the start node, D* performs a backward search from the target area, utilizing a priority queue to plan an optimal path efficiently. Its core mechanism dynamically adjusts the search process by recording both nodes' minimum and current path costs, enabling rapid adaptation to environmental changes. In agricultural scenarios, when soil becomes muddy or unexpected obstacles appear, D* can promptly respond by recalculating only the affected segments of the path, thereby significantly reducing computational time and resource consumption. The D* Lite algorithm (Koenig and Likhachev 2002), introduced by Sven Koenig in 2002, builds upon the Label-Correcting Algorithm and employs reverse search combined with incremental updates to enhance replanning efficiency. While sharing similar functionality with D*, D* Lite offers a more streamlined implementation and is widely adopted in robot navigation tasks. The Anytime Repairing A* (ARA*)

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TABLE 5 | Some application scenarios and influences of A* algorithm.

Actual scenarios	Concrete impacts
Precision pesticide spraying coverage in farmland (Uchiyama and Tsuchiya 2024)	The farmland is divided into dynamic grid nodes, and the A* algorithm plans a path with minimized turning. By incorporating weight adjustments based on pesticide spraying ranges, the approach effectively reduces redundant coverage and minimizes pesticide solution waste.
Greenhouse multilayer planting inspection (Qu et al. 2022)	In the vertical planting rack structure, the A* algorithm generates multilayer inspection routes that prioritize visiting areas with a high incidence of diseases. This approach enhances the timeliness of monitoring and ensures equipment safety during inspections.
Fruit-picking robot (M. Wang, Xu, Zhang, et al. 2024)	Upon visually identifying and locating the apple, the A* algorithm is utilized to determine the shortest trajectory for the robotic arm from its present position to the target fruit.
The automatic navigation system in the farmland (Chatzisavvas, Dossis, et al. 2024)	Global positioning system and sensors acquire farmland terrain and obstacle information in real-time. Subsequently, the A* algorithm is utilized to plan an optimal path that ensures nonoverlapping coverage of the entire farmland.
John Deere's see and spray intelligent spraying system (Avent et al. 2025)	Camera systems identify crops and weeds in the farmland in real-time, and the spraying path is planned by integrating the A* algorithm. This approach ensures effective crop and obstacle avoidance during the spraying process.
The agricultural unmanned vehicle of Naio Technologies (Greenhouses 2025)	The A* algorithm, in combination with the farmland map and obstacle information, plans the optimal patrol route for the uncrewed vehicle.

TABLE 6 | Challenges and solutions of A* algorithm in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① During the growth stage of tall crops, preset paths may result in collisions due to inadequate clearance height.	① Integrate a growth prediction model to update the height parameters of map nodes dynamically.
② When multiple seeders operate simultaneously in the same field, their paths may intersect, leading to traffic congestion.	② Employ the distributed A* algorithm to allocate independent subgraphs with buffer zones for each robot, ensuring path independence and reducing interference.
③ Rainy conditions degrade the performance of visual sensors, causing inaccurate node positioning and path deviations.	③ Fuse millimeter-wave radar and infrared sensors to enhance environmental modeling under adverse weather conditions, improving sensor robustness.
④ Irregular field boundaries can lead to omitting edge areas in gridded path planning.	④ Adopt nonuniform grid partitioning, prioritizing increased node density in boundary areas to ensure comprehensive coverage.
⑤ Strong winds introduce noise into LiDAR data, compromising the accuracy of path generation.	⑤ Implement redundant verification at key nodes to dynamically correct positioning offsets, enhancing localization accuracy.

Abbreviation: LiDAR, light detection and ranging.

algorithm (Likhachev et al. 2003), developed by M. Likhachev in 2003, integrates weighted A* with incremental planning techniques. It first generates a suboptimal solution quickly and refines iteratively toward optimality. By progressively relaxing the heuristic function's weight during multiple rounds of optimization, ARA* allows termination at any point while returning the best available solution up to that moment. Lastly, the Jump Point Search (JPS) algorithm (Harbor and Grastien 2014), proposed by D. Harabor and A. Grastien in 2011, leverages the symmetry properties of grid maps to skip unnecessary node expansions and focus solely on "jump points." By preprocessing the map structure, JPS reduces computational overhead and is well-suited for fast navigation in structured farmland environments.

The A* algorithm is anticipated to advance in three key areas within the agricultural domain. First, by leveraging neural heuristic networks and graph neural networks (GNNs) for

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optimization, a GNN-based model will be developed to learn environment-path mappings. The attention mechanism will be employed to dynamically adjust the weights of the heuristic function, while transfer learning will be integrated to reduce computational costs. Global and local path planning will be enhanced through multiresolution grid hierarchical planning. Second, multiobjective Pareto optimization with dynamic weight allocation will be implemented. Multiobjective evolutionary algorithm (MOEA) will be utilized to address conflicting objectives and generate Pareto-optimal solution sets. Fuzzy logic (FL) and reinforcement learning (RL) will be incorporated to dynamically allocate and adaptively adjust weights, ensuring flexibility in meeting diverse operational requirements. Third, online continuous learning and environmental adaptation capabilities will be strengthened. Neural networks (NNs) will be updated via an online learning framework, enabling adaptive strategy adjustments based on real-time sensor data. Knowledge distillation techniques will be applied to achieve model lightweight, ensuring efficient real-time updates in field operations.

3.4 | Dynamic Window Approach

The DWA is a local path-planning algorithm based on the velocity space, devised explicitly for real-time obstacle avoidance of mobile robots in dynamic environments. Initially proposed by Dieter Fox, Wolfram Burgard, and Sebastian Thrun in 1997, it mainly encompasses three steps: generation of the dynamic window, trajectory prediction, and trajectory evaluation function. The generation of the dynamic window is predicated on the robot's kinematic model and environmental constraints, sampling velocity values within the defined state space to define the range of velocities that the robot can safely execute at the current instance. Trajectory prediction involves generating motion trajectories to prognosticate the robot's state over a specific period. The evaluation function assesses each generated motion trajectory. Once the evaluation is accomplished, based on the robot's preset parameters and environmental conditions, a window containing motion trajectories with higher evaluation scores is screened out, representing the feasible pre-trajectories for the robot in the current situation. Eventually, the trajectory with the highest evaluation score is selected from the screened trajectories as the robot's next motion strategy, directing the robot to move along this trajectory.

1. Generation of the Dynamic-Window: The dynamic window is generated by intersecting the following three constraints:
   i. Kinematic Constraint Window V_m: It is determined by the maximum/minimum linear and angular velocities constrained by the robot's hardware.

$$V_m = \{(v, \omega) | v \in [v_{min}, v_{max}], \omega \in [\omega_{min}, \omega_{max}]\}. \quad (9)$$

   Specifically, v is the linear velocity of the robot (m/s), w is the angular velocity of the robot (rad/s), v_{min}, v_{max} are the minimum and maximum permissible linear velocities of the robot, and $\omega_{min}, \omega_{max}$ and the minimum and maximum permissible angular velocities of the robot.
   ii. Acceleration Constraint Window V_a: Taking into account the maximum acceleration limit of the robot to ensure smooth variations in speed.

$$\begin{aligned} v &\in [v_c - \dot{v}_{max}\Delta t, v_c + \dot{v}_{max}\Delta t], \\ \omega &\in [\omega_c - \dot{\omega}_{max}\Delta t, \omega_c + \dot{\omega}_{max}\Delta t]. \end{aligned} \quad (10)$$

   Specifically, v_c, ω_c are the current linear velocity and angular velocity, respectively; $\dot{v}_{max}, \dot{\omega}_{max}$ are the maximum linear acceleration and angular acceleration (m/s^2 and rad/s^2), and Δt is the control cycle time (s).
   iii. Safety Braking Window V_s: To ensure that the robot can stop promptly when it detects an obstacle.

$$\begin{aligned} v_{brake} &= \sqrt{2 \cdot d_{obs} \cdot \dot{v}_{max}}, \\ \omega_{brake} &= \sqrt{2 \cdot d_{obs} \cdot \dot{\omega}_{max}}, \\ V_s &= \{(v, \omega) | v \le v_{brake}, \omega \le \omega_{brake}\}. \end{aligned} \quad (11)$$

   Specifically, d_{obs} is the distance from the robot to the nearest obstacle (m).

   The dynamic window is the intersection of the three ones:

$$V_d = V_m \cap V_a \cap V_s. \quad (12)$$

   As depicted in Figure 10.

2. Trajectory Prediction: For each velocity pair (v, ω) within the dynamic window, predict the motion trajectory within a future period of time T. Supposing that v and ω are constant, the trajectory model is as follows:

   When it $\omega \neq 0$ occurs (arc motion):

$$\begin{aligned} x(t) &= x_0 + \frac{v}{\omega}[\sin(\theta_0 + \omega t) - \sin(\theta_0)], \\ y(t) &= y_0 - \frac{v}{\omega}[\cos(\theta_0 + \omega t) - \cos(\theta_0)], \\ \theta(t) &= \theta_0 + \omega t. \end{aligned} \quad (13)$$

   When it occurs (linear motion):

$$\begin{aligned} x(t) &= x_0 + vt\cos(\theta_0), \\ y(t) &= y_0 + vt\sin(\theta_0), \\ \theta(t) &= \theta_0. \end{aligned} \quad (14)$$

   Specifically, x_0, y_0, θ_0 are the initial position (m, m) and heading angle (rad) of the robot, t is the time variable (s), within the range $t \in [0, T]$, and T is the duration of trajectory prediction, as depicted in Figure 11.

3. Trajectory Evaluation Function: For each trajectory, the multiobjective score is calculated, typically encompassing the following four aspects: heading toward the target, obstacle clearance, velocity, and target distance.

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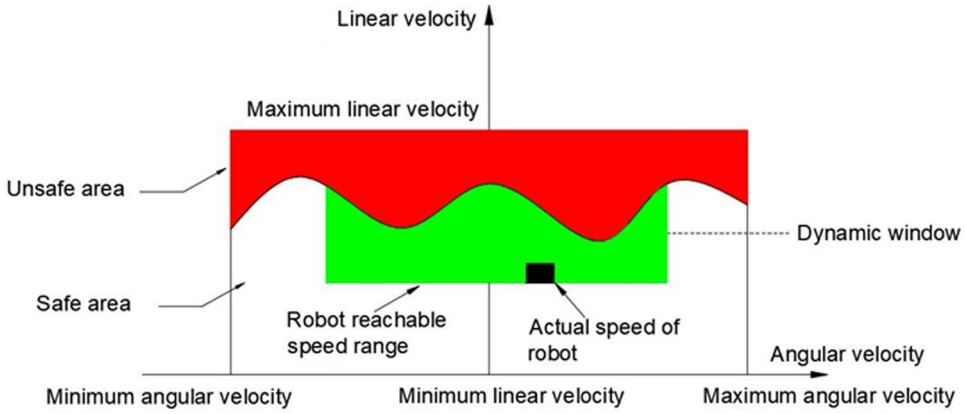


FIGURE 10 | Schematic diagram of the dynamic window approach.

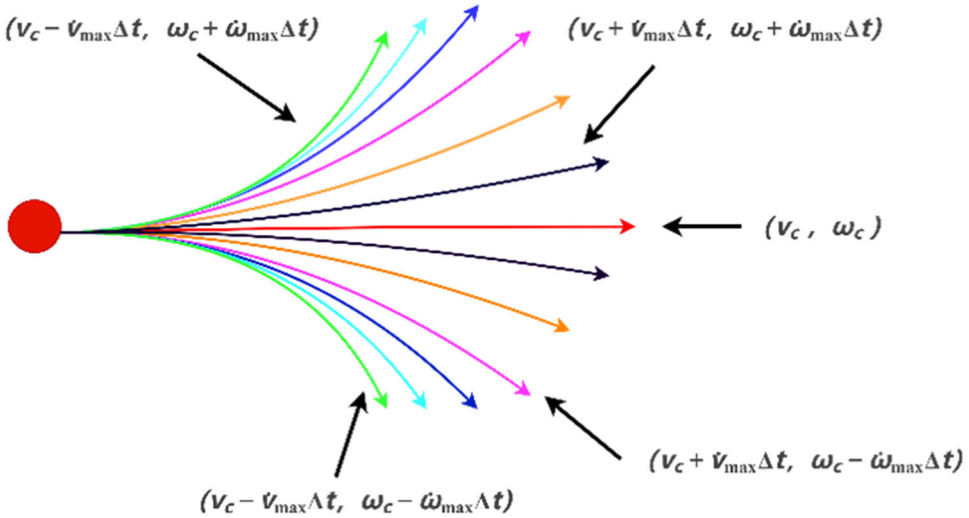


FIGURE 11 | Dynamic trajectory prediction.

Therefore, the comprehensive score is the weighted sum:

$$Score(v, \omega) = \alpha \cdot Heading + \beta \cdot Clearance + \gamma \cdot Velocity + \delta \cdot Distance. \quad (15)$$

Specifically, $\alpha, \beta, \gamma, \delta$ are the weighting coefficient, which requires adjustment based on the scenario, typically $\alpha + \beta + \gamma + \delta = 1$.

DWA constructs a feasible velocity space by considering the robot's kinematic and dynamic constraints in the context of path planning for agricultural robots. Candidate trajectories are generated through sampling velocity combinations, and the optimal path is determined using a multiobjective evaluation function. Meanwhile, real-time environmental information is obtained via sensors such as LiDAR and vision systems, enabling dynamic adjustment of the velocity window and evaluation weights. This ensures that the robot can safely and efficiently perform tasks like sowing and harvesting in complex farmland environments while balancing operational accuracy and crop protection. as detailed in Tables 7 and 8.

DWA exhibits strong real-time obstacle-avoidance capabilities in the path planning of agricultural robots. Sampling feasible velocity regions and optimizing trajectories under environmental constraints eliminates the need for a global map while enabling rapid response, making it particularly suitable for low-speed dynamic scenarios. However, the DWA has notable limitations. It is prone to being trapped in local optima, which may result in suboptimal paths that fail to achieve global optimality. Its reliance on sensor data introduces potential inaccuracies, increasing the risk of planning errors. Furthermore, its applicability is primarily restricted to local obstacle avoidance, necessitating integration with other algorithms for effective global planning. Without such integration, achieving distant targets becomes significantly challenging.

Cailian et al. (2021) proposed a real-time path-planning (RPP) method for greenhouse robots by integrating the improved A* algorithm with DWA. Specifically, they utilized DWA to construct a global evaluation function and employed ultrasonic sensors for local obstacle avoidance, thereby establishing a multiscale framework. Tianyi et al. (2023) restructured the traditional DWA evaluation function to meet the obstacle-avoidance requirements of electric tractors and established a multiobjective optimization model to determine the optimal obstacle-avoidance trajectory. Y. Li, Li, et al. (2022) suggested building maps of densely planted jujube orchards using LiDAR and combined the improved A* algorithm with DWA to achieve

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TABLE 7 | Some application scenarios and influences of dynamic window approach (DWA).

Actual scenarios	Concrete impacts
Navigation of rice paddy transplanting robots (Yong et al. 2024)	The DWA enables real-time adjustment of the robot's motion parameters to adapt to the slippery terrain and fluctuating water levels in paddy fields. This facilitates the generation of smooth transplanting trajectories while effectively reducing the risks of slippage and rollover.
Inspection of dense vegetable greenhouses (H. Guo et al. 2024)	In the narrow aisles of greenhouses, DWA integrated with real-time obstacle detection rapidly generates optimal obstacle-avoidance trajectories. This ensures that the inspection robot efficiently performs pest and disease monitoring and data collection, even in confined spaces.
Fertilizer applicators under the Trimble Green Seeker system in Australia (Mirzakhaninafchi et al. 2021)	In the wheat belt, variable-rate fertilization is conducted by incorporating the three-dimensional (3D) DWA model for *Z*-axis vibration compensation. The fertilizer applicator's 3D obstacle avoidance is achieved through the fusion of multispectral sensor data and anemometer measurements.
The John Deere X9 combine harvester is harvesting in a large wheat field (Goltyapin et al. 2021)	The technicians integrated the DWA algorithm into the AutoTrac system, setting parameters based on engine torque, adjusting weight coefficients using radar data, and optimizing the harvesting trajectory and obstacle-avoidance performance.
Ag Leader's crop monitoring vehicle equipped with OptRx sensors (Ristorto et al. 2015)	The primary window controls the speed using a dual-DWA collaborative framework, while the secondary window adjusts the sensor angle. Additionally, the RGB-D camera is utilized to identify flower stems for dynamically adjusting the path.

TABLE 8 | Challenges and solutions of DWA in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① In some regions of the paddy field, subsidence has occurred, causing the robot to tilt or become entrapped.	① Ground hardness sensors are integrated to adjust the parameters of the dynamic window in real-time dynamically.
② When the slope exceeds 30°, the driving wheels experience skidding, resulting in substantial path deviation.	② A slope-adaptive control algorithm is employed to dynamically limit the maximum speed and steering angle, enhancing stability on inclined surfaces.
③ The loose soil texture exacerbates the instability of the robot's movement.	③ IMU data are utilized to continuously correct the robot's pose in real-time, improving its antiskid performance.
④ High humidity causes fogging on the LiDAR lens, leading to missed obstacle detections.	④ Infrared cameras and ultrasonic sensors are fused to compensate for deficiencies in visual data, ensuring reliable obstacle detection.
⑤ Strong gusts of wind cause the outdoor-operating robot to deviate from its preset path.	⑤ Wind speed sensors are integrated to adjust the antiwind weighting factor in the motion model dynamically, improving path tracking accuracy under windy conditions.

Abbreviations: DWA, dynamic window approach; IMU, inertial measurement unit; LiDAR, light detection and ranging.

global and real-time local path planning, effectively reducing navigation deviation on specific paths. K. Zhang (2024) developed a path-planning method for robots in densely planted jujube orchards by integrating the improved A* algorithm with DWA to optimize global paths and collaborative real-time obstacle avoidance. Additionally, they introduced a smoothness index into DWA to enhance precise navigation. S. Wang, Hu, et al. (2023) designed an adaptive algorithm that integrates Deep Deterministic Policy Gradient (DDPG) and DWA to solve the obstacle-avoidance problem in complex spaces. They leveraged DDPG for autonomous learning strategies and constructed a reward function combined with experience replay to optimize DWA further.

DWA is expected to evolve in two primary directions. First, it will integrate predictive control with kinematic constraints by leveraging Model Predictive Control (MPC) for motion trajectory prediction, optimizing decision-making through the kinematic model of agricultural robots, and employing Moving Horizon Estimation to correct errors. This integration aims to

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reduce slippage risks and improve trajectory accuracy. Second, DWA will enhance its coupling with cognitive reasoning and operational tasks by translating operational goals into motion constraints via a cognitive architecture. The assistance of semantic segmentation networks and Temporal Convolutional Networks will enable the planning of paths that meet operational requirements while ensuring practical obstacle avoidance and adherence to operational time windows.

3.5 | Comparison of Traditional Classical Algorithms

The APF method, the Dijkstra algorithm, the A* algorithm, and the DWA each possess distinct advantages and are suited to specific application scenarios. The choice of an appropriate algorithm is contingent upon the characteristics of the agricultural environment. For instance, the APF method is well-suited for addressing simple robot navigation problems; the Dijkstra algorithm is effective for determining the shortest path in unweighted graphs; the A* algorithm is widely employed for shortest path planning in static environments; and the DWA is particularly advantageous for real-time path updates in dynamic environments. Table 9 provides a comprehensive comparison of these four classical algorithms.

To comprehensively evaluate the performance of the APF method, the Dijkstra algorithm, the A* algorithm, and the DWA in agricultural environment path planning, this study conducts a systematic comparison across six dimensions: path optimality, real-time obstacle avoidance, cost-effectiveness, computational efficiency, environmental adaptability, and scalability. The comparison results are visualized using a radar chart, and the performance of each algorithm under different metrics is quantified through a scoring system ranging from 1 to 5 (from weak to strong), as illustrated in Figure 12.

4 | Modern Intelligent Bionic Algorithms

Modern intelligent bionic algorithms for path planning in agricultural robots are computational intelligence algorithms inspired by natural biological systems' behaviors, structures, or physiological characteristics. By simulating collaboration, evolutionary mechanisms, and individual perception and learning processes observed in organisms, these algorithms translate biological principles into mathematical models and computational methods to solve complex optimization, search, and pattern recognition problems. They exhibit self-organization, self-adaptation, and self-learning capabilities, enabling them to identify optimal or near-optimal solutions within intricate problem spaces, as shown in Figure 13.

Bionic algorithms play a critical role in the path planning of agricultural robots. For instance, by mimicking the collaborative behavior of fish schools, robots can dynamically adjust their trajectories based on environmental feedback and peer path information, effectively avoiding obstacles and irrigation infrastructure. Additionally, precise localization can be achieved by leveraging the navigation strategies of migratory birds and integrating sensor data with environmental features. These biologically inspired approaches transform natural behaviors into mathematical frameworks and computational techniques, allowing robots to adapt to complex agricultural terrains and improve operational automation swiftly. Current mainstream algorithms include the ant colony algorithm (ACA) based on intelligent optimization (Dorigo et al. 2006), the genetic algorithm (GA) (Haldurai et al. 2016), and the particle swarm optimization (PSO) algorithm (Kennedy and Eberhart 1995), among others.

4.1 | Ant Colony Algorithm

In 1992, the Italian scholar Dorigo put forward the ACA (ACO). It is an intelligent optimization strategy inspired by the foraging habits of natural ant colonies. It ingeniously mimics the communication among ants while exploring food sources by releasing and following a chemical substance known as a "pheromone." The path with a higher concentration of secreted pheromone will likely be the shortest or optimal between the nest and the food source, as depicted in Figure 14.

The core steps are as follows, as shown in Figure 15:

1. Pheromone Initialization: At the initial stage, the pheromone concentrations of all paths are set to the identical value:

$$\tau_{ij}(0) = \tau_0 \quad (\forall\, i, j\,). \tag{16}$$

   Specifically, pheromone concentration on t path (i, j) at time $\tau_{ij}(t)$.

2. Formula for Path Selection Probability: The probability that ant k selects the next node j at node i:

$$P_{ij}^{k}(t) = \begin{cases} \dfrac{[\tau_{ij}(t)]^{\alpha} \cdot [\eta_{ij}]^{\beta}}{\sum l \in allowedk\,[[\tau_{il}(t)]^{\alpha} \cdot [\eta_{il}]^{\beta}]} & \text{if } j \in allowedk, \\ 0 & \text{otherwise.} \end{cases} \tag{17}$$

   Specifically, η_{ij} is the heuristic information, typically the distance $\eta_{ij} = 1/d_{ij}$ of path (i, j) (where d_{ij} denotes the distance between two nodes). α is the pheromone weight factor ($\alpha \geq 0$), which governs the impact of pheromone concentration on selection. β is the heuristic information weight factor ($\beta \geq 0$), which regulates the influence of heuristic information. *allowedk* is the set of nodes currently accessible to ant k.

3. Rule for Pheromone Update:

   After each round of iteration is accomplished, the pheromone of all paths is updated in accordance with the following rules:

$$\tau_{ij}(t+1) = (1-\rho) \cdot \tau_{ij}(t) + \Delta\tau_{ij}(t). \tag{18}$$

   Specifically, ρ is the pheromone evaporation rate $0 < \rho < 1$, which is utilized to simulate the natural evaporation of pheromones. $\Delta\tau_{ij}(t)$ is the total quantity of pheromone released by all ants on path (i, j) in this round of iteration.

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TABLE 9 | Comparison of traditional classical algorithms.

Name	Computational efficiency	Cost-effectiveness	Case studies
APF	It is fast in simple scenarios but prone to inefficiency in complex ones.	The pre-calculation cost is low, but additional treatment is necessary when trapped in a local optimum. The scenario significantly affects the overall benefit.	The path planning efficiency of inspection robots using the improved APF method has increased by 37%, the average operation time has been shortened to 2.3 h per occasion, and the success rate of obstacle avoidance has reached 98.6%. This study provides theoretical support for the autonomous operation of agricultural robots in facility agriculture (János and Matijevics 2010).
Dijkstra's algorithm	The time complexity is high and the efficiency is low for large-scale maps.	Although the computational cost is high, it can guarantee the discovery of the globally optimal path and is applicable to scenarios with high path accuracy demands.	For the path planning of transplanter replanting, when the improved bidirectional Dijkstra algorithm is employed for path planning, the average path movement distance is decreased by 14.74%, the average number of turns is reduced by 8%, and the average movement time is lowered by 13.41% (Luo and Zhang 2007).
A* algorithm	It is typically more efficient than Dijkstra's algorithm, particularly in large-scale maps, where its superiority is more pronounced.	The computational cost is relatively low, and the benefit is favorable if the path quality is guaranteed.	With the application of small intelligent pesticide spraying robots, the maximum distance deviation in path planning has been reduced by 31.8%, the maximum heading angle deviation by 28.6%, and the safety distance has been increased by 30% simultaneously (Y. Wang, Fu, et al. 2024).
DWA	Real-time computation can respond promptly to environmental variations, but the computational amount rises when the search space is large.	The calculation cost is moderate. It can effectively handle dynamic environments, guarantee robot safety and mobility, and be relatively effective.	The intelligent lawn mowing robot, via path planning based on the dynamic window method, achieves a 100% row-by-row traversal coverage rate and a 6.56% traversal repetition rate. It effectively solves the issue of operation dead points caused by obstacles in the environmental map and accomplishes the task of complete traversal of the working area (L. Liu, Liu, et al. 2024).

Abbreviations: APF, artificial potential field; DWA, dynamic window approach.

Calculation of pheromone increment $\Delta\tau_{ij}(t)$:

$$\Delta\tau_{ij}(t) = \sum_{k=1}^{m} \Delta\tau_{ij}^{k}(t). \tag{19}$$

The contribution $\Delta\tau_{ij}^{k}(t)$ of a single ant k is typically defined as

$$\Delta\tau_{ij}^{k}(t) = \begin{cases} \frac{Q}{L_k} & \text{if the path traversed by ant}k\ (i,j), \\ 0 & \text{otherwise.} \end{cases} \tag{20}$$

Specifically, Q is the pheromone intensity constant and L_k the total cost of the path of ant k in this round.

In the agricultural domain, the ACO algorithm is critical in facilitating path planning for agricultural robots. The algorithm iteratively enriches the pheromone information associated with high-quality paths by simulating "virtual ants" that traverse potential operation paths and select movement directions based on pheromone concentration. This process enables the identification of optimal or near-optimal paths from the starting point to the endpoint. The ACO algorithm is particularly suited for global planning tasks in complex farmland environments with multiple obstacles and targets. Leveraging distributed parallel computation, it effectively addresses uncertainties in dynamic environments and demonstrates significant advantages in multirobot collaborative scenarios, as detailed in Tables 10 and 11.

ACO exhibits several limitations in the path planning of agricultural robots. First, its slow convergence rate hinders the ability to meet real-time requirements in large-scale farmland applications. Second, the algorithm's sensitivity to the pheromone evaporation coefficient and heuristic factor increases the risk of premature convergence or entrapment in local optima when suboptimal parameter tuning. Third, in multirobot

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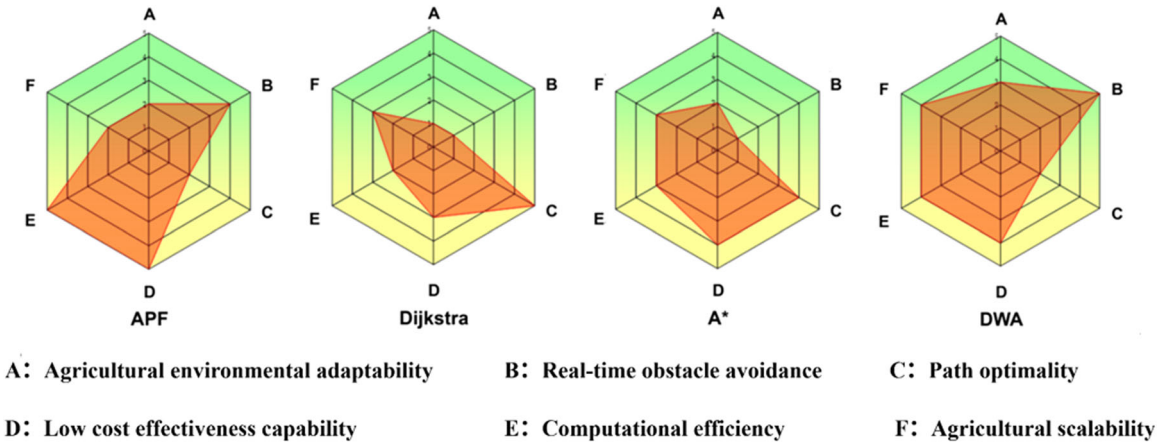


FIGURE 12 | Radar chart of performance comparison of traditional classical algorithms. A, agricultural environmental adaptability; B, real-time obstacle avoidance; C, path optimality; D, low cost-effectiveness capability; E, computational efficiency; F, agricultural scalability. APF, artificial potential field; DWA, dynamic window approach.

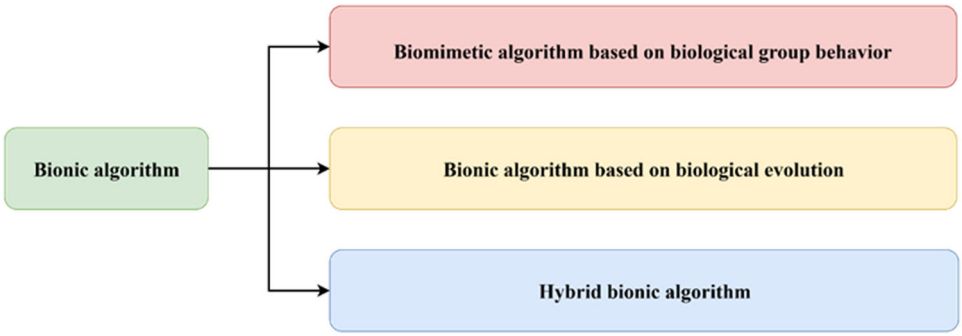


FIGURE 13 | Schematic diagram of bionic algorithm.

collaborative scenarios, the distributed nature of the algorithm can lead to path conflicts, necessitating a robust coordination mechanism for collision avoidance. Lastly, the generated paths often lack smoothness, requiring interpolation optimization to ensure compliance with dynamic constraints.

Q. Wu, Chen, et al. (2024) proposed a hybrid intelligent algorithm integrating ACO and PSO for agricultural information collection robots. By leveraging the global search capability of PSO and the local obstacle-avoidance feature of ACO within an adaptive framework, they developed dynamic pheromone update and coevolution strategies, providing a novel technical solution for autonomous navigation. Nguyen et al. (2017) introduced a simulation and optimization framework for agricultural irrigation and fertilization scheduling, with ACO as the core algorithm to enhance the efficiency of solving complex agricultural systems. Haichen et al. (2023) presented an improved ACO framework incorporating harmony search to meet the mechanization requirements of vegetable production. They effectively overcame local optima by optimizing the pheromone update strategy using the sigmoid function and integrating the harmony search strategy. Liang et al. (2023) designed a hybrid framework combining whale optimization algorithms (WOAs) to address the limitations of traditional ACO. They employed strategies such as reverse learning to construct a collaborative model and adaptively tuned ACO parameters, offering an efficient approach for intelligent path planning in agricultural equipment. Tian et al. (2023) developed a collaborative framework integrating WOA to overcome the drawbacks of traditional ACO in path planning. By introducing strategies like reverse learning, they constructed a hybrid model, formulated a multiobjective function considering path geometry, turning smoothness, and energy consumption, and adaptively adjusted ACO parameters. This improvement primarily focused on optimizing turning paths to achieve smoother turns.

ACO presents three key future development directions in the agricultural domain. First, it integrates bionic chemical signals with multimodal information to optimize pheromone updates and enhances exploration and guidance capabilities by leveraging environmental and visual data. Second, it incorporates edge computing and lightweight algorithms, adapting to edge devices through event-driven mechanisms and pruning strategies while enabling multimachine collaboration via federated learning. Third, it leverages digital twin pretraining and online correction to construct a pre-generated path library in a virtual environment, combining RL and filtering techniques to improve robustness in dynamic scenarios.

4.2 | Genetic Algorithm

John Holland of the United States introduced the GA in the 1970s. GA is an optimization algorithm that emulates

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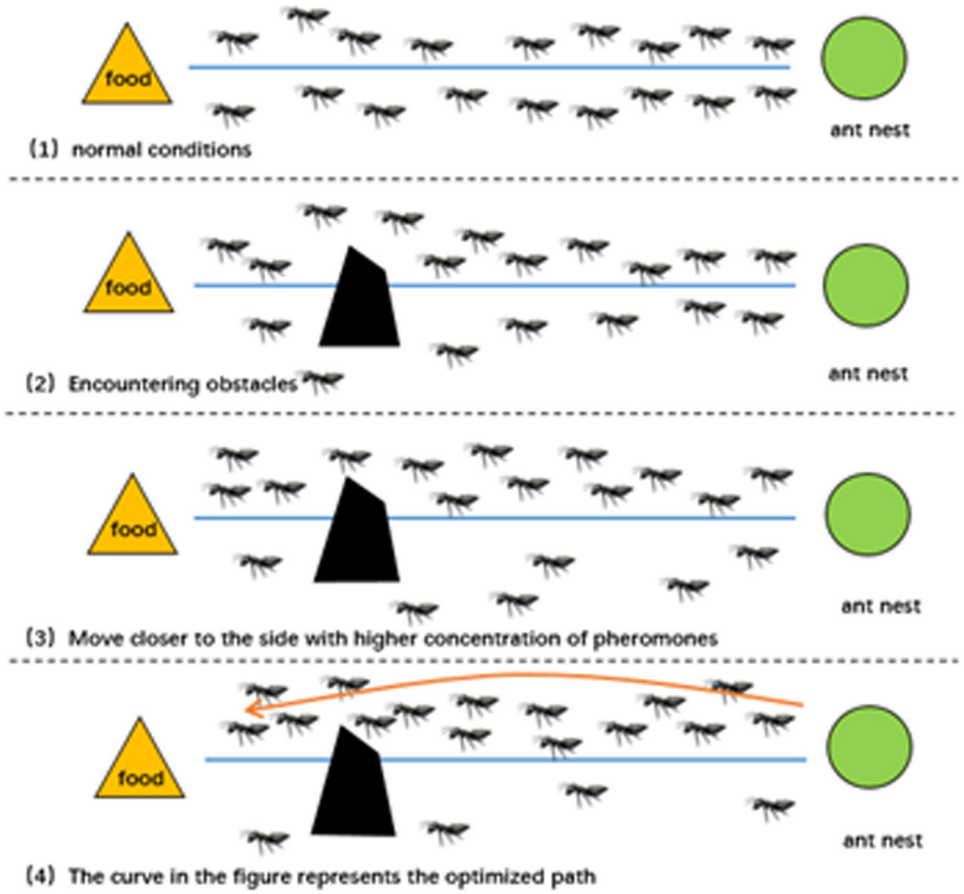


FIGURE 14 | Schematic diagram of the ant colony optimization algorithm.

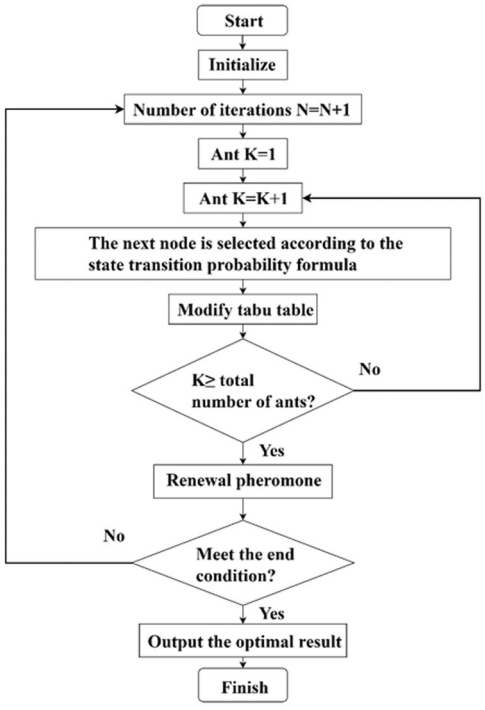


FIGURE 15 | Flowchart of the ant colony optimization algorithm.

the process of natural biological evolution. Rooted in Darwin's theory of natural selection and Mendel's principles of genetics, GA iteratively improves a population through operations such as selection, crossover, and mutation to converge toward the optimal solution of the problem. The core steps of GA include initializing the population, evaluating fitness, selecting individuals, performing crossover and mutation, forming a new generation, and assessing termination conditions, as illustrated in Figure 16.

1. Initialize the Population: Generate a set of initial candidate solutions.

   Randomly generate N individuals, each represented by a chromosome X_i. The encoding method can be binary, real number, or permutation encoding. For example, the length of a binary-encoded chromosome is L:

$$X_i = (x_1, x_2, ..., x_L), \quad x_j \in \{0, 1\}, i = 1, 2, ..., N. \tag{21}$$

   Specifically, N is the population size, usually taken as $50 \leq N \leq 200$.

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TABLE 10 | Some application scenarios and influences of ACO.

Actual scenarios	Concrete impacts
Coordinated transplanting of rice in terraced fields (N. Yang et al. 2010)	Simulating the pheromone mechanism of ant colonies, the coverage paths of multiple rice transplanters are optimized to prevent repetitive operations and missed planting areas and to accommodate the irregular terrain of terraced fields.
Full-coverage picking in orchards (C. Zhang, Wang, et al. 2023)	By leveraging the accumulation of pheromones, robots are guided to systematically traverse unharvested areas while dynamically adjusting path weights to avoid regions with dense vines. This approach reduces fruit omission rates, balances the workload among multiple machines, and effectively shortens overall operation time.
AGCO Fendt Xaver Small Autonomous Seeding Robot (Pejaković and Gronauer 2023)	Interrow planning for corn sowing is accomplished via multimachine collaboration employing the ACA. LiDAR is utilized to construct three-dimensional maps, and soil moisture information is bound as pheromones for distributed path optimization.
Yanmar YV01 Unmanned Spraying Robot (Milburn et al. 2023)	The ACA is applied to pesticide spraying in rice fields, integrating multispectral cameras to identify pests, diseases, and weeds. This approach prioritizes spraying in severely affected areas while preventing redundant spraying and enhancing efficiency and resource utilization.
The Harvest Automation HV-100 nursery handling robot in the United States (Misiuk 2023)	An enhanced ACO algorithm is integrated with laser radar perception capabilities in the context of automated handling for nursery potted plants. This system incorporates task-priority mechanisms, enabling multiple robots to mark operational areas using pheromone-based signaling. Consequently, path planning achieves greater efficiency while minimizing risks associated with navigation and coordination.
Farm Wise Titan FT-35 Fully Automatic Weeding Robot (Bogue 2024)	An AI camera is employed to identify seedling density, with the ideal plant spacing as a heuristic objective. pheromone concentration increases during thinning, enabling the robot to prioritize overly dense regions while minimizing potential damage to retained plants.

Abbreviations: ACA, ant colony algorithm; ACO, ant colony optimization; LiDAR, light detection and ranging.

TABLE 11 | Challenges and solutions of ACO in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① Rainfall-induced scouring of farmland results in the degradation and invalidation of ground pheromones.	① A virtual pheromone mechanism is introduced to synchronize path status in real-time through wireless communication, ensuring seamless coordination among multiple robots.
② Arc-shaped or serrated field ridges lead to incomplete path coverage, compromising the effectiveness of the pheromone-based guidance system.	② The pheromone release strategy is dynamically adjusted to prioritize the increased density of boundary nodes, enhancing coverage and accuracy at field edges.
③ Weeds at the edges of fields are not adequately incorporated into the weeding paths, potentially leading to suboptimal weed control.	③ Aerial data from uncrewed aerial vehicles are integrated to refine the connectivity weights of farmland boundaries, improving the robustness of path planning.
④ Path overlap causes some robots to experience an excessive workload, leading to inefficiencies in task distribution.	④ A load balancing factor is incorporated to modulate the intensity of pheromone release, optimizing resource allocation and preventing task overload.
⑤ Frequent path queries result in system response delays, affecting real-time operation and coordination.	⑤ A pre-generated main path library is utilized, and the ACO algorithm is employed to rapidly refine local paths, ensuring efficient and adaptive route optimization.

Abbreviation: ACO, ant colony optimization.

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2. Selection: The selection process involves identifying high-quality individuals as parents, typically employing methods such as roulette wheel selection and tournament selection.

   Crossover: The crossover operation recombines the genes of selected parents to generate offspring, with commonly used techniques including single-point crossover and uniform crossover.

3. Mutation: Random disturbances are introduced to maintain the diversity of the population.

   Binary encoding:

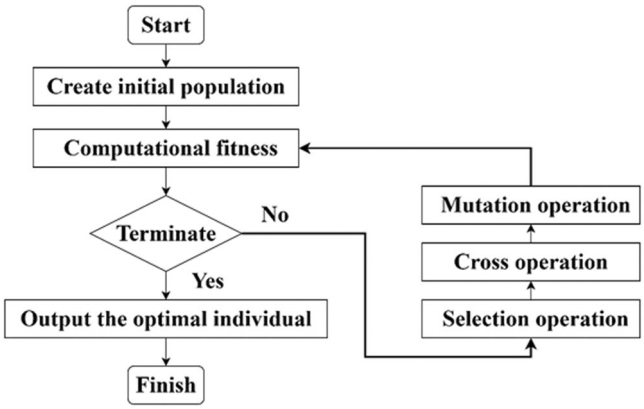


FIGURE 16 | Flowchart of genetic algorithm.

$$x_{j'} = \begin{cases} 1 - x_j & \text{with a probability of } P_m, \\ x_j & \text{with a probability of } 1 - P_m, \end{cases} \quad (22)$$
$$j = 1,2, ..., L.$$

4. Forming a New Population: Merge the parent and offspring generations and retain the elite individuals.

$$\text{New population} = X_{\text{best}} \cup \left(\bigcup_{k=1}^{N-1} Progeny_k \right). \quad (23)$$

GA models the paths of agricultural robots as individual entities within a population, encoding these paths into genotypes that incorporate information such as turning points. Fitness evaluation is used to identify superior paths, while crossover and mutation operations generate new combinations to explore the solution space systematically. Through multiple generations of evolution, the algorithm converges on an optimal path that satisfies the requirements of agricultural operations. This enhances the precision of robot operations, minimizes crop damage, optimizes water and fertilizer management, and balances resource efficiency with ecological benefits, thereby supporting intelligent path planning for sustainable agriculture. The detailed results are presented in Tables 12 and 13.

GAs, which simulate natural selection and genetic mechanisms, optimize path populations via crossover and mutation operations. These algorithms are well-suited for global path search and multiobjective optimization in high-dimensional and complex

TABLE 12 | Some application scenarios and influences of GA.

Actual scenarios	Concrete impacts
Emergency harvest of farmlands following typhoons (L. Liu, Liu, et al. 2024)	GAs generate disaster emergency routes through multiple generations of iterations, prioritizing avoiding lodged crops and waterlogged areas and balancing the efficiency of emergency harvesting and equipment safety.
Multilevel operations in vertical farms (De Ocampo and Dadios 2017)	In the three-dimensional planting rack, the GA optimizes the movement sequence of the mechanical arm by minimizing idle travel distance, thereby improving the efficiency of water and fertilizer spraying and monitoring.
Cross-regional multiobjective sowing (J. Ji et al. 2021)	In the context of scattered farmland plots and multiple crop varieties, the GA generates the globally optimal seeding path, balancing distance, soil adaptability, and seeding priority and reducing the no-load rate of the equipment.
Claas Lexion harvester (Stawicki and Sędłak 2015)	The harvester leverages the CEMOS automatic navigation system to encode the farmland information via the GA, establish a dual fitness function for iterative optimization, and mitigate repetitive, empty running.
John Deere 8 R 410 Tractor (Uberti et al. 2025)	To achieve fully automatic contour tillage in complex terrain farmland for soil erosion prevention, the system utilizes a GA to process terrain data, encodes path genes, generates curved paths based on specific indicators, and selects the optimal solution for soil and water conservation.
AGCO Fendt 936 Vario high-horsepower tractor (Heisler et al. 2009)	By utilizing the HerdPath system, collaborative scheduling encodes the progress of operations, vehicle capacity, and silo positions into a dynamic gene matrix. To minimize travel distance and energy consumption, the GA optimizes the sequence of harvesting and transportation while dynamically adjusting routes in real-time.

Abbreviation: GA, genetic algorithm.

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TABLE 13 | Challenges and solutions of GA in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① In the event of a sudden weather-induced path failure, the replanning process using the GA is time-consuming, leading to delays in emergency response.	① Introduce catastrophe operators to forcibly escape from local optima, enhancing global search capability.
② Sudden changes in crop lodging connectivity result in insufficient convergence speed of the population, hindering timely optimization.	② Leverage real-time sensor data to dynamically update the fitness function, accelerating iterative response and improving adaptability.
③ The allocation of multiobjective weights is highly subjective, potentially compromising the robustness and reliability of the optimization results.	③ Incorporate an energy consumption model as a core parameter in the fitness function to optimize resource utilization.
④ In farm path optimization, individuals with similar structural characteristics tend to dominate the population prematurely, limiting diversity and exploration of the solution space.	④ Develop a diversity preservation strategy to maintain population diversity and prevent premature convergence.
⑤ Significant differences in farmland accessibility between rainy and dry seasons render a fixed fitness function inadequate for universal application, necessitating adaptive adjustments.	⑤ Design a seasonally adaptive fitness function that dynamically adjusts terrain and crop parameters to accommodate varying environmental conditions.

Abbreviation: GA, genetic algorithm.

agricultural environments and have been widely adopted. However, they also exhibit certain limitations. The design of encoding schemes and fitness functions is sophisticated, and improper encoding may lead to deviations from practical requirements. Additionally, the algorithm heavily depends on the initial population size and crossover and mutation probabilities. Inappropriate parameter settings can result in premature convergence or inefficient resource utilization. Achieving global optimization and balancing multiobjective weights remain particularly challenging in high-dimensional and complex scenarios.

The improvement of GAs requires optimizing encoding schemes and enhancing search efficiency. First, variable-length chromosome encoding is employed, with dynamic adjustment of path segmentation granularity to adapt to complex terrains. Second, local search strategies are integrated to refine the mutation operator and strengthen local optimization capabilities. Rahman et al. (2025) developed a hybrid intelligent framework combining PSO and GA for precise crop planting prediction based on soil classification. They designed a coevolution mechanism to leverage the strengths of both methods for decision tree parameter tuning. J. Han et al. (2024) proposed an enhanced algorithm framework integrating Markov chains for robot full-coverage path planning, addressing the limitations of traditional GAs. They constructed a state transition model to guide individual mutations, improving population quality and boosting search efficiency. Noguchi and Terao (1997) introduced a path optimization method for agricultural mobile robots by integrating NNs and GA. They applied GA to optimize control parameters such as turning angle time series to achieve global optimality in path planning. Mahmud et al. (2019) presented a multiobjective optimization framework for planning a greenhouse pesticide spraying robot path. Using the NSGA-III algorithm, they designed specialized operators and strategies to optimize paths effectively. Zou et al. (2017) proposed a nonlinear programming-based GA optimization framework for motion control of watermelon-picking robotic arms. On the basis of the D-H model, they established a kinematic model and solved the inverse kinematics problem using a nonlinear GA to optimize joint angles. Cao et al. (2019) developed an optimized rapidly exploring random trees (RRT) algorithm for obstacle avoidance in litchi-picking robots. Adjusting sampling and expansion rules improved search efficiency and incorporated an improved GA for smoothing path nodes.

In the future, GAs are expected to develop in three key directions within the agricultural domain. First, MOEAs combined with dynamic weight allocation will be employed to resolve multiple conflicting objectives in agriculture. By integrating adaptive strategies and FL systems, weights can be dynamically adjusted to generate Pareto-optimal solution sets. Second, research into multirobot coevolution and distributed optimization will be expanded. A framework will be developed to enable efficient information exchange and collaborative planning among multiple robots, thereby optimizing resource allocation. Third, advancements in hardware acceleration and ASIC design will be pursued. Technologies such as FPGAs will be utilized for parallel computing, coupled with NN compression techniques, to meet the real-time operational demands of agricultural applications.

4.3 | Particle Swarm Optimization

American scholars Kennedy and Eberhart introduced the PSO algorithm in 1995. PSO is a global optimization algorithm rooted in swarm intelligence inspired by the foraging behavior of bird flocks or fish schools. The core principle of PSO is to represent potential solutions as particles in a multidimensional search space, where each particle possesses a position and velocity. Particles iteratively update their velocities and positions based on their historical best positions and the best positions discovered by other particles in the swarm. This iterative process allows particles to explore the search space collaboratively, sharing information to converge toward optimal or near-optimal solutions. The key mechanisms of PSO include

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velocity update, position update, and fitness evaluation, as illustrated in Figure 17.

1. Initialize the particle swarm:

   Randomly generate N particles, each of which encompasses an initial position $\boldsymbol{x}_i$ and velocity $\boldsymbol{v}_i$.

   ① Position initialization:

$$x_i^{(0)} = x_{\min} + r_1 \odot (x_{\max} - x_{\min}). \tag{24}$$

   Among them, $x_{\max}, x_{\min}$ are the vectors of the upper and lower bounds of the solution space. r_1 is a uniformly distributed random vector. $\odot$ is the elementwise multiplication.

   ② Initialization of speed:

$$v_i^{(0)} = v_{\min} + r_2 \odot (v_{\max} - v_{\min}). \tag{25}$$

   Among them, $v_{\max}, v_{\min}$ are the upper and lower bound vectors of speed.

2. Compute fitness:

   The fitness of each particle is assessed via the objective function $f(x)$:

$$f_i = f\left(x_i^{(t)}\right). \tag{26}$$

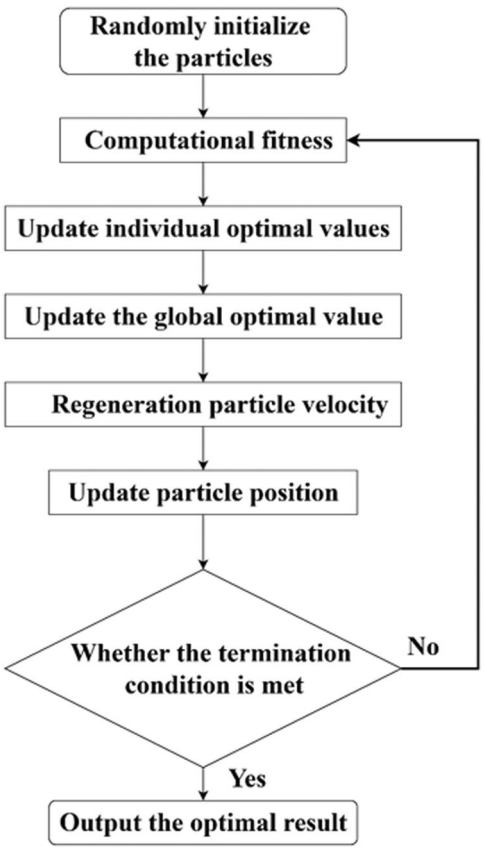


FIGURE 17 | Flowchart of the particle swarm optimization algorithm.

3. Update the individual optimum: Compare the current fitness with the historical optimum and retain the superior solution.

$$p_i^{(t+1)} = \begin{cases} x_i^{(t)} & \text{if } f\left(x_i^{(t)}\right) \text{ outperforms } f\left(p_i^{(t)}\right), \\ p_i^{(t)} & \text{otherwise.} \end{cases} \tag{27}$$

   Among them, $p_i^{(t)}$ is the individual historical optimal location of particle i in the t generation.

4. Update the group optimum: Select the global optimum from all individual optima.

$$g^{(t)} = \underset{p_i^{(t)}}{\operatorname{argmin}} f\left(p_i^{(t)}\right) \quad \text{(minimization issue)} \tag{28}$$

   Or

$$g^{(t)} = \underset{p_i^{(t)}}{\operatorname{argmax}} f\left(p_i^{(t)}\right) \quad \text{(maximization issue).} \tag{29}$$

5. Update the velocities and positions of the particles:

   ① Velocity update formula:

$$v_i^{(t+1)} = w \cdot v_i^{(t)} + c_1 r_1 \odot \left(p_i^{(t)} - x_i^{(t)}\right) + c_2 r_2 \odot \left(g^{(t)} - x_i^{(t)}\right). \tag{30}$$

   Among them, w is the inertia weight, c_1, c_2 are the individual learning factor and social learning factor, and $r_1, r_2$are the uniformly distributed random vectors.

   ② Position update formula:

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)}. \tag{31}$$

6. Boundary treatment: Impose constraints on positions or velocities that exceed the solution space.

   ① Position clipping:

$$x_{ij}^{(t+1)} = \min\left(\max\left(x_{ij}^{(t+1)}, x_{\min,j}\right), x_{\max,j}\right). \tag{32}$$

   ② Speed clipping:

$$v_{ij}^{(t+1)} = \min\left(\max\left(v_{ij}^{(t+1)}, v_{\min,j}\right), v_{\max,j}\right). \tag{33}$$

In the path planning of agricultural robots, PSO utilizes factors such as path length as fitness evaluation criteria to search for the optimal path iteratively. In precision agriculture, PSO can optimize multiple variable parameters and rapidly converge to a balanced allocation of resources and yields by leveraging the synergy among particles. Moreover, its parallel search capability makes it particularly suitable for addressing large-scale complex nonlinear problems in farmland scenarios, as shown in Tables 14 and 15.

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TABLE 14 | Some application scenarios and influences of PSO.

Actual scenarios	Concrete impacts
Multirobot cooperative pesticide spraying (L. Fan 2023)	The PSO algorithm optimizes the coverage paths of multiple sprayers through collaborative optimization, minimizing overlapping areas and balancing workload distribution. Adapting to irregular field shapes enhances pesticide utilization efficiency and overall operational performance.
Dynamic weather-adaptive path planning (Du et al. 2017)	By integrating real-time meteorological data, the PSO algorithm dynamically adjusts path weights to avoid muddy or waterlogged areas, thereby reducing the risk of equipment malfunction and ensuring continuous operation under adverse weather conditions.
The Dino autonomous weeding robot from Naio Technologies (Epée Missé et al. 2020)	For interrow weeding in French vineyards, a multimodal optimization framework based on PSO is developed to jointly plan the robotic arm's trajectories and chassis. This approach achieves a weeding coverage rate of 98% in plots with steep slopes.
Valmont Valley 8000 Series Center Pivot Irrigation System (Andrade et al. 2020)	In variable irrigation for soybean fields, the sprinkler irrigation system utilizes the PSO algorithm to integrate multisource data and predictive models, optimizes the encoding of particle swarm parameters, and formulates dynamic irrigation strategies while using the water deficit index as a constraint condition.
Lemken Solitair air-seeder (Yi et al. 2024)	In the context of no-till seeding in sunflower fields, the PSO algorithm fuses radar-based terrain data with historical yield records to optimize the configuration of seeding parameters, leading to a substantial improvement in seedling emergence rates.
TORUM 785 combine harvester (Kastidi 2021)	In harvesting rye fields, the PSO algorithm optimizes the rotational speed of the threshing drum, the sieve opening degree, and the traveling speed by utilizing the load data of the threshing drum. Through the fitness function, it achieves an optimal balance between minimizing grain breakage and ensuring adequate straw crushing.

Abbreviation: PSO, particle swarm optimization.

The PSO mimics the foraging behavior of bird flocks in the path planning of agricultural robots. It updates the positions of particles based on individual and global optimal information, enabling rapid generation of suboptimal paths. While PSO is well-suited for dynamic environments due to its simplicity in computation and minimal parameter requirements, it demonstrates strong capabilities in rapidly searching complex terrains. However, notable limitations exist. The algorithm is prone to converging to local optima, particularly in complex scenarios, where improper inertia weight tuning may lead to entrapment in local extremums. Additionally, PSO is sensitive to population size and iteration count; unreasonable parameter settings can degrade search performance or increase computational overhead. Furthermore, the lack of a mechanism for memorizing and reusing historical paths limits its ability to improve efficiency during repetitive tasks.

Researchers have introduced several key enhancements to balance global and local search. First, an adaptive inertia weight mechanism is incorporated to dynamically adjust the trade-off between global exploration and local exploitation based on population diversity. Second, quantum-behaved PSO enhances the algorithm's ability to escape from local optima through quantum mechanics-inspired mechanisms. Third, the differential evolution (DE) strategy is integrated to maintain population diversity via mutation operations. Khalid et al. (2021) addressed the challenge of optimal scheduling under conflicting constraints by deeply integrating PSO with the Nawaz-Enscore-Ham (NEH) heuristic algorithm and designing a coevolutionary strategy to balance search efficiency and exploitation effectiveness. Zheng et al. (2022) proposed an improved PSO framework for agent model parameter optimization. They introduced an adaptive jump function to regulate the inertia weight and designed a nonlinear strategy for the adaptive evolution of the inertia factor. Z. Wang et al. (2022) developed a hybrid optimization framework capable of adaptively escaping from local extrema by incorporating the simulated annealing (SA) temperature parameter adjustment strategy. The study adopted the Metropolis criterion to accept inferior solutions, thereby maintaining population diversity and achieving collaborative optimization of global exploration and local exploitation. L. Chen (2023) proposed a hybrid framework combining PSO and the A* algorithm to solve the path-planning problem for fruit and vegetable-picking robots. They first constructed a global model using the A* algorithm and Manhattan distance to improve efficiency, then applied PSO for secondary optimization with dynamic particle step size adjustment, and finally designed a spline interpolation smoothing algorithm to ensure connectivity and smoothness. Z. Yan and Jingtao (2019) proposed an enhanced weighted particle swarm optimization-particle filter (WPSO-PF) algorithm framework for combined navigation in agricultural machinery. By constructing a collaborative mechanism that integrates the global search advantages of PSO with the state estimation capabilities of PF, they achieved

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TABLE 15 | Challenges and solutions of particle swarm optimization in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① The farmland environment is highly complex and dynamic, leading to the frequent emergence of locally optimal solutions.	① By integrating the simulated annealing algorithm, the positions of selected particles are periodically reset to enhance population diversity.
② Strong winds disrupt the positional and orientational stability of robots, significantly increasing path deviation.	② An antiwind interference factor is introduced into the particle velocity update mechanism to improve the robustness of the movement trajectory.
③ During multirobot collaborative operations, communication delays result in asynchronous convergence toward the global optimal solution.	③ Edge computing nodes are utilized for local path data processing, thereby reducing reliance on cloud resources and minimizing latency.
④ The operational range and precision of robotic arms are constrained by mechanical and environmental factors.	④ Height constraints are incorporated into the fitness function to prioritize low-risk channels, ensuring safer navigation.
⑤ Robots operating in mountainous terrain have limited battery capacity, and traditional paths exhibit high energy consumption.	⑤ The energy consumption model is embedded within the particle fitness function to generate Pareto-optimal paths that balance efficiency and resource utilization.

rapid and precise positioning. Guohai et al. (2023) proposed a solution for autonomous navigation and full-coverage operation of high-clearance unmanned sprayers in convex fields. They obtained the initial set of operational rows by equidistantly translating the field's geometric center baseline and optimized the operational row direction angle using an improved particle swarm algorithm. Additionally, they designed a U-turn-based path connection algorithm to generate a full-coverage path.

In the future, GAs will achieve breakthroughs in three key areas within the agricultural domain. First, by leveraging biomechanical models to optimize motion equations, incorporating bird flight models for constructing particle motion equations, and dynamically adjusting particle update rules based on terrain data while aligning with robotic kinematic models, the naturalness and executability of paths can be significantly improved. Second, dynamic neighborhood structures and adaptive environmental features will be realized by switching search modes according to obstacle density and designing directional structures with ridge distributions, thereby balancing search efficiency with crop protection. Third, exploration capabilities will be enhanced through generative adversarial networks (GANs), where a GAN framework integrated with cloud-edge-end collaboration forms an adversarial closed loop between the generator and discriminator, and RL is employed to strengthen global search capabilities using virtual path samples.

4.4 | Other Bionic Algorithms

4.4.1 | Bionic Algorithms Based on the Behavior of Biological Groups

This category of algorithms draws inspiration from the collaborative behaviors of biological groups in nature, such as foraging, migration, and cooperation, transforming these phenomena into mathematical models and computational strategies. When applied to agricultural robotics, these algorithms enable dynamic path adjustment based on real-time environmental information. Through group-based information interaction, they facilitate obstacle avoidance and route planning, enabling robots to accurately and efficiently perform diverse tasks in complex farmland terrains, such as sowing and irrigation. Table 16 summarizes additional bionic algorithms inspired by biological group behaviors.

4.4.2 | Bionic Algorithms Based on Biological Evolution

Bionic algorithms based on biological evolution simulate natural selection and genetic variation mechanisms, treating the farmland environment as the evolutionary context and the initial path as the population's starting point. Through selection, crossover recombination, and mutation, these algorithms explore optimized paths, adapt to complex farmland terrains, and design energy-efficient obstacle-avoidance paths, providing critical technical support for agricultural automation. Specific algorithms are detailed in Table 17.

4.4.3 | Hybrid Bionic Algorithms

Hybrid bionic algorithms contribute to innovative solutions in the path planning of agricultural robots by integrating the strengths of various bionic algorithms into a composite system. This approach treats farmland as a dynamic system for multisource information collection, enabling real-time environmental perception by simulating biological group interactions. Combining evolutionary optimization and iterative strategies generates optimal paths that balance efficiency, safety, and energy conservation in complex farmland scenarios, thereby promoting the intelligent upgrading of agriculture. Specific algorithms are detailed in Table 18.

4.5 | Comparison of Modern Intelligent Bionic Algorithms

The ACO algorithm, by mimicking the foraging behavior of ants, is well-suited for solving path optimization problems in

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TABLE 16 | Summary of bionic algorithms based on biological group behaviors.

Algorithm name		Time	Applications in agriculture
Artificial fish swarm algorithm (Xiaolei et al. 2002)	AFSA	2002	Aquaculture environment monitoring and optimization of farmland irrigation systems
Artificial bee colony algorithm (Karaboga and Akay 2009)	ABC	2005	Optimization of farmland irrigation scheduling and path planning for seeders
Cat swarm optimization algorithm (S-C et al. 2006)	CSO	2006	Task allocation for agricultural monitoring unmanned aerial vehicle (UAV) swarms and parameter optimization for agricultural image recognition
Firefly algorithm (X.-S. Yang 2009)	FA	2009	Optimization of greenhouse environmental parameters and path planning for pest monitoring
Cuckoo search algorithm (X.-S. Yang and Deb 2009)	CS	2009	Deployment of the farmland pest and disease monitoring network and localization of sensor nodes
Bat algorithm (X.-S. Yang 2010)	BA	2010	Dynamic adjustment of UAV flight paths in precision agriculture
Fruit fly optimization algorithm (W.-T. Pan 2012)	FOA	2012	Parameter optimization of agricultural climate prediction models and regulation of agricultural product storage
Gray wolf optimization algorithm (Mirjalili et al. 2014)	GWO	2014	Division of crop planting areas and optimization of paths for collaborative operation of agricultural machinery
Whale optimization algorithm (Mirjalili and Lewis 2016)	WOA	2016	Path planning for combine harvesters and optimization of soil sampling points in farmland
Harris hawk optimization algorithm (Heidari et al. 2019)	HHO	2019	Optimization of plant protection paths for agricultural drones and dynamic scheduling of agricultural machinery
Butterfly optimization algorithm (Arora and Singh 2019)	BOA	2019	Optimization of agricultural planting patterns
Sparrow search algorithm (Xue and Shen 2020)	SSA	2020	Obstacle-avoidance navigation for greenhouse robots and task allocation for farmland monitoring
Sand cat swarm optimization algorithm (Seyyedabbasi and Kiani 2023)	SCSO	2022	Dynamic adjustment of obstacle-avoidance paths for agricultural robots
Egret swarm optimization algorithm (Z. Chen et al. 2022)	ESOA	2022	Path planning for multirobot fruit picking in orchards and energy-saving optimization of vehicle routes in farmland transportation
The crown porcupine optimization algorithm (Abdel-Basset et al. 2024)	CPO	2024	Path planning for multiobjective monitoring in farmland and collaborative transportation paths for agricultural robots

complex environments and demonstrates exceptional performance in handling large-scale data. The GA, grounded in the principles of biological evolution, employs iterative optimization through operations such as selection, crossover, and mutation, making it ideal for addressing path-planning problems that require global optimization. The particle swarm algorithm, which simulates swarm intelligence, excels in dynamic and complex environments, enabling rapid identification of optimal paths in scenario-based path planning, as summarized in Table 19.

To comprehensively evaluate the performance of the ACO algorithm, GA, and particle swarm algorithm in agricultural environment path planning, this study conducts a systematic comparison across six dimensions: path optimality, real-time obstacle avoidance, cost-effectiveness, computational efficiency, environmental adaptability, and scalability. The comparison results are visualized using a radar chart, and the performance of each algorithm under different metrics is quantified through a scoring system ranging from 1 to 5 (from weak to strong), as illustrated in Figure 18.

5 | Planning Algorithms Based on Sampling

Sampling-based planning algorithms constitute advanced solutions for tackling the path-planning challenges of agricultural robots in high-dimensional or continuous state spaces, with RRT (Lavalle 1998) and probabilistic roadmap (PRM) (Kavraki et al. 2002) serving as representative examples. These algorithms explore potential paths within the extensive state space of agricultural operational environments through random sampling, generate candidate paths, and evaluate their suitability based on the specific requirements of agricultural tasks. Using iterative sampling and assessment, they progressively converge toward the optimal path, enabling efficient and accurate trajectory planning for robot operations in complex agricultural scenarios, as illustrated in Figure 19.

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TABLE 17 | Summary of bionic algorithms based on biological evolution.

Algorithm name		Time	Applications in agriculture
Evolutionary strategy (Beyer and Schwefel 2002)	ES	1965	Simple straight-line path planning for farmland
Differential evolution algorithm (Storn and Price 1997)	DE	1997	Real-time adjustment of unmanned aerial vehicle spraying paths in dynamic farmland environments
Covariance matrix adaptation evolution strategy (Hansen and Ostermeier 2001)	CMA-ES	2001	Path generation for agricultural machinery in complex terrain of hilly areas and path optimization for precision fertilization robots
Nondominated sorting genetic algorithm (Deb et al. 2001)	NSGA	2001	Multiobjective path planning for greenhouse robots
Multiobjective evolutionary algorithm based on decomposition (Q. Zhang and Li 2007)	MOEA/D	2007	Multiobjective optimization of vehicle routing for farmland transportation
Adaptive differential evolution algorithm (J. Zhang and Sanderson 2009)	JADE	2009	Dynamic obstacle-avoidance path optimization for orchard picking robots

TABLE 18 | Summary of hybrid bionic algorithms.

Algorithm name		Combining principle
Hybrid particle swarm–gravitational search algorithm (Khan and Ling 2021)	PSO-GSA	Combining the velocity-position update mechanism of PSO with the mass-gravity simulation of the GSA enhances the global search capability through the gravitational forces among particles, avoiding being trapped in local optima.
Artificial bee colony–differential evolution hybrid algorithm (Xiang et al. 2014)	ABC-DE	The division mechanism of employed bees and onlooker bees in the ABC is integrated with DE mutation and crossover operations, leveraging ABC's local fine-searching ability and DE's global diversity preservation capability.
Cuckoo search–firefly hybrid algorithm (Elkhechafi et al. 2018)	CS-FA	The Levy flight random exploration in CS is combined with the luminance attraction mechanism in the Firefly Algorithm (FA). The search scope is expanded via Levy flight, and local convergence is realized through luminance attraction.
Hybrid gray wolf optimization–particle swarm optimization algorithm (Singh and Singh 2017)	GWO-PSO	The $\alpha/\beta/\delta$ hierarchical leadership mechanism in GWO combines the individual-group historical optimal guidance in PSO, balancing social collaboration and individual experience, and enhancing adaptability in dynamic environments.
Ant colony–simulated annealing hybrid algorithm (Stodola et al. 2020)	ACO-SA	The pheromone positive feedback mechanism of the ACO algorithm is combined with the probabilistic acceptance strategy of inferior solutions in SA to avoid premature convergence of path search and improve adaptability to complex terrains.
Whale optimization algorithm–genetic hybrid algorithm (J. Xu and Peng 2021)	WOA-GA	The spiral-encircling predation strategy of the WOA is combined with the crossover and mutation operations of the Genetic Algorithm (GA), leveraging WOA's strong local exploitation capacity and GA's global exploration characteristic.
Harris Hawk optimization–cuckoo hybrid algorithm (Pani et al. 2023)	HHO-CS	The diving-encirclement predation strategy of HHO and the nest parasitism elimination mechanism of CS are combined, with HHO reinforcing local exploitation and enhancing global exploration efficiency.
Particle swarm optimization–simulated annealing hybrid algorithm (X. Pan et al. 2019)	PSO-SA	The group velocity update of PSO + the Metropolis criterion of SA balances the exploration and exploitation capabilities.

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TABLE 19 | Comparison of modern intelligent bionic algorithms.

Name	Computational efficiency	Cost-effectiveness	Case studies
ACO	Convergence is slow, especially for large-scale problems. The early search is random, and multiple rounds of iterations are required for optimization, with a high time complexity.	The exploration cost in the early stage is high. It relies on multiple iterations for optimization. After finding a relatively better solution, the cost is reduced. It is effective in dealing with complex problems.	The improved algorithm, in contrast to the traditional ACO algorithm, reduces the path-planning time by 38%–45%, decreases the number of turns by 33%–58%, and reduces the standard deviation of the path by 27%, significantly improving the path smoothness and operational efficiency (K. Feng et al. 2022).
GA	The convergence rate is affected by factors such as population size and genetic operators. Under reasonable settings, it can converge relatively rapidly.	The initial population and iterations consume resources, but they are good at solving complex constraints and show outstanding benefits when the global optimum is found.	By incorporating chromosome pair encoding and a multiobjective balanced fitness function, the proposed algorithm achieved a 38.54% reduction in repetitive operation areas, a 35.00% decrease in U-turns, and a 7.82% savings on average energy consumption during simulation experiments (X. Wu et al. 2023).
PSO	It converges quickly and approaches the optimal solution quickly in the early stage. However, it is prone to being trapped in local optima in the later stage and has a low computational complexity.	The computational cost is low, and it is easy to implement. It can rapidly find the optimal solution for small-scale problems with good efficacy. However, when encountering complex problems, multiple parameter adjustments might be required.	The path-planning time is consistently maintained between 0.2 and 0.4 s per instance, enabling effective avoidance of moving obstacles with speeds up to 1.5 m/s. Additionally, the path smoothness has been improved by 42%, and the objective function value (path length + steering angle) has been optimized by 28.7% (L. Xu et al. 2023).

Abbreviations: ACO, ant colony optimization; GA, genetic algorithm; PSO, particle swarm optimization.

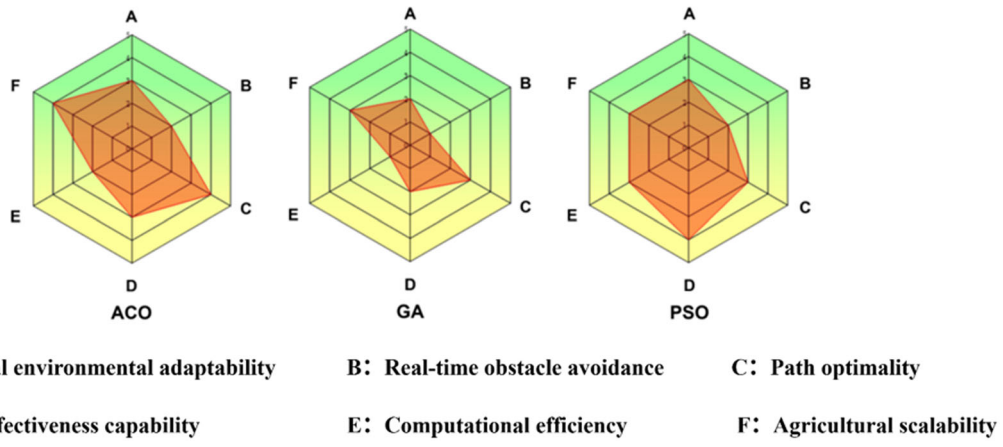


FIGURE 18 | Radar chart of performance comparison of modern intelligent bionic algorithms. A, agricultural environmental adaptability; B, real-time obstacle avoidance; C, path optimality; D, low cost-effectiveness capability; E, computational efficiency; F, agricultural scalability. ACO, ant colony optimization; GA, genetic algorithm; PSO, particle swarm optimization.

5.1 | RRT Algorithm

In 1998, Steven M. LaValle and James J. Kuffner Jr. (2001) introduced the RRT algorithm, a sampling-based path-planning method designed for high-dimensional spaces and environments with complex obstacles. RRT constructs a tree structure by randomly sampling points in the configuration space to explore feasible paths from the start to the goal point. The

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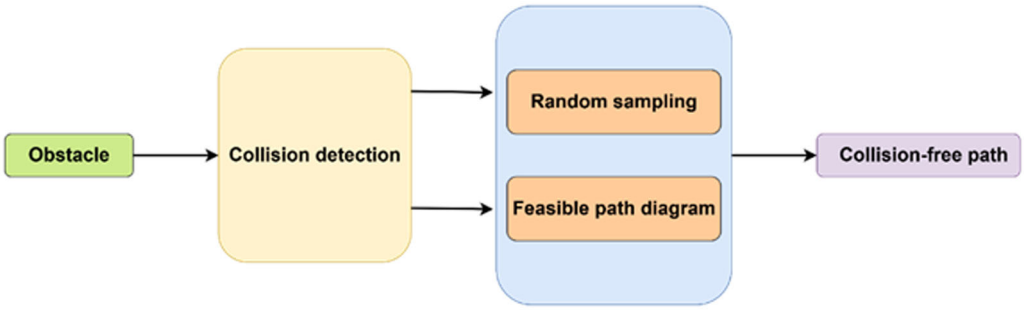


FIGURE 19 | Schematic diagram of planning algorithms based on sampling.

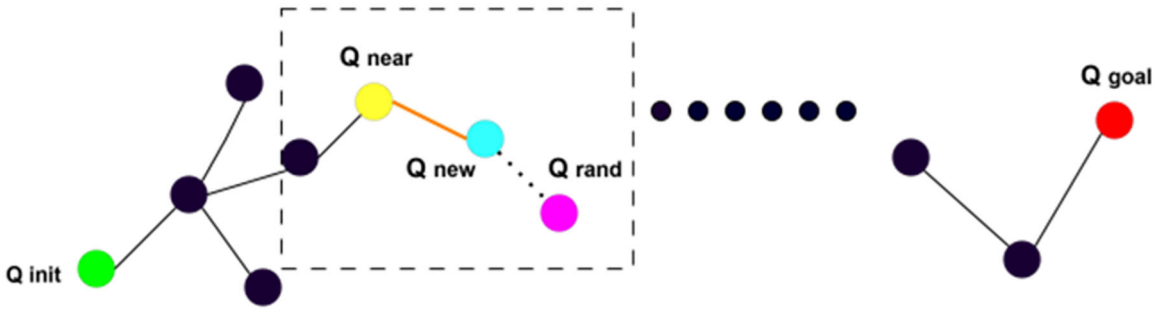


FIGURE 20 | Schematic diagram of rapidly exploring random trees algorithm.

algorithm begins at the initial node and iteratively generates new nodes in the free space, attempting to connect each new node to the nearest existing node in the tree. The tree expands progressively through repeated iterations until it reaches the target point or meets a predefined termination condition, as illustrated in Figures 20 and 21.

1. Initialization.

$$V = \{q_{\text{init}}\}, \quad E = \varnothing. \tag{34}$$

Among them, V is the set of nodes in the tree, initially including only the starting point q_{init}. E is the set of edges in the tree, initially being empty. q_{init} is the starting point.

2. Random sampling.

$$q_{\text{rand}} = Sample(X). \tag{35}$$

Among them, q_{rand} is the points randomly sampled in the state space X. X is the complete configuration space of the system.

3. Nearest-neighbor search.

$$q_{\text{near}} = \arg\min_{q \in V} d(q, q_{\text{rand}}). \tag{36}$$

Among them, q_{near} is the node in the tree that is nearest q_{rand} and $d(q, q_{\text{rand}})$ the distance metric between two points.

4. Expand new nodes.

$$q_{\text{new}} = q_{\text{near}} + \min\left(\epsilon, \frac{q_{\text{rand}} - q_{\text{near}}}{\|q_{\text{rand}} - q_{\text{near}}\|}\right). \tag{37}$$

Among them, q_{new} is the new node that expands from q_{near} in the direction of q_{rand} and ϵ the maximum expansion step length (controlling the growth rate of the tree).

5. Collision detection.

$$Collision\ \ Free(q_{\text{near}}, q_{\text{new}}) = \begin{cases} True & \text{If line segment } q_{near} \to q_{new} \\ & \subset X_{free}, \\ False & \text{Otherwise.} \end{cases} \tag{38}$$

Among them, X_{free} is the free space without collision.

6. Node addition.

$$\text{If} Collision\ \ Free(q_{\text{near}}, q_{\text{new}}) : V \leftarrow V \cup \{q_{\text{new}}\}, \quad E \leftarrow E \cup \{(q_{\text{near}}, q_{\text{new}})\}. \tag{39}$$

Among them, $\leftarrow$ is the operation of updating the set.

Simple RRT includes both single-tree and double-tree types. The single-tree RRT algorithm initializes a random tree structure with the planning origin as the starting point. It continuously samples points randomly in the state space and integrates those sampled points that meet specific conditions into the tree as new nodes. This process expands the random tree until it covers the target region or satisfies predefined termination criteria. Through this mechanism, the algorithm explores and determines a feasible trajectory from the start point to the goal point, as illustrated in Figure 22.

The dual-tree RRT algorithm constructs two random trees simultaneously, with the start and goal points as root nodes. It follows expansion principles such as random sampling and

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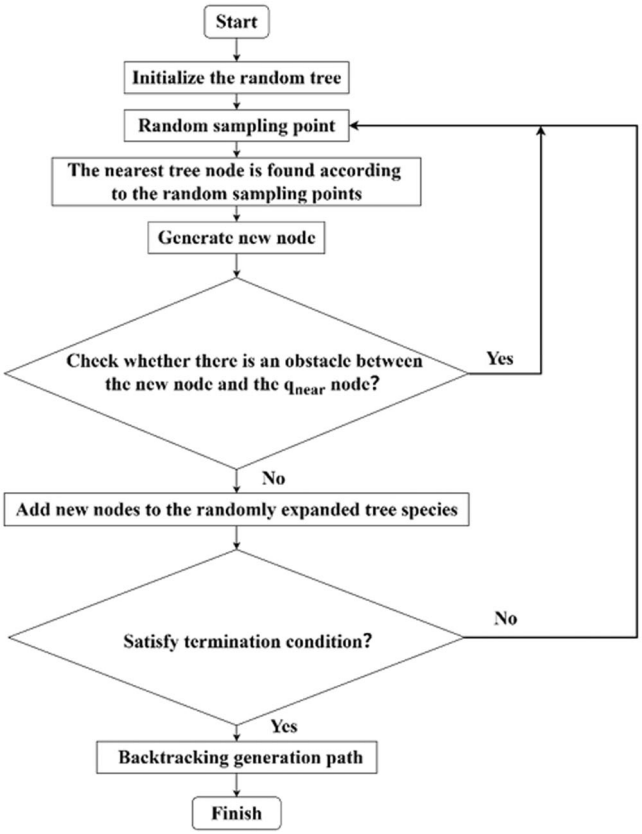


FIGURE 21 | Flowchart of the rapidly exploring random trees algorithm.

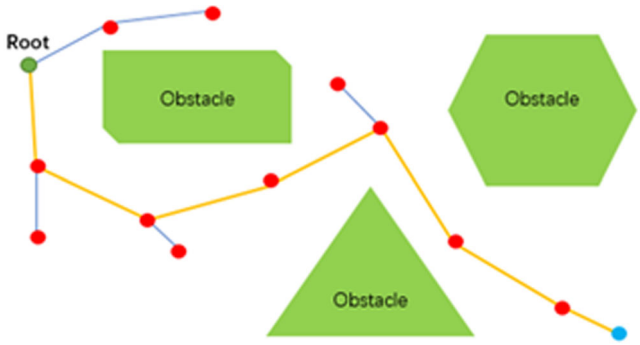


FIGURE 22 | Schematic diagram of a single-tree type of rapidly exploring random trees algorithm.

collision detection to facilitate the rapid convergence of the two trees. Compared with the single-tree RRT, this approach performs parallel searches from both ends, thereby reducing the search space and enabling faster identification of feasible paths, especially in complex or high-dimensional environments. This algorithm can effectively address dynamic obstacles and nonholonomic constraints, making it a critical method for path planning in agricultural robotics. Its high efficiency in complex scenarios is demonstrated in Figure 23.

In agricultural path planning, the RRT algorithm can address complex farmland environments. It selects randomly sampled points as targets, identifies the nearest node within the tree for

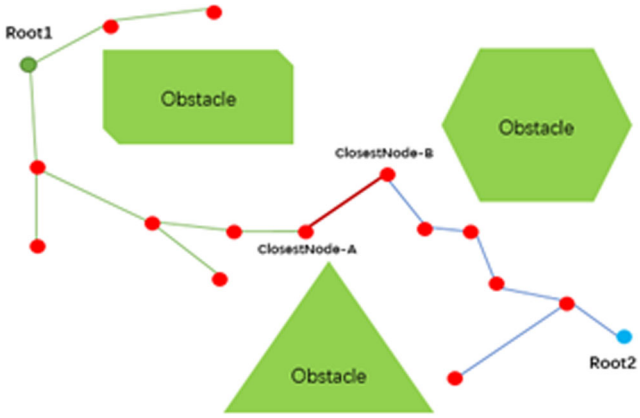


FIGURE 23 | Schematic diagram of the dual-tree type of RRT algorithm.

expansion, and validates the path's feasibility through collision detection. Through iterative processes, it efficiently explores the free space, connects target points, and constructs feasible paths tailored for operations such as weeding, sowing, or harvesting, as detailed in Tables 20 and 21.

In the path planning for agricultural robots, the RRT algorithm efficiently explores high-dimensional spaces via random sampling and search tree expansion. It is well-suited for unknown farmland environments as it does not rely on a global map, exhibits strong scalability, and can handle dynamic obstacles, making it widely applicable. However, the algorithm requires discretization of the input space. Insufficient sampling may degrade path quality, while excessive sampling increases computational complexity and compromises real-time performance. Moreover, the generated paths may contain redundant nodes. To address these limitations, researchers have proposed various improved algorithm versions. The relevant details are summarized in Table 22.

To enable mobile robots to navigate autonomously and conduct precise path planning in complex orchard environments, Ye et al. (2023) proposed the IKB-RRT algorithmic framework to address the challenges of path planning for mobile robots in complex orchards. This framework integrates a dual-tree connection mechanism for path smoothing and optimization, utilizing heading angle constraints and cubic spline interpolation to eliminate path discontinuities. Experimental results demonstrate that compared with similar algorithms, IKB-RRT exhibits significant advantages in path quality and computational efficiency. Castro et al. (2023) introduced an innovative online adaptive path-planning strategy by integrating RRT and deep reinforcement learning (DRL) algorithms. This approach enables drones to autonomously generate trajectories and achieve precise control during olive fly trap detection, plan reliable routes, reach inspection points, capture images, and avoid obstacles, thereby constructing an intelligent system adaptable to complex environments. C. Liu, Feng, et al. (2022) proposed the time-optimal rapidly-exploring random tree (TO-RRT) algorithm to enhance the search efficiency of random tree algorithms in complex environments. They established a target offset model, adjusted the target bias probability, and combined it with priority search to escape potential field interference.

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TABLE 20 | Some application scenarios and influences of RRT.

Actual scenarios	Concrete impacts
Unmapped mountain terrain exploration (Takemura and Ishigami 2017)	The RRT algorithm rapidly explores complex terrains by extending random trees, avoids collapse risk areas, and generates safe exploration paths.
Emergency inspection after flood disasters (Kalisiak et al. 2006)	After the flood, the farmland's terrain changed abruptly. The RRT algorithm dynamically expanded the random tree, avoiding waterlogged areas and soft ground, and quickly generated a passage path for the rescue robot.
The Titan FT-35 intelligent weeding robot of FarmWise Company (Lenain et al. 2021)	The complete coverage path-planning algorithm based on RRT generates spiral-shaped weeding paths in the early stage of crop growth. The expansion step length is dynamically adjusted to adapt to the plant spacing.
Virgo strawberry harvesting robot (C. Wang et al. 2020)	In the automated harvesting of strawberries in high-rise cultivation, the RRT algorithm plans the gripper's three-dimensional trajectory. Random sampling is employed for obstacle avoidance, and "stepwise expansion" is adopted to conform to the fruit closely.
Burro Company's mobile robot (Lehmann et al. 2022)	In vineyard and orchard environments, autonomous material transportation is achieved by applying the RRT algorithm for planning cooperative paths among multiple vehicles. A collision detection mechanism is integrated to prevent potential deadlocks during operation, ensuring smooth task execution. The system demonstrates outstanding performance metrics in practical deployments.
The laser weeder from carbon robotics (Hu et al. 2024)	The RRT algorithm dynamically generates the laser head's movement trajectory in real-time, ensuring 5 cm precision for interrow coverage of crops. By employing "nearest neighbor search" to locate weed clusters accurately, a collision-free path can be planned within 0.2 s, thereby enhancing operational efficiency.

Abbreviation: RRT, rapidly exploring random trees.

TABLE 21 | Challenges and solutions of RRT in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① The support structures obstruct the sensor's line of sight, limiting the random tree's feasible expansion directions.	① A biased sampling strategy is introduced to prioritize the expansion of the random tree toward open channels, enhancing exploration efficiency in cluttered environments.
② The complexity of mountainous terrain leads to a significantly reduced convergence rate for the traditional RRT algorithm.	② The bidirectional RRT (Bi-RRT) algorithm accelerates path generation, significantly reducing convergence time.
③ Muddy or sandy regions render the planned paths nontraversable, compromising the effectiveness of path planning.	③ Ground hardness sensor data are integrated to dynamically eliminate infeasible nodes, ensuring that only viable paths are considered.
④ The robot becomes immobilized in soft ground, rendering it unable to extricate itself without external assistance.	④ The fitness function incorporates a ground load-bearing coefficient to preferentially select hardened paths, thereby improving the robustness and reliability of the planned routes.
⑤ Competition for computing resources results in system delays, impacting real-time performance and operational efficiency.	⑤ A distributed RRT framework is designed, leveraging edge computing nodes to coordinate global path planning, optimizing computational resource allocation, and system performance.

Abbreviation: RRT, rapidly exploring random trees.

Additionally, they introduced an adaptive step size mechanism to optimize directional search and proposed a regression superposition algorithm to improve the exploration of unknown spaces. Wencheng et al. (2020) developed an intelligent framework integrating improved RRT and Dijkstra algorithms to overcome the limitations of manual spraying in orchards. By constructing a dynamic perception model, this framework leverages improved RRT's global search capability and Dijkstra's

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TABLE 22 | Comparison of improved versions of the RRT algorithm.

Algorithm name		Improvement principle	Applications in agriculture
Rapidly exploring random tree star algorithm	RRT*	Introduce a cost function and employ the "rewiring" strategy to eliminate redundant nodes, gradually optimize the path, and generate a progressively optimal path.	The orchard pruning robot optimizes the mechanical arm's pruning path to minimize energy consumption.
Dynamic adaptive RRT algorithm	Dynamic RRT	Update the environmental map in real-time. The historically effective nodes are retained during replanning with the dynamic obstacle prediction model.	The uncrewed aerial vehicle for crop monitoring dynamically avoids birds and temporary obstacles during flight.
Quantum behavior RRT* algorithm	QB-RRT*	A quantum potential field model is introduced to probabilistically bias random sampling toward the target direction, thereby reducing path oscillations caused by randomness.	The intelligent transplanter plans stable and efficient transplanting paths in six-dimensional pose space, ensuring high-quality planting.
Multiobjective optimization RRT algorithm	MO-RRT	Multiple cost functions are supported, and the Pareto frontier is used to filter out the optimal compromise path.	The feed distribution vehicle for livestock farms simultaneously optimizes travel time and feed spreading uniformity, enhancing operational efficiency.

Abbreviation: RRT, rapidly exploring random trees.

local optimization to achieve obstacle avoidance and path optimization. Adaptive sampling was designed to increase target orientation, and smoothing techniques were introduced to ensure motion stability. Huang et al. (2025) constructed an optimized inform-RRT* algorithmic framework to collaboratively optimize robotic arms' path-planning efficiency and motion stability during picking. They introduced target deviation guidance to enhance orientation accuracy, designed an adaptive step size adjustment to balance efficiency and quality, and proposed redundant path pruning to optimize structure and reduce unnecessary movements. Lv et al. (2023) proposed an improved P-RRT algorithmic framework to address the limitations of the traditional P-RRT algorithm in uncrewed aerial vehicle (UAV) path-planning. They constructed a greedy expansion strategy based on APF direction guidance, expanding the search tree according to the vector synthesis direction. Furthermore, they designed an overhead threshold filtering mechanism to compare costs and eliminate high-cost nodes, thereby improving convergence efficiency.

In the future, the RRT algorithm is anticipated to achieve advancements in the agricultural domain through three key directions. First, by leveraging measure theory for optimization and ensuring probabilistic optimality, variants of the RRT* algorithm will be developed. These variants will employ probability density functions to evaluate path quality, integrate MCTS with adaptive sampling strategies to mitigate suboptimal path issues, and enhance exploration efficiency in high-dimensional spaces. Second, kinematic constraints will be incorporated into the planning process, and trajectory smoothness will be improved. This will involve embedding the robot's kinematic model during tree expansion and applying spline interpolation techniques to optimize path curvature, ensuring that the planned paths align with the robot's actual motion capabilities. Third, multimodal sampling guided by environmental features will be implemented. Specifically, GMM will be utilized to cluster environmental characteristics, while semantic segmentation networks will identify operational areas, enabling a deep integration of path planning with agricultural tasks and improving overall operational efficiency.

5.2 | PRM Algorithm

The PRM algorithm was introduced by Kavraki and Svestka in 2002. It is a sampling-based method for path planning in high-dimensional configuration spaces, particularly suited for robot motion planning. PRM consists of two main phases: the construction and query phases. The core principle involves randomly generating sample points (landmarks) within the configuration space and establishing collision-free connections between these points to construct a PRM. This roadmap effectively captures the robot's workspace's geometric characteristics and the obstacles' distribution. Once the roadmap is constructed, the PRM algorithm utilizes graph search techniques to identify an optimal or feasible path from the start to the goal point, as illustrated in Figure 24.

1. Initialization.

   Define the configuration space: Suppose the degrees of freedom of the robot is D, the configuration space is $c \subseteq \mathbb{R}^D$, the configuration space occupied by obstacles is c_{obs}, and the free space is

   $$c_{\text{free}} = c \backslash c_{\text{obs}}. \tag{40}$$

   Set parameters: Sample number N (total number of nodes).

   Neighboring node selection strategy: k-nearest neighbor (k-NN) or radius r.

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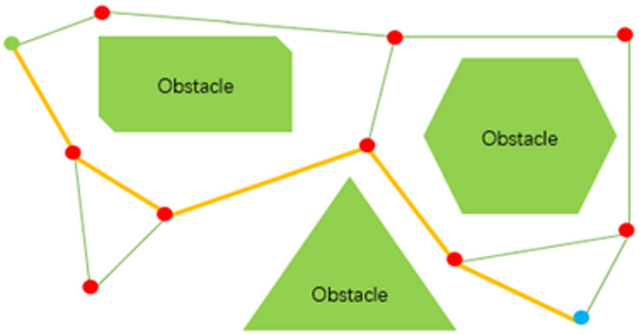


FIGURE 24 | Schematic diagram of probabilistic roadmap algorithm.

Local path resolution Δt (discretization step size for collision detection).

2. Random sampling.

   Generate N candidate configuration points uniformly and randomly in c_{free}:

$$\{q_i\}_{i=1}^{N}, \quad q_i \in c_{\text{free}}. \tag{41}$$

3. Collision detection.

   ① Node collision detection: Eliminate the sampling points colliding with obstacles and retain the set of valid nodes:

$$\mathcal{V} = \{q_i | Collision(q_i) = False\}. \tag{42}$$

   ② Edge collision detection: When local paths are being connected, it is necessary to check whether the entire path is within c_{free}.

$$Collision(q) = \begin{cases} 1 & \text{if } q \in c_{\text{obs}}, \\ 0 & \text{otherwise.} \end{cases} \tag{43}$$

4. Selection of neighboring nodes.

   For each node $q_i \in V$, the set of neighboring nodes $\mathcal{N}(q_i)$ is selected. Two common strategies are adopted:

   ① k-NN:

$$\mathcal{N}(q_i) = \{q_j | q_j \text{ constitutes the } k \text{ nodes that are closest to } q_i \text{ within } \mathcal{V}\}. \tag{44}$$

② Radius neighborhood.

$$\mathcal{N}(q_i) = \{q_j | \|q_j - q_i\| \le r\}. \tag{45}$$

Distance formula: Commonly, the Euclidean distance is employed:

$$\|q_j - q_i\| = \sqrt{\sum_{d=1}^{D} \left(q_j^{(d)} - q_i^{(d)}\right)^2}. \tag{46}$$

5. Local path connection.

   For each pair of adjacent nodes (q_i, q_j), an attempt is made to generate a collision-free path:

   ① Parametric path:

$$q(t) = q_i + t(q_j - q_i), \quad t \in [0, 1]. \tag{47}$$

   ② Discretization detection: The path is discretized with a step size of Δt, and it is examined whether there are collisions at each intermediate point:

$$\forall\ t_k = k\Delta t, \quad k = 0, 1, ..., \frac{1}{\Delta t}, \quad Collision(q(t_k)) = 0. \tag{48}$$

   ③ Add edge: If the path is collision-free, then incorporate edge e_{ij} into the edge set $\mathcal{E}$.

$$\mathcal{E} = \mathcal{E} \cup \{e_{ij}\}. \tag{49}$$

6. The roadmap has been completed.

Ultimately, an undirected graph $G = (v, \varepsilon)$ is yielded, wherein v constitutes the set of collision-free nodes and ε constitutes the set of collision-free edges.

When the PRM algorithm is utilized for path planning in agricultural robotics, random sampling is first performed within the free space of the agricultural operational environment. After collision detection, valid nodes are retained to form a node set. Subsequently, neighboring nodes are identified for each node based on a predefined neighborhood criterion. If no collisions occur during the connection process, edges are established between these nodes, constructing a PRM. The robot's start and goal positions are then incorporated into the graph as additional nodes and collision-free connections are established with existing nodes if feasible. Finally, a graph search algorithm, such as Dijkstra's or A*, is employed to identify a collision-free path within the roadmap, enabling the robot to navigate obstacles and efficiently complete its task. Common strategies for addressing complex obstacles and optimizing sampling distributions are summarized in Tables 23 and 24.

In the context of agricultural robot path planning, the PRM algorithm constructs a PRM through random sampling, effectively addressing high-dimensional path-planning challenges and demonstrating suitability for static complex environments. Its reusable global path library enhances online planning efficiency and enables broad applicability. However, the algorithm exhibits certain limitations. It depends on obstacle distribution, and sparse sampling in environments with dense or irregular obstacles may lead to an incomplete PRM. The preprocessing phase incurs significant computational costs and struggles to adapt to dynamic environments. Moreover, its handling of kinematic constraints is relatively inadequate. To address these issues, potential improvements include adopting D-PRM for temporary obstacle management, integrating Voronoi diagrams for optimized path selection, and implementing a hierarchical PRM framework to separately address global and local path-planning problems, thereby improving adaptability in dynamic scenarios.

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TABLE 23 | Some application scenarios and influences of probabilistic roadmap (PRM).

Actual scenarios	Concrete impacts
Large-scale farmland global inspection (Gautam et al. 2024)	The PRM algorithm constructs a PRM through pre-sampling, enabling efficient global path planning for farmlands spanning thousands of acres. It effectively avoids fixed obstacles such as field ridges and ditches while reducing the real-time computational load.
Inspection of fixed facilities in orchards (Canto 2022)	For orchard irrigation pipelines and support structures, PRM pre-generates a path network for fixed facilities, allowing robots to plan inspection routes rapidly based on the PRM.
Navigation for vertical planting in greenhouses (Mahmud et al. 2017)	In multilayer vertical planting racks, PRM builds a three-dimensional (3D) PRM to optimize the movement trajectories of robotic arms in 3D space and minimize empty travel distances.
DJI Agras T30 drone (Kholodiuk and Tokarchuk 2022)	During pesticide spraying in hilly terraced fields, the PRM algorithm generates a connectivity graph based on 3D terrain and obstacle data, using the height difference of terraced fields as a constraint to screen safe nodes and generate feasible paths.
SDF Group DEUTZ-FAHR C9306 combine harvester (Dugonjić 2021)	When harvesting in fallen rice fields, the PRM algorithm incorporates stem strength into the configuration space and generates antisinking paths via stratified sampling in a six-dimensional space.
Lely Astronaut A5 milking robot (Simões Filho et al. 2020)	In dynamic cattle shed environments, the PRM algorithm constructs a two-layer roadmap: the static layer pre-samples fixed facilities, while the dynamic layer updates the coordinates of moving obstacles in real-time.

TABLE 24 | Challenges and solutions of probabilistic roadmap in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① The limited number of pre-sampling nodes in farmlands results in discontinuous and suboptimal path planning.	① Adaptively increase the density of nodes at the boundaries of obstacles to improve path-planning accuracy and robustness.
② Inadequate coverage of narrow passages in orchards leads to redundant detours and inefficient navigation.	② Dynamically supplement key area nodes using uncrewed aerial vehicle aerial photography data to ensure comprehensive coverage of complex terrains and dynamic environments.
③ Post-rain muddy areas are not incorporated into the static roadmap, potentially causing unplanned obstacles.	③ Introduce a dynamic node elimination mechanism to identify and mark impassable regions, thereby preventing ineffective path-planning.
④ The rugged terrain of terraced fields introduces challenges in accurately calculating connection costs between nodes.	④ Adjust the connection weights of nodes based on terrain elevation data to enhance the accuracy of path planning in rugged terrains.
⑤ Crop rotation induces positional changes in fixed facilities, necessitating dynamic updates to the path planning model.	⑤ Periodically reconstruct the roadmap by integrating historical operation data and satellite imagery to adapt to the dynamic changes in agricultural environments.

Y. Li, Wang, Fan, et al. (2022) proposed the IPRM algorithm framework to address the energy consumption issue in multi-UAV collaborative path planning for agricultural applications. They developed an adaptive sampling strategy to optimize the sampling space by integrating density-weighted sampling and Voronoi diagram partitioning, thereby enhancing uniformity and spatiotemporal efficiency. Bohlin and Kavraki (2000) introduced the Lazy PRM algorithm, which assumes that the vertices and edges of the roadmap are initially collision-free and defers collision detection until the path search stage. This approach involves first generating candidate paths and subsequently verifying them for collisions. If collisions are detected, conflicting elements are removed, the roadmap is updated, and the search process is repeated. Hongwei et al. (2024) presented the I-NPRM algorithm for uncrewed surface vehicle (USV) path planning. They established a high-smoothness framework by combining bounding box analysis with Catmull-Rom spline interpolation to ensure safe USV operation. Xiaolu et al. (2022), aiming to overcome the limitations of traditional PRM, proposed a bidirectional search PRM global path-planning algorithm. They implemented a bidirectional search strategy with an alternating expansion

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mechanism for both forward and reverse directions. Jiyue (2022) addressed the challenge of UAV swarm trajectory planning by proposing a lightweight sampling strategy based on an improved PRM algorithm. They optimized the sampling density and update mechanism while adopting a hierarchical architecture to support task allocation and information sharing, ensuring trajectory connectivity and improving path-planning efficiency.

In the future, the PRM algorithm will have two primary development directions in the agricultural domain. First, a spatiotemporal dynamic roadmap will be constructed by developing a dynamic PRM framework based on spatiotemporal graphs. This involves assigning timestamps to nodes, predicting obstacle trajectories using Kalman filtering, dynamically updating connection relationships, and reducing computational costs through an event-triggered mechanism. Second, a hierarchical semantic roadmap can be generated by dividing the farmland into multiple levels. Macro-level planning enables coarse-grained global path planning, while micro-level processing focuses on fine-grained local adjustments. Semantic information is annotated via knowledge graph integration to enhance decision-making for path planning.

5.3 | Comparison of Planning Algorithms Based on Sampling

In the path-planning algorithms for agricultural robots, the selection of RRT and PRM algorithms depends on the characteristics of the agricultural environment and the specific requirements of the path-planning task. For instance, the RRT algorithm, with its robust spatial exploration capability and incremental construction of the path tree, is particularly well-suited for path planning in complex and partially unknown farmland environments. In contrast, the PRM algorithm constructs a PRM during the preprocessing stage, enabling rapid path queries and making it more suitable for known environments or scenarios requiring repeated traversals. Both algorithms play significant roles in the navigation and operation of agricultural robots, as summarized in Table 25.

To comprehensively evaluate the performance of the RRT and PRM algorithms in agricultural environment path planning, this study conducts a systematic comparison across six dimensions: path optimality, real-time obstacle avoidance, cost-effectiveness, computational efficiency, environmental adaptability, and scalability. The comparison results are visualized using a radar chart, and the performance of each algorithm under different metrics is quantified through a scoring system ranging from 1 to 5 (from weak to strong), as illustrated in Figure 25.

6 | Machine Learning Algorithms

Machine learning is an interdisciplinary field that enables the automation of intelligent decision-making through data-driven approaches. By leveraging methods such as supervised learning,

TABLE 25 | Comparison of planning algorithms based on sampling.

Name	Computational efficiency	Cost-effectiveness	Cost-effectiveness
RRT algorithm	The construction is rapid, and an initial path in a high-dimensional complex space can be acquired within a short period. However, this path is often tortuous and not optimal.	The cost of constructing the initial tree is relatively low, allowing for a feasible solution to be obtained efficiently. However, optimizing the path requires additional computational resources, which increases the overall computational cost.	The target-biased and greedy strategies are integrated into the improved algorithm in response to the complex three-dimensional canopy structure in apple orchards. Experimental results show that, compared with the traditional RRT algorithm, the path length is shortened by 14%, the search time is decreased by 55%, the planning time is only 5.74 s, and the completeness of path coverage reaches 100% (Y. Li, Li, et al. 2022).
PRM algorithm	The mapping stage entails a significant computational load due to extensive random sampling and collision detection processes. However, once the mapping is completed, path search operations are executed rapidly, making it well-suited for repeated path-planning tasks.	The cost of map establishment is high; however, when it is utilized multiple times subsequently, the cost of each path planning is low. In the long term, the efficacy is better. The average cost is relatively low for scenarios, where frequent path planning is needed.	In the simulated greenhouse setting, the average planning time of the enhanced algorithm is less than 1 s, the replanning time for dynamic obstacles is shortened to 0.02 s, and the path length optimization rate amounts to 23.7%, significantly augmenting the adaptability of the robot in complex facility agriculture scenarios (C. Liu et al. 2023).

Abbreviations: PRM, probabilistic roadmap; RRT, rapidly exploring random tree.

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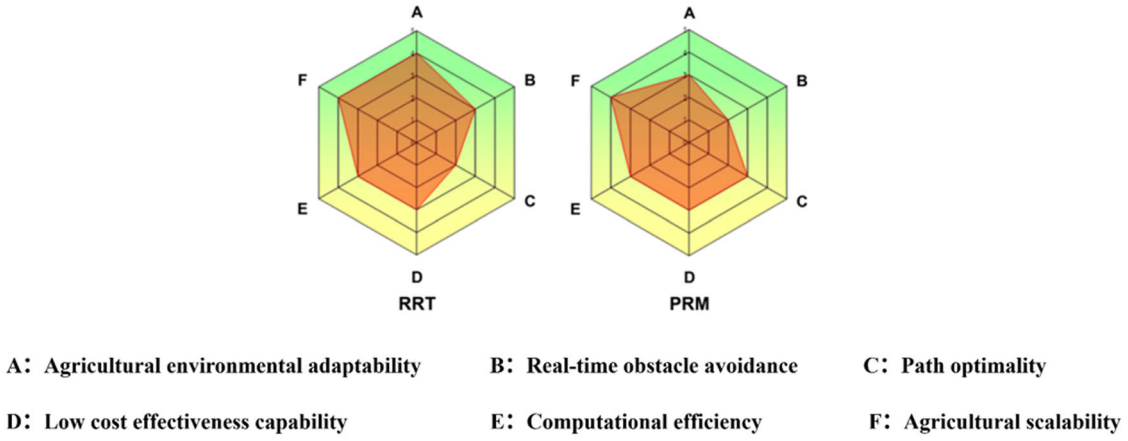


FIGURE 25 | Radar chart comparing the performance of sampling-based planning algorithms. A, agricultural environmental adaptability; B, real-time obstacle avoidance; C, path optimality; D, low cost-effectiveness capability; E, computational efficiency; F, agricultural scalability. PRM, probabilistic roadmap; RRT, rapidly exploring random tree.

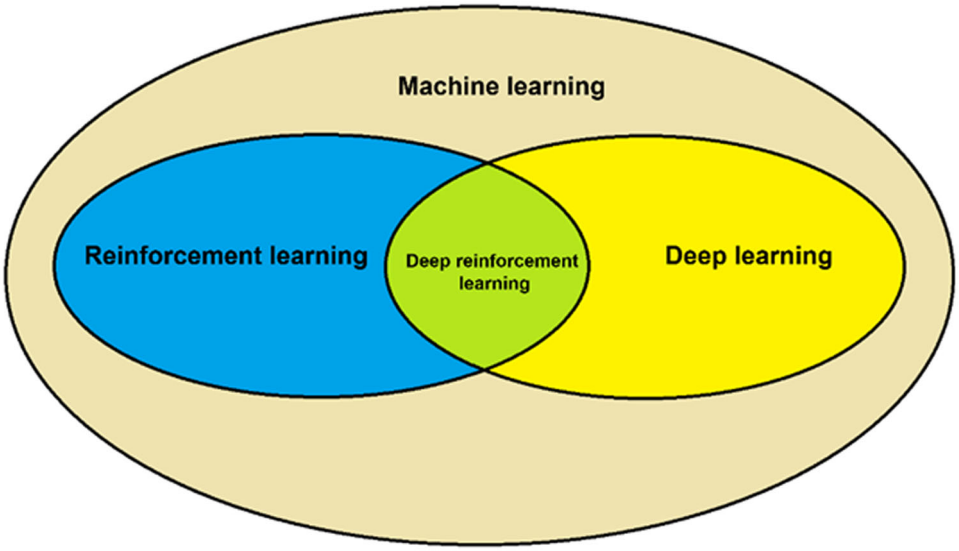


FIGURE 26 | Schematic diagram of machine learning algorithms.

unsupervised learning, and RL, it equips systems with the ability to generalize and reason about environmental features. In path planning, machine learning investigates the mapping relationship between environmental attributes and path quality, replacing traditional methods with data-driven models. This facilitates the generation of feasible paths in complex and dynamic scenarios. Leveraging its strengths in handling environmental noise, uncertainty, and novel situations, machine learning offers innovative solutions for intelligent navigation in unstructured environments.

Deep learning and RL represent two distinct paradigms in machine learning. Deep learning extracts knowledge from extensive datasets by iteratively optimizing network weights to perform tasks such as classification and regression. This static learning approach requires training models on pre-prepared datasets before applying them to recognize new data. In contrast, RL relies on continuous interaction with the environment, adjusting behavioral strategies based on feedback received during each interaction. Its goal is to maximize cumulative rewards by learning the optimal strategy in real-time through environmental feedback. DRL combines the strengths of both approaches, utilizing deep neural networks (DNNs) to directly extract features from raw signals and output decision actions without prior experience. This method exhibits superior learning and decision-making capabilities in complex and unknown environments, demonstrates strong generalization ability, and offers broad application potential, as illustrated in Figure 26.

6.1 | RL Algorithms

RL models agricultural robot path planning using Markov decision processes (MDPs), transforming path optimization into a dynamic interaction problem between the agent and the farmland environment. The robot selects actions based on state perception. It optimizes its policy network according to a predefined reward function, gradually learning to generate adaptive paths that balance obstacle-avoidance efficiency, energy optimization, and operational coverage. A key advantage of RL

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is its ability to achieve autonomous decision-making in dynamic environments through online trial-and-error learning without requiring prior maps. Typical applications include optimizing branch-avoidance paths for orchard-picking robotic arms, generating antislip trajectories for uncrewed tractors in muddy fields, and planning antidrift flight paths for plant protection drones under 3D wind field interference, as illustrated in Figure 27.



FIGURE 27 | Schematic diagram of reinforcement learning algorithm.

At present, some of the mainstream RL algorithms are presented in Table 26 as follows:

This paper mainly discusses the application of the Q-Learning algorithm in RL algorithms in the domain of agricultural robot path planning.

6.1.1 | Q-Learning Algorithm

British computer scientist Chris Watkins introduced the Q-learning algorithm (Watkins and Dayan 1992) in 1989. This algorithm operates within the RL framework, optimizing decision-making strategies through iterative state-action-value function (Q value) updates. During the interaction with the environment, the agent selects actions based on the current state and receives immediate rewards. The Q-table is dynamically updated using the Bellman equation. The algorithm progressively converges to an optimal policy by balancing exploration and exploitation, as illustrated in Figure 28.

TABLE 26 | Comparison of reinforcement learning algorithms.

Algorithm name		Improvement principle	Applications in agriculture
Q-Learning	QL	Model-independent algorithms based on value iteration maintain a Q-table and employ the Bellman equation to refine the optimal policy iteratively.	Greenhouse mobile robot navigation
SARSA	SARSA	Online temporal-difference learning algorithms select actions according to the current policy, updating Q values to optimize the policy in dynamic environments.	Real-time obstacle-avoidance path adjustment for combine harvesters in lodged crop fields
Dyna-Q	Dyna-Q	Hybrid architectures integrate model-based learning and planning, leveraging actual and simulated experiences to update Q values and enhance sample efficiency.	Global path planning for tractors in complex farmland terrain
Monte Carlo method	MC	MC methods sample complete episodes to estimate the state-action-value function through average returns without requiring an environmental model, making them suitable for episodic tasks.	Full-coverage path optimization for orchard spraying robots
Temporal difference learning	TD Learning	TD learning combines elements of MC methods and dynamic programming, updating the value function incrementally to facilitate online learning and policy adjustment in continuous tasks.	Real-time flight path correction for plant protection drones in windy environments
Policy gradient method	PG	Policy gradient methods directly optimize policy parameters by calculating gradients to update the network, maximizing cumulative rewards, and are particularly suited for continuous high-dimensional action spaces.	Dynamic path planning for greenhouse mobile robots
Double Q-learning	Double QL	Double Q-learning decomposes Q value updates into two independent estimators, alternating their updates to mitigate the overestimation bias inherent in traditional Q-learning, thereby improving policy reliability.	Path decision-making for intelligent weeding robots in high-weed farmland

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In the path planning for agricultural robots, the Q-learning algorithm uses the state of the farmland environment as input. The robot selects and executes actions to receive feedback rewards. By employing the update formula, which integrates rewards, discount factors, and the maximum Q value of the new state, the Q values are iteratively updated. This continuous process of optimization and exploration enables the robot to learn an optimal path strategy, thereby allowing it to efficiently and safely complete picking tasks in complex scenarios such as orchards, as demonstrated in Tables 27 and 28.

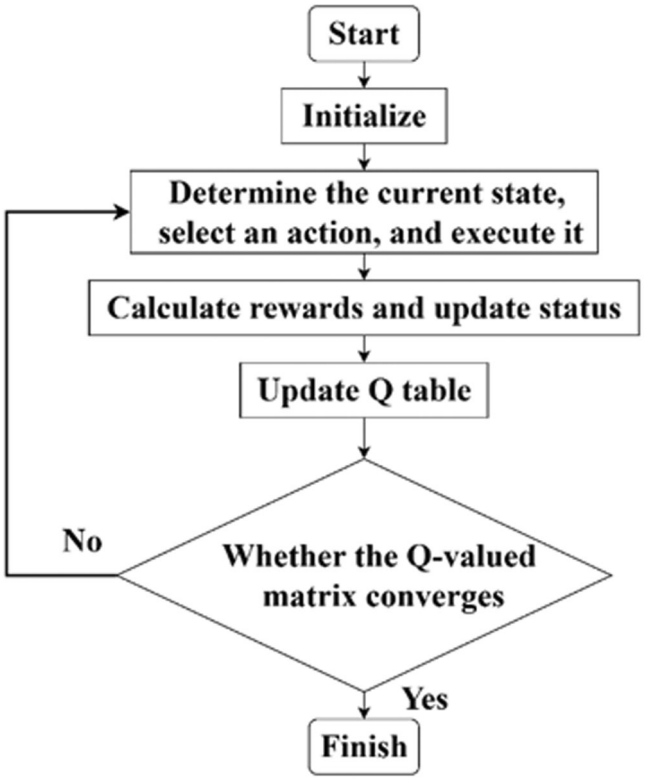


FIGURE 28 | Schematic diagram of Q-learning algorithm.

The Q-learning algorithm incorporates an RL mechanism for path planning in agricultural robots. It enables the robot to autonomously explore the optimal path in an unknown environment using a reward function without requiring prior knowledge. This makes it suitable for dynamic farmland environments and multitask switching scenarios. However, the algorithm exhibits notable limitations: its slow convergence rate necessitates extensive trial-and-error processes in complex environments, making it less effective for emergency tasks; it is susceptible to the design of the reward function, which may lead to deviations from practical requirements; and discretization of the state space can result in the loss of critical information, thereby affecting the precision of path planning.

The current mainstream improvement directions aim to accelerate learning and generalization: first, the Prioritized Experience Replay (PER) technique is adopted to prioritize the training of high-value samples; second, the double Deep Q-Network (DQN) architecture is introduced to decouple action selection from evaluation, thereby reducing overestimation; third, curriculum learning based on domain knowledge is designed, gradually transitioning from simple scenarios to complex farmland environments. Y. Chen, Tao, et al. (2024) proposed an intelligent framework for visual servo control. They constructed a multimodal visual perception system that

TABLE 27 | Some application scenarios and influences of Q-learning algorithm.

Actual scenarios	Concrete impacts
Dynamic weed distribution area weeding (Sengupta et al. 2012)	Q-learning learns the real-time distribution of weed density, prioritizes navigation to high-weed-density areas, dynamically optimizes path weights, and reduces redundant coverage and pesticide waste.
Seasonal terrain adaptive navigation (X. Wang, Xu, Wang, et al. 2024)	For muddy farmlands during the rainy season and hardened soils in the dry season, Q-learning dynamically adjusts path strategies to balance the robot's energy consumption and operational safety.
Multirobot collaborative fertilization (Mei et al. 2023)	On the basis of a shared Q-table, multiple fertilization robots coordinate their paths to prevent task overlap and equipment collisions, thereby improving the efficiency of large-scale farmland operations and ensuring balanced fertilizer application.
Harvest CROO robotics strawberry picking robot (Bogue 2020)	Q-learning optimizes the robotic arm's picking path and grasping force in greenhouse strawberry harvesting. The model is trained using a reward function to adapt to the physical characteristics of different strawberry varieties.
XAG agricultural drone (Borysenko et al. 2023)	Q-learning optimizes flight paths and spraying parameters during plant protection and sowing. The algorithm dynamically adjusts real-time flight altitude and liquid flow rate to ensure uniform pesticide distribution and obstacle avoidance.
Netafim's AI Irrigation Controller (Negara et al. 2023)	The Q-learning algorithm dynamically selects irrigation periods and pump power levels to optimize irrigation cycles and water volumes. The reward function considers both water conservation and crop yield maximization.

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TABLE 28 | Challenges and solutions of Q-learning algorithm in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① The complexity of the farmland environment significantly hinders learning efficiency, leading to suboptimal performance of reinforcement learning algorithms.	① State clustering techniques aggregate similar states, reducing the state space and improving learning efficiency.
② Unforeseen weather events, such as hailstorms, can render preset Q-tables ineffective, necessitating adaptive strategies for real-time updates.	② Real-time meteorological data are integrated into the reward function for dynamic updates, enabling adaptive responses to changing environmental conditions.
③ When multiple robots share the same Q-table, conflicts in path decision-making may arise, potentially causing inefficiencies and collisions.	③ A distributed Q-learning framework is designed to maintain an independent Q-table for each robot, with periodic synchronization to ensure consistent policy updates across multiple agents.
④ Intense light or dust can cause significant visual positioning deviations, requiring frequent updates to the Q-table to maintain accurate navigation.	④ Multimodal sensor fusion, incorporating millimeter-wave radar and IMU data, is employed to correct environmental measurements and enhance localization accuracy.
⑤ High humidity levels can degrade the accuracy of LiDAR sensors, leading to deviations in path execution and necessitating compensatory measures.	⑤ A robust Q-learning algorithm is introduced to tolerate partial noise in sensor data and dynamically adjust Q values, ensuring reliable performance in noisy environments.

Abbreviations: ACA, ant colony algorithm; ACO, ant colony optimization; LiDAR, light detection and ranging.

integrates target localization, self-localization, and acquisition strategies, enabling precise prediction of motion trajectories. Additionally, they introduced a regulatory mechanism and scheduling strategy to enhance performance. Y. Li, Wang, Fan, et al. (2022) proposed an improved algorithm framework to address the limitations of traditional Q-learning. They developed a Q-table initialization strategy based on the WOA and designed a goal-oriented strategy to adjust exploration priorities according to quantified spatial relationships. Huynh et al. (2021) addressed the issue of low search efficiency in traditional DE algorithms caused by fixed parameters. They constructed a collaborative mechanism between RL and evolutionary algorithms, allowing Q-learning to be integrated into DE as an adaptive parameter controller. Parameters are dynamically adjusted during iterations based on the population state. C. Yan and Xiang (2018) proposed an enhanced Q-learning algorithm for UAV path planning in unknown adversarial environments. They constructed an RL model based on the MDP and designed a dynamic reward function to quantify multiple objectives. J. Li, Xiao, et al. (2020) proposed the DMQL-CS algorithm, which integrates Q-Learning with genetic operators. They established an RL step size optimization framework, iteratively calculating the optimal step size through the Q-function value to improve the balance between exploration and exploitation.

In the future, the Q-learning algorithm is expected to exhibit two primary development directions in the agricultural domain. First, meta-learning-driven rapid adaptation can be achieved by integrating model-agnostic meta-learning (MAML) with Q-learning. This approach enables the transfer of path-planning strategies using a limited number of samples, reduces trial-and-error processes through gradient matching, and enhances generalization capabilities via Bayesian optimization. Second, to prevent catastrophic forgetting, a continuous learning framework based on Elastic Weight Consolidation (EWC) can be constructed. This framework dynamically preserves network weights and leverages GANs to generate virtual samples, thereby assisting robots in accumulating experience and adapting to environmental changes.

6.2 | Deep Learning Algorithms

In the context of agricultural robot path planning, deep learning algorithms construct multilayer NNs to automatically extract abstract features from multimodal sensor data in farmland environments, thereby achieving end-to-end path decision optimization. These algorithms leverage the hierarchical structure of NNs to learn high-level and low-level features and generate optimal strategies via reinforcement or supervised learning. For instance, convolutional neural networks (CNNs) can be employed for boundary identification, while recurrent neural networks (RNNs) are suitable for path prediction. This data-driven approach eliminates manual feature engineering, enabling adaptation to unstructured environments and enhancing robots' autonomous navigation and operational efficiency. Some mainstream deep learning algorithms currently used in this domain are summarized in Table 29.

This literature primarily discusses the applications of the CNN and RNN algorithms within deep learning algorithms in agricultural robot path planning.

6.2.1 | Convolutional Neural Network

The CNN algorithm (Lecun et al. 1989), introduced by Yann LeCun and colleagues in 1989, operates on the principle of performing convolution operations through sliding convolution kernels over input data in the convolutional layers to extract local features automatically. The pooling layer compresses and downsamples these extracted features, reducing the data dimensionality while preserving key characteristics. The fully connected layer

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TABLE 29 | Summary of deep learning algorithms.

Algorithm name		Improvement principle	Applications in agriculture
Convolutional neural network	CNN	Multilayer convolutional layers are utilized to extract spatial features from the input data. The model reduces parameter complexity by leveraging local perception and weight sharing while maintaining robust feature extraction. Pooling operations further enhance translation invariance, ultimately outputting path-planning information.	Farmland terrain, generation of static obstacle-avoidance paths.
Recurrent neural network	RNN	On the basis of the recursive structure of time-series data, hidden states transmit historical information to capture dynamic changes (such as the trajectories of agricultural machinery). This enables the prediction of path points or adjustment of strategies, facilitating dynamic and continuous path optimization.	Dynamic path adjustment in farmland.
Long short-term memory network	LSTM	As an advanced version of RNN, LSTM introduces input, forget, and output gates to address the vanishing gradient problem, thereby improving long-term dependency learning.	Continuous operation path planning for large-scale farmland.
Graph convolutional network	GCN	The farmland is abstracted into a graph structure, where graph convolutional layers aggregate information from neighboring nodes. This approach facilitates the learning of global topological relationships, enabling the generation of globally optimal paths that cover complex farmland structures.	Global path optimization for irregular plots.
Generative adversarial network	GAN	A GAN framework, comprising a generator and a discriminator, simulates and distinguishes farmland data. Through adversarial training, this framework enhances the generalization ability and robustness of the path-planning model.	Virtual farmland environment generation.
U-Net	U-Net	Utilizing a symmetric encoder-decoder architecture with skip connections, U-Net fuses multiscale features to achieve pixel-level semantic segmentation. This provides high-precision semantic maps for accurate path planning.	Fine navigation in greenhouse scenarios and obstacle-avoidance path planning for robotic arms.

then integrates the processed features to perform final classification or regression tasks. Additionally, CNN employs the back-propagation algorithm to compute errors and update network weights, optimizing the model toward minimizing the loss function. This process enables effective processing and pattern recognition of input data, as illustrated in Figure 29.

With the advancement of deep learning, integrating CNNs with machine learning algorithms has been widely adopted for path planning in agricultural robots. CNNs exploit their local perception and multilevel feature extraction capabilities to process multimodal data from farmland environments. The convolutional layers autonomously learn spatial features such as terrain, crops, and obstacles, reducing computational complexity through weight sharing while adapting to environmental changes. The pooling layers perform downsampling to preserve critical information and improve model robustness. Subsequently, fully connected layers or deconvolutional structures map the extracted features into path decisions, enabling real-time obstacle avoidance and efficient path generation. Compared with traditional methods, CNNs achieve end-to-end integration of perception and decision-making, significantly enhancing navigation accuracy and adaptability in complex farmland scenarios, as shown in Tables 30 and 31.

When CNNs are employed in the path planning of agricultural robots, they process visual data to extract obstacle features and generate navigation maps. This makes them suitable for real-

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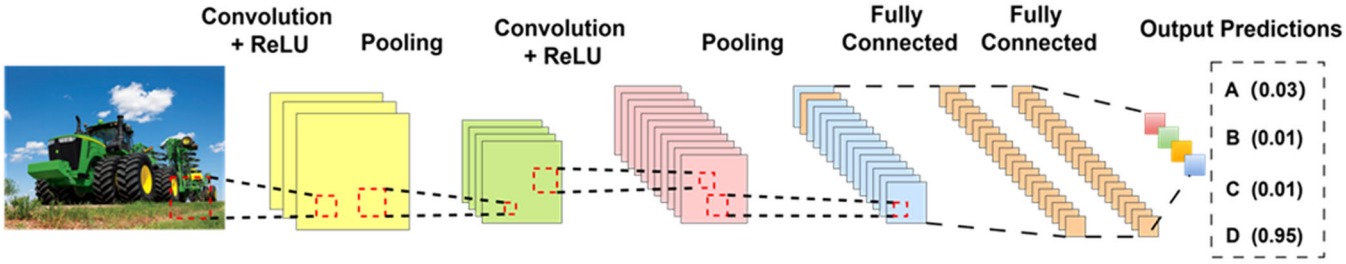


FIGURE 29 | Schematic diagram of a convolutional neural network algorithm. ReLU, rectified linear unit.

TABLE 30 | Some application scenarios and influences of CNN.

Actual scenarios	Concrete impacts
Crop row visual navigation (Foroughi et al. 2021)	Employ CNN to parse farmland RGB images in real-time, identify crop rows and field ridge boundaries precisely, and generate straight navigation paths to minimize crop damage during sowing or harvesting operations.
Priority inspection for disease-affected areas (Lu et al. 2021)	Extract disease features of crop leaves using CNN to guide robots in prioritizing inspections of high-risk areas. This will enhance the early detection rate of diseases and improve the response speed for prevention and control measures.
Complex lighting environment positioning (Jiang et al. 2022)	In orchards with alternating intense light and shadows, utilize CNN to fuse multispectral images for enhanced feature extraction, accurately identify tree trunks and low-hanging fruits, and generate obstacle-avoidance paths.
AgGPS autopilot agricultural autonomous driving system (Biswas and Goswami 2025)	The system employs high-resolution thermal imaging cameras and utilizes Mask R-CNN to identify the conditions of forage, livestock, and fences. It integrates TCN to predict livestock movement and dynamically plans inspection paths while marking locations for fence repairs.
Ecorobotix AVO all-terrain autonomous robot (Lochan et al. 2024)	The robot adopts the DeepLabV3 + CNN model to segment aerial images, combines ToF data to construct three-dimensional (3D) maps, and plans S-shaped paths in real-time to avoid trellis supports.
XAG P100 agricultural drone (Gheorghe et al. 2024)	High-resolution multispectral cameras and lightweight CNNs identify farmland information and integrate LiDAR data for path planning and obstacle avoidance. The model demonstrates transferability and adaptability across multiple varieties, maintaining precise identification even under rainy or cloudy weather conditions.
SITIA VineScout robot (Farella et al. 2024)	The robot is equipped with a hyperspectral imager, utilizes U-Net-CNN and 3D-CNN to analyze grape plants, and optimizes pruning and inspection paths using the particle swarm optimization algorithm.

Abbreviations: CNN, convolutional neural network; LiDAR, light detection and ranging; R-CNN, region-based convolutional neural network; TCN, telecommunications consultant; ToF, time-of-flight.

time perception and decision-making in unstructured environments while enabling precise identification of complex terrains. However, CNNs have notable limitations: their reliance on labeled data means that insufficient data can reduce generalization capabilities; their computational complexity imposes high hardware requirements, limiting embedded deployment; and their inability to reason effectively in dynamic environments necessitates collaboration with other algorithms to address challenges such as data noise and latency.

Enhancements to CNNs focus on improving data efficiency and robustness. Specifically, self-supervised learning (SSL) can pretrain models using unlabeled farmland data. Attention mechanisms, such as SE-Net, can be incorporated to emphasize key obstacle features. Furthermore, a federated learning framework can be developed to share privacy-preserving training data among multiple robots. C. Wang, Li, et al. (2023) proposed a deep learning-based intelligent perception framework to address the issue of failed obstacle avoidance in lychee-picking robots. They constructed a CNN-driven model integrated with a multimodal vision fusion system, employing monocular vision for picking point localization and binocular vision for obstacle reconstruction. Additionally, they designed a multiscale network and attention mechanism to enhance performance. Z. Zhang, Dai, et al. (2023) introduced a path-planning framework for tomato-harvesting robotic arms that

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TABLE 31 | Challenges and solutions of CNN in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① In high-density corn fields, the obscuration of ridges causes CNN to misjudge navigation boundaries.	① Integrate LiDAR point cloud data to construct a multimodal environmental perception model for improved scene understanding.
② The coverage of ground features by vine crops results in deviations between the generated paths and the specified requirements.	② Incorporate an attention mechanism to enhance the CNN's ability to extract and focus on features in key regions.
③ Rain and fog lead to image blurring, thereby reducing the recognition accuracy of CNN.	③ Leverage defogging algorithms and image enhancement techniques to preprocess input data, thereby improving the quality of visual information.
④ During the seedling stage, the morphological similarity between young crops and weeds induces classification errors in CNN, which in turn leads to incorrect path avoidance.	④ Utilize transfer learning to pretrain the model on a diverse farmland data set, thereby enhancing its generalization performance.
⑤ The embedded devices on the robot lack sufficient computational resources to support the operation of complex CNN models.	⑤ Quantitatively evaluate the model's accuracy and optimize its inference speed for efficient deployment.

Abbreviations: CNN, convolutional neural network; LiDAR, light detection and ranging.

combines 3D-CNN with RRT. They developed a heuristic cost model based on PPM and integrated fully convolutional neural network (FCN) to generate a 3D voxel map containing critical information, thereby improving planning efficiency. Y. Liu, Zheng, et al. (2022) proposed an RPP framework named Res-Planner for UAVs, which integrates traditional algorithms with DRL. They employed ResNet to build an end-to-end policy model and designed a multiscale module and curriculum learning strategy to enhance performance. Y. Zhang and Ge (2022) presented an improved 13-layer CNN architecture to address the challenge of target recognition for underwater robots under complex lighting conditions. J. Wu, Li, et al. (2024) proposed a point configuration optimization framework based on CNN to facilitate robot navigation. They constructed an intelligent decision-making model centered on CNN to analyze spatial and motion features of point configuration data for optimizing navigation layouts. Siyue et al. (2025) proposed a robot obstacle-avoidance path-planning method based on an improved CNN. They combined bilinear interpolation with the DWA to calculate target point coordinates, evaluate functions, and determine expansion distances. Real-time pose estimation was used to update coordinates, and a function was introduced to achieve effective obstacle avoidance.

In the future, CNNs in the agricultural domain will advance in two primary directions: few-shot learning and edge intelligence. First, leveraging few-shot learning techniques such as prototype networks can significantly reduce reliance on labeled data. This approach will address model generalization challenges across diverse environments while lowering labeling costs in conjunction with domain adversarial NNs and SSL. Second, lightweight models developed through neural architecture search will be combined with quantization and pruning techniques to enable efficient deployment on embedded devices, thereby meeting real-time navigation requirements in field operations. Additionally, a federated learning framework will be integrated to promote parameter sharing among multiple robots, enhancing the overall operational efficiency of agricultural robots.

6.2.2 | RNN and Long Short-Term Memory (LSTM) Network

American physicist Jeffrey L. Elman (1990) introduced the RNN (Medsker and Jain 2001). Its core principle lies in its ability to process sequential data by incorporating hidden states and recurrent connections within the network architecture. This allows information to be transmitted and shared across different time steps, meaning that the output at any given time depends on the current input and the hidden state from the previous time step. Consequently, RNNs can model long-term dependencies in sequences, leveraging past information to influence current decisions and outputs. This capability makes RNNs indispensable in natural language processing, speech recognition, and time-series prediction.

However, traditional RNNs often suffer from the vanishing gradient problem, where the gradients used for weight updates diminish significantly during backpropagation through time. As a result, RNNs struggle to capture long-term dependencies because they prioritize recent information over older data, leading to a loss of historical context. Sepp Hochreiter and Jürgen Schmidhuber proposed the LSTM (Hochreiter and Schmidhuber 1997) network in 1997 to address this limitation. LSTM introduces three gating mechanisms—the input gate, forget gate, and output gate—to control the flow and retention of information. Specifically, the input gate regulates how much new information is stored in the memory cell, the forget gate determines which parts of the previous memory should be retained or discarded, and the output gate decides what information to output based on the current memory state and input. By selectively remembering and forgetting information, LSTMs effectively mitigate the vanishing gradient problem and enhance the modeling of long-term dependencies in sequential data, as illustrated in Figure 30.

LSTM is utilized in the path planning of agricultural robots to address the limitations of RNN. Equipped with three memory units that control the retention of historical information, LSTM

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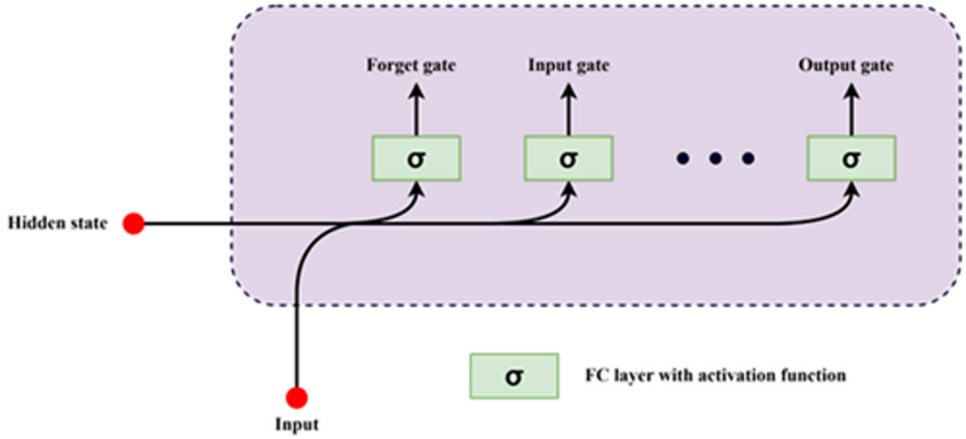


FIGURE 30 | Schematic diagram of the long short-term memory algorithm.

can effectively memorize the layout of fruit trees in orchards and predict changes in fruit ripening zones. Combined with real-time obstacle data, it enables adaptive adjustment of obstacle-avoidance strategies. Compared with RNN, LSTM alleviates the vanishing gradient problem through its gating mechanism, supports multimodal fusion and closed-loop optimization, and facilitates autonomous navigation in unstructured farmland environments, as demonstrated in Tables 32 and 33.

LSTMs demonstrate significant advantages in the path planning of agricultural robots. The gating mechanism allows for capturing long-term dependencies in dynamic environments, thereby providing forward-looking guidance for path planning. LSTM can integrate time-series data from multiple sensors and adapt to complex farmland scenarios. It supports online learning for model updates, enhancing the robustness of decision-making processes. Combined with RL and CNNs, it enables the construction of end-to-end systems to improve operational efficiency. However, LSTM also has limitations. The modeling of long-term dependencies is constrained by network depth and data volume, which may lead to cumulative errors in long-term predictions. It imposes high requirements on synchronizing time-series data, where deviations or omissions in data can negatively affect planning accuracy. Additionally, training is resource-intensive, parameter tuning is complex, and the real-time inference capability of edge devices is limited, necessitating model compression and optimization techniques.

The improvement directions of LSTM networks are as follows: First, the traditional LSTM can be replaced with a Transformer to capture long-term dependencies via self-attention mechanisms. Second, integrating visual, LiDAR, and global positioning system data enhances environmental understanding capabilities. Third, EWC can be employed to prevent catastrophic forgetting and continuously accumulate experience. Gao and Zhao (2024) proposed a trajectory optimization framework that integrates LSTM with the traditional ACA to address the limitations of ACA. They developed a dynamic pheromone update mechanism based on LSTM to leverage its ability to learn temporal features for optimizing path search strategies. N. Guo et al. (2021) introduced a collaborative path-planning algorithm framework that combines LSTM, FL Control, and RL. They constructed an long short-term memory with fine-tuning (LSTM-FT) network model, where FL generates low-dimensional features and pretraining resolves data dependency issues. J. Zhang, Guo, et al. (2024) presented an RPP algorithm for UAVs (RPP-LSTM) based on LSTM. It integrates RNN temporal modeling with DRL to overcome the adaptability challenges of traditional single-step decision-making models. H. Zhang, Xing, et al. (2023) proposed an intelligent fire escape path-planning framework that combines LSTM with an improved A* algorithm to handle path failures caused by sensor anomalies under high-temperature conditions. They developed a dynamic situation prediction model based on LSTM to learn real-time fire data and compensate for sensor deficiencies.

In the future, RNNs will advance in three key areas within the agricultural domain. First, they will enhance long-sequence dynamic modeling and environmental prediction capabilities. Specifically, LSTM will integrate agricultural time-series data to construct long-term dependency models, predict environmental changes using an encoder-decoder architecture, and incorporate attention mechanisms to improve the timeliness of planning, such as preplanning detour routes for heavy rain and waterlogging scenarios. Second, they will strengthen multimodal data fusion and decision-making. LSTM will aggregate multisource data, extract critical information based on environmental contexts, optimize group trajectories in conjunction with spatiotemporal graph convolutional networks, and leverage multimodal models to predict movement directions for obstacle avoidance. Third, they will refine online learning and adaptive path adjustment. LSTM will establish an online learning framework, quantify uncertainties through Bayesian deep learning, progressively increase task complexity via curriculum learning, and achieve real-time path optimization by leveraging edge computing to minimize communication latency.

6.3 | DRL Algorithms

In the domain of agricultural robot path planning, the DRL algorithm (Y. Li 2017) integrates DNNs with the RL framework, enabling robots to autonomously process high-dimensional environmental data and learn optimal path strategies through interaction with dynamic farmland environments. The DRL approach leverages DNNs for feature extraction, employs the policy network to generate actions, and optimizes parameters based on environmental rewards to achieve closed-loop

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TABLE 32 | Some application scenarios and influences of LSTM.

Actual scenarios	Concrete impacts
Seasonal farmland path prediction (Dewangan and Vij 2024)	LSTM analyzes historical meteorological and operational data to predict muddy areas during the rainy season or hardened zones during the dry season, thereby enabling the avoidance of impassable regions.
Avoidance of livestock herd movement trends (D. Wu et al. 2021)	On the basis of sequential livestock activity data, LSTM predicts animal congregation areas, preplan detour routes, and reduces the risk of robot entrapment.
Adaptation to crop growth cycles (Bhimavarapu et al. 2023)	By integrating the sequential crop growth model, LSTM dynamically adjusts robotic arm trajectories to avoid mature plants, reducing damage rates and ensuring adaptability for long-term monitoring and harvesting tasks.
Valtra Versu intelligent skidder tractor (Zvěřina et al. 2022)	In Finnish coniferous forest logging and transportation, sequential terrain data collected via LiDAR and IMU is modeled using an LSTM network to assess long-term safety risks of transport paths, enabling dynamic route planning to avoid high-risk areas.
DJI Agras T40 agricultural drone (Miru and Munir 2024)	In Chinese citrus orchard experiments, an Attention-based RNN model analyzes sequential leaf reflectance data to dynamically plan flight paths and pesticide concentrations for optimal spray operations.
Kinze 4900 autonomous seeder (Moreda et al. 2016)	Data integration with the Attention-RNN model predicts field carbon sequestration potential. This allows for dynamic agronomic practice adjustment, significantly enhancing carbon sequestration and seedling emergence rates.

Abbreviations: IMU, inertial measurement unit; LiDAR, light detection and ranging; LSTM, long short-term memory; RNN, recurrent neural network.

TABLE 33 | Challenges and solutions of LSTM in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① The extended time horizon for predicting herd movement exacerbates the vanishing gradient problem in LSTM, leading to cumulative prediction errors.	① Introduce an attention mechanism to emphasize the features of critical time steps, thereby enhancing the model's ability to capture salient temporal patterns.
② The accuracy of cross-seasonal predictions in crop growth models diminishes over time due to changing environmental conditions.	② Integrate gated recurrent units (GRUs) to streamline the network architecture and improve long-term memory retention, addressing the limitations of traditional LSTM models.
③ Synchronous processing of multivariate time-series data, including meteorological and soil variables, presents significant complexity.	③ Design a multichannel LSTM architecture to independently process different modal data streams, ensuring specialized handling of diverse input types.
④ Sudden weather changes or equipment malfunctions can disrupt the established time-series patterns, affecting the reliability of predictive models.	④ Incorporate a real-time anomaly detection module to trigger online fine-tuning of the LSTM, enabling adaptive adjustments in response to unexpected events or system anomalies.
⑤ High-temperature environments cause computing units to reduce their operating frequency, resulting in insufficient real-time performance.	⑤ Develop an edge-cloud collaborative framework to offload complex prediction tasks to cloud resources, optimizing computational efficiency and real-time performance.

Abbreviation: LSTM, long short-term memory.

optimization. This method obviates the need for manual feature engineering, adapts effectively to unstructured scenarios, and utilizes the exploration–exploitation trade-off mechanism to refine planning. Consequently, it enhances agricultural robots' autonomous decision-making capabilities and operational efficiency, as illustrated in Figure 31.

The current mainstream DRL algorithms are shown in Table 34 as follows:

This paper focuses on the application of the DQN algorithm, an RL method, in agricultural robot path planning.

6.3.1 | DQN Algorithm

The DQN algorithm was first introduced by Volodymyr Mnih et al. in 2013 and further refined in 2015. The core principle of DQN lies in integrating DNNs with the Q-learning algorithm

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FIGURE 31 | Schematic diagram of deep reinforcement learning algorithm. DL, deep learning; RL, reinforcement learning.

from RL. Specifically, it uses a DNN to approximate the state-action-value function Q, leveraging its powerful nonlinear fitting capabilities to handle high-dimensional state spaces. To address data correlation and improve sample efficiency, DQN incorporates an experience replay mechanism that stores interaction experiences between the agent and the environment in a replay buffer and samples them randomly during training. Additionally, DQN employs a target network technique, periodically updating the target network parameters to generate stable target Q values, mitigating oscillations caused by fluctuating targets and ensuring model convergence. Ultimately, DQN enables autonomous learning of complex decision-making strategies without the need for manual feature engineering. The detailed process is illustrated in Figures 32 and 33.

1. Initialization of the network and parameters.

   Initialize two NNs of the same structure: The leading network (for predicting Q values) and the target network (for providing stable target Q values). In the beginning, the parameters of the two networks are identical.

   $$\text{The main network, } Q(s, a; \theta); \quad \text{the target network, } Q(s, a; \theta^-) \quad (\text{Initial, } \theta^- = \theta). \tag{50}$$

2. Action selection (ε-Greedy).

   With a probability of ϵ, explore the environment randomly, and with a probability of $1 - \epsilon$, select the optimal action predicted by the current network, thereby balancing exploration and exploitation.

   $$a_t = \begin{cases} \text{Randomized action}, & \text{Probability } \epsilon, \\ \arg\max_a Q(s_t, a; \theta), & \text{Probability } 1 - \epsilon. \end{cases} \tag{51}$$

3. Interaction with the environment and storage of experience.

   Execute action a_t, observe reward r_t, and the next state s_{t+1}, and deposit the transition sample in the experience pool D. $done_t$ marks whether to terminate.

   $$D \leftarrow D \cup \{(s_t, a_t, r_t, s_{t+1}, done_t)\}. \tag{52}$$

4. Experience replay and sampling.

   Randomly extract N samples from the experience pool to break the data correlation and improve the training stability.

   $$\text{Sampling of small batches:} \{(s_i, a_i, r_i, s_{i+1}, done_i)\}_{i=1}^{N} \sim D. \tag{53}$$

5. Compute the temporal difference target.

   If the next state is a terminal state, the target value merely consists of the immediate reward; otherwise, the future maximum Q value predicted by the target network is incorporated.

   $$y_i = \begin{cases} r_i & \text{if} done_i = 1, \\ r_i + \gamma \cdot \max_{a'} Q(s_{i+1}, a'; \theta^-) & \text{otherwise.} \end{cases} \tag{54}$$

   Among them, γ is the discount factor, which measures the significance of future rewards ($0 \le \gamma < 1$), and θ^- the parameters of the target network, fixed to mitigate target fluctuations.

6. Compute the loss and update the primary network.

   The disparity between the predicted Q value and the target value is gauged by the mean square error, and the parameters of the main network are updated through backpropagation and gradient descent.

   $$L(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - Q(s_i, a_i; \theta)\right)^2, \quad \theta \leftarrow \theta - \alpha \cdot \nabla_\theta L(\theta). \tag{55}$$

   Among them, α is the learning rate, which controls the step size of parameter updates.

The DQN integrates Q-Learning with DNNs to analyze multimodal data and construct high-dimensional state-action mappings in agricultural robot path planning. Leveraging experience replay disrupts sample correlations, thereby improving training stability. Using a target network minimizes the bias in Q value estimation during training, while the ε-Greedy strategy facilitates an effective balance between exploration and exploitation. This enables the robot to adapt dynamically to environmental changes in farmland settings and meet multitask requirements. DQN achieves low latency and highly robust continuous decision-making through end-to-end learning, which significantly enhances the accuracy of dynamic obstacle avoidance and operational energy efficiency, as demonstrated in Tables 35 and 36.

The DQN is utilized in the path planning of agricultural robots, combining DNNs with RL. It effectively handles high-dimensional state spaces and autonomously learns optimal strategies, making it suitable for unknown or semi-structured farmland environments. By leveraging experience replay and

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TABLE 34 | Summary of deep reinforcement learning algorithms.

Algorithm name		Improvement principle	Applications in agriculture
Deep Q-network	DQN	Integrating Q-learning with deep neural networks stores data through experience replay and random sampling. A target network stabilizes training, solving the problem of approximating in high-dimensional spaces.	Static path planning for farmland in discrete action spaces.
Deep deterministic policy gradient	DDPG	On the basis of the Actor-Critic framework, the Actor outputs precise continuous actions while the Critic evaluates their value. The Actor parameters are updated according to the policy gradient, making them suitable for continuous control of obstacle avoidance of agricultural machinery.	Dynamic path adjustment for agricultural machinery in continuous action spaces.
Proximal policy optimization	PPO	Limiting the policy's update amplitude and combining trust region optimization to balance exploration and exploitation enhance the training stability. The system is adapted to multimodal sensor inputs for adaptive path planning in farmlands.	Adaptive path decision-making in multisensor fusion environments.
Twin delayed deep deterministic policy gradient	TD3	Improved upon DDPG uses dual Critic networks to take the minimum Q value to alleviate overestimation, introduces delayed policy updates and action noise smoothing to enhance robustness, and is adapted to high-noise farmland sensor data for precise path tracking.	Precise path tracking under high-noise conditions.

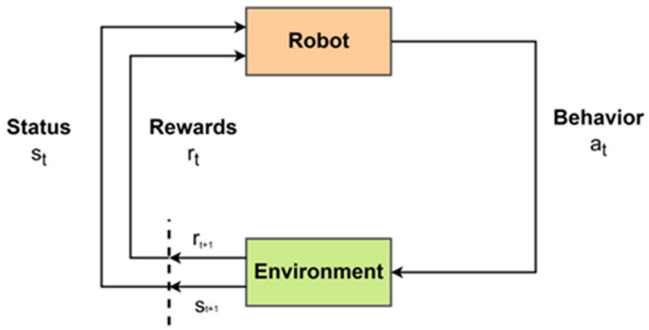


FIGURE 32 | Schematic diagram of deep Q-network.

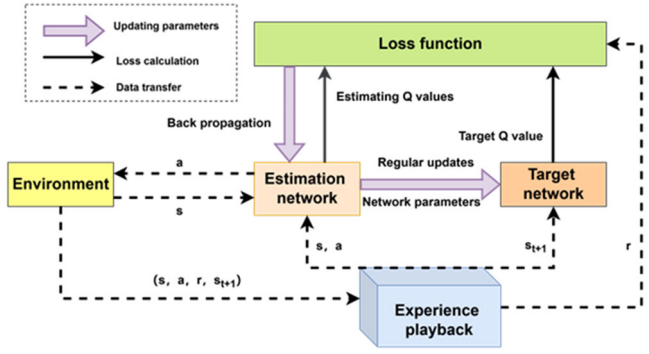


FIGURE 33 | Flowchart of deep Q-network.

target network techniques, DQN stabilizes the training process and demonstrates competence in multiobjective dynamic obstacle avoidance. However, certain limitations exist. DQN may become overly reliant on prior experiences or engage in excessive exploration, negatively impacting operational efficiency. Additionally, its performance is sensitive to the architecture of the network and parameter settings, where improper configurations may lead to instability during training. Furthermore, end-to-end learning in DQN lacks explicit incorporation of domain-specific agricultural knowledge, potentially resulting in neglecting critical constraints relevant to farming operations.

The current mainstream directions for improvement include: First, by designing more rational reward functions and introducing prioritized sampling mechanisms, the efficiency of sample utilization is enhanced, thereby accelerating the algorithm's convergence. Second, variant algorithms such as Double DQN are adopted, utilizing a dual-network structure to reduce overestimation issues and improve the stability and robustness of the algorithm. Third, a probability-based state exploration DQN algorithm is proposed, adjusting the Q value calculation method to avoid being trapped in local optima. Kong et al. (2023) proposed the B-spline-artificial potential field-deep Q-network (B-APFDQN) framework for agricultural robot path planning, integrating prior knowledge of the APF. They constructed a multiobjective reward function based on APF, transformed the single-output DQN into a multioutput architecture, and introduced APF potential field information to assist training, strengthening collaborative optimization capabilities.

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TABLE 35 | Some application scenarios and influences of DQN.

Actual scenarios	Concrete impacts
Dynamic navigation in multicrop mixed planting areas (Fakhrezi et al. 2025)	DQN optimizes the interrow traversal path by enabling real-time perception of different crop regions and avoiding shaded areas caused by tall crops.
Emergency escape route in case of sudden fire (W. Liu, Lyu, et al. 2024)	In the event of a farmland fire, DQN rapidly generates the shortest escape route by leveraging thermal imaging data, avoids areas with spreading fire, and simultaneously coordinates the evacuation paths of multiple robots.
Coordination of multirobot water resource competition (W. Xu et al. 2021)	In arid regions, DQN coordinates the trajectories of multiple irrigation robots, dynamically prioritizes water resource allocation, and prevents path overlap and equipment collisions.
Claas Lexion 8000 intelligent combine harvester (Redenius et al. 2019)	In the combined harvesting of Canadian canola fields, sensor-collected data are integrated into the DQN algorithm to optimize both the path and operational parameters.
Raven industries autonomous grain transport platform (Bahmutsky et al. 2025)	On the basis of satellite remote sensing normalized difference vegetation index (NDVI) time-series and soil nitrogen content sensor data, the DQN algorithm establishes a dynamic mapping between fertilizer application rates and corn growth stages, generating a "honeycomb grid" path to enhance the uniformity of fertilizer distribution simultaneously.
Fendt Rogator 600 sprayer (Niestrawski 2019)	The sprayer utilizes three-dimensional vision and inertial navigation systems, employing the DQN algorithm for risk modeling and generating a "serpentine" path to optimize nozzle activation timing, thereby reducing pesticide usage by 28%.
Lely Juno automatic feed pushing robot (Nabokov et al. 2020)	In Dutch pasture path optimization, the DQN algorithm models the relationship between robot operating parameters and forage grass regrowth, generating a "wavy" path to increase forage grass yield while protecting root systems.

Abbreviation: DQN, deep Q-network.

Y. Chen et al. (2025) proposed the reinforced deep Q-network (Re-DQN) algorithm framework based on DRL to address lawn mowing robots' comprehensive coverage path planning problem. They developed an exploration strategy combining an intrinsic reward mechanism for state novelty with a dynamic input structure. C. Chen, Yu, et al. (2024) proposed a DRL framework integrating multiple technologies to solve the energy consumption optimization problem in agricultural wireless sensor networks. They built an architecture based on Dueling DQN, separating value and advantage functions to enhance decision-making efficiency, introduced PER to optimize samples, and improved stability. X. Han et al. (2021) proposed a DDPG RL framework for mixed-variable optimization to handle the joint optimization problem of transmission power and channel allocation in communication networks. They constructed an end-to-end model based on DDPG, modeled the policy as a mixed action space, used deep networks to approximate Q and functions, designed a black-box optimization reward function incorporating multiple objective indicators, and stabilized training through experience replay and target network updates.

The DQN algorithm is anticipated to achieve significant advancements in the agricultural domain across three key areas. First, offline RL will be utilized, leveraging historical data for model training. Stability will be maintained through conservative policy iteration, while contrastive learning will be incorporated to enhance training performance, effectively addressing the low sample efficiency inherent in online learning. Second, a hierarchical decision-making architecture will be developed, decomposing path-planning tasks via hierarchical DQN. Curriculum learning will be employed to improve the handling of complex tasks, and an attention mechanism will be introduced to allocate task priorities dynamically. Third, physical simulation and domain randomization techniques will be enhanced, constructing a virtual environment capable of generating diverse scenarios. This will facilitate robustness training under extreme conditions, and virtual-real collaborative optimization will be achieved through digital twin technology, thereby improving the effectiveness of actual deployment.

6.4 | Comparison of Machine Learning Algorithms

In the path planning of agricultural robots, the application of Q-learning, CNNs, RNNs, and DQNs depends on the characteristics of the agricultural environment and the requirements of path planning. On the basis of RL, Q-learning updates the action-value function through interaction with the environment, making it suitable for dynamic and stochastic farmland

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TABLE 36 | Challenges and solutions of DQN in some actual agricultural scenarios.

Problems in specific agricultural scenarios	Solution measures
① In farmland environments, the sparsity of effective action rewards leads to slow model convergence.	① Design a coverage-based incremental reward mechanism to stimulate the exploration of new areas.
② Sudden weather changes cause abrupt environmental state transitions, resulting in delayed Q value updates.	② Integrate real-time meteorological sensor data for the dynamic update of the environmental state.
③ The objectives of minimizing path length and energy consumption often conflict, and a single Q value cannot adequately balance these competing goals.	③ Design a multiobjective DQN framework to output the Pareto-optimal solution set.
④ The computational units available in field machines cannot support real-time inference of deep neural networks.	④ Adopt lightweight networks to compress model parameters.
⑤ Models trained in simulation environments exhibit limited generalization performance when deployed in real-world farmlands.	⑤ Employ domain randomization to enhance the diversity of simulation data.

Abbreviation: DQN, deep Q-network.

scenarios, where robots can learn optimal strategies via trial-and-error processes. CNNs, with their strong image feature extraction capabilities, effectively process visual information and exhibit superior performance in path-planning tasks that rely heavily on visual perception, such as those in orchards and greenhouses. RNNs excel at handling sequential data and capturing time-series information like weather conditions and crop growth patterns, which makes them well-suited for farmland environments with temporal dynamics and enables forward-looking decision-making for robots. DQNs, combining the strengths of DNNs and Q-learning, can address complex environmental states and learn optimal path-planning strategies, making them ideal for large-scale, high-dimensional, and complex agricultural settings while ensuring efficient navigation and operation of robots. These algorithms are all essential for agricultural robots' navigation and operational effectiveness, as summarized in Table 37.

To comprehensively evaluate the performance of Q-learning, the CNN, RNNs, and DQNs in agricultural environment path planning, this study conducts a systematic comparison across six dimensions: path optimality, real-time obstacle avoidance, cost-effectiveness, computational efficiency, environmental adaptability, and scalability. The comparison results are visualized using a radar chart, and the performance of each algorithm under different metrics is quantified through a scoring system ranging from 1 to 5 (from weak to strong), as illustrated in Figure 34.

7 | Discussion

In agricultural robot path planning, scholars domestically and internationally have extensively explored and applied various algorithms, achieving significant advancements. Table 38 summarizes the current mainstream path-planning algorithms. Traditional classical algorithms, such as Dijkstra and A*, are theoretically mature and highly efficient in path solving, making them widely used for global path planning in agricultural robots. However, high-precision planning is computationally intensive, and balancing accuracy with efficiency remains challenging, leading to their everyday use in preliminary or auxiliary planning stages. Modern intelligent bionic algorithms, including GAs and ACO, simulate natural optimization mechanisms and exhibit strong adaptability and global search capabilities, effectively addressing uncertainties in farmland environments. Scholars continue to explore novel algorithms or hybrid approaches that integrate these modern methods with classical algorithms to enhance overall performance. Sampling-based algorithms like the PRM and RRT series demonstrate unique advantages in local path planning and avoiding real-time obstacles for agricultural robots. Although they demand significant computational resources, optimizing sampling strategies and heuristic functions can improve planning efficiency and path quality. Additionally, various improvement strategies have been developed to meet the specific requirements of agricultural environments. Machine learning algorithms can adapt to dynamic environmental changes and be applied to tasks such as dynamic obstacle recognition. However, challenges remain, including high computational resource consumption, necessitating the integration of traditional algorithms with lightweight technologies to achieve a balance between efficiency and robustness.

Research on agricultural robot path-planning algorithms is evolving toward diversification and intelligence. Traditional classical algorithms, modern intelligent bionic algorithms, sampling-based planning algorithms, and machine learning each possess distinct advantages. Future research will focus on leveraging complementary strengths, deep integration among these algorithms, and customized optimization for agricultural scenarios to advance agricultural robot technology.

8 | Conclusions

This paper conducts an in-depth analysis of traditional classical algorithms, modern intelligent algorithms, sampling-based planning algorithms, and machine learning algorithms, evaluating their strengths, weaknesses, and improvement methods. The main conclusions are as follows:

1. Most existing path-planning algorithms are designed based on static simulation environments. However, real

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TABLE 37 | Comparison of machine learning algorithms.

Name	Computational efficiency	Cost-effectiveness	Case studies
Q-learning algorithm	The convergence rate in the initial learning phase is relatively slow, requiring extensive exploration and numerous iterations to identify an optimal strategy. The computational complexity increases significantly when dealing with large state and action spaces.	The initial exploration phase incurs high costs due to the necessity of extensive experimentation with different actions for data collection. However, once effective strategies are learned, tasks can be executed efficiently, yielding considerable long-term benefits.	The improved algorithm reduces the path step count by 39.29% compared with the traditional multi-layer perceptron (MLP)-DDQN, decreases the repeat coverage rate by 7.06%, and shortens the total operation time by 22.45%. This study provides robust technical support for dynamic path planning of agricultural unmanned aerial vehicles in complex terrains (J. Li et al. 2024).
Convolutional neural network (CNN)	The computational demand is high, especially for processing high-resolution images. However, GPU acceleration can substantially enhance computational efficiency.	Training is resource-intensive, requiring substantial labeled data and computational resources. Nevertheless, a well-trained model can process image data with high efficiency and accuracy, leading to significant performance improvements in tasks such as image recognition.	A semantic segmentation approach based on CNNs is proposed for extracting navigation lines of agricultural robots in the field. The data indicates that the lightweight network ENet demonstrates the best performance in five field scenarios, with an average segmentation accuracy of 84.94%, a navigation line pixel error of merely 10.73, and a processing speed reaching 16.8 frames per second, which significantly surpasses traditional visual algorithms (Yu et al. 2023).
Recurrent neural network	When handling long sequences, vanishing or exploding gradients pose challenges, leading to difficult training and reduced computational efficiency. Advanced RNN variants, such as LSTM and GRU, can effectively mitigate these issues.	Training is complex and costly, particularly when handling sequential data. Despite this, RNNs excel in processing time-series information, providing notable advantages in applications like time-series prediction.	A path-planning method based on the long short-term memory (LSTM) network was proposed to optimize obstacle-avoidance efficiency in dynamic environments through temporal feature learning. Experimental results demonstrate that, in a simulated greenhouse environment, the improved LSTM algorithm reduces path-planning time by 62% compared with the traditional A* algorithm, increases the accuracy of dynamic obstacle recognition to 94.3%, and enhances path smoothness by 41%. This study validates the feasibility and effectiveness of RNN-based approaches in real-time path planning for agricultural robots (Nair and Supriya 2020).
Deep Q-network (DQN) algorithm	Training is complex and costly, particularly when handling sequential data. Despite this, RNNs excel in processing time-series information, providing notable advantages in	Training requires robust computing infrastructure and large datasets, making it resource-intensive. However, DQNs can learn optimal strategies in complex environments, offering significant benefits in	The bidirectional long short-term memory-deep Q-network (BL-DQN) algorithm was proposed based on the DQN algorithm and integrating the bidirectional LSTM structure. On this basis, a path-planning framework for pest and disease control using agricultural

(Continues)

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TABLE 37 | (Continued)

Name	Computational efficiency	Cost-effectiveness	Case studies
	applications like time-series prediction.	challenging agricultural scenarios.	drones was devised. Through simulation experiments, the BL-DQN algorithm achieved a 41.68% increase in coverage compared with the traditional DQN algorithm. The repeated coverage rate of BL-DQN was 5.56%, lower than 9.78% of the DQN algorithm and 31.29% of the depth-first search algorithm (Fu et al. 2024).

Abbreviations: CNN, convolutional neural network; DQN, deep Q-network; DDQN, double deep Q-network; GPU, graphics processing unit; GRU, gated recurrent unit; LSTM, long short-term memory; RNN, recurrent neural network.

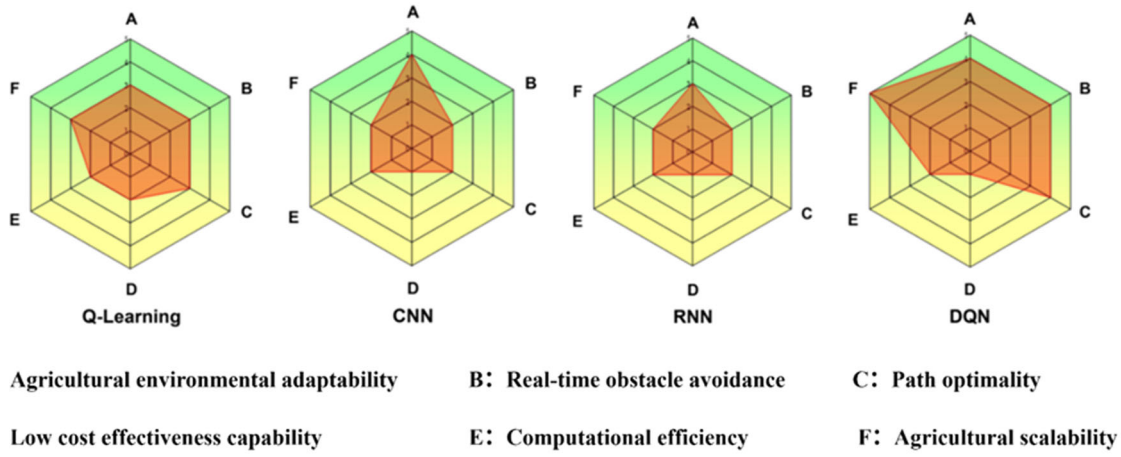


FIGURE 34 | Radar chart of performance comparison of machine learning algorithms. A, agricultural environmental adaptability; B, real-time obstacle avoidance; C, path optimality; D, low cost-effectiveness capability; E, computational efficiency; F, agricultural scalability. CNN, convolutional neural network; DQN, deep Q-network; RNN, recurrent neural network.

agricultural operation environments involve static obstacles such as trees and utility poles, dynamic disturbances like moving livestock, fluctuating crop heights, and unpredictable soil conditions. Enhancing the dynamic search capabilities of these algorithms is critical for improving their adaptability across diverse environments and enabling a smooth transition from simulation to practical agricultural scenarios.

2. In the face of variable farmland environments, unpredictable obstacles, and seasonal variations in crop growth, a single-path-planning algorithm struggles to meet the diverse and dynamic operational requirements of agricultural robots, which demand enhanced environmental adaptability and intelligent decision-making capabilities. Consequently, algorithm fusion technology has become an inevitable trend and a critical focus for future development. For instance, integrating global path-planning algorithms with local obstacle-avoidance algorithms ensures efficient robot operation across large-scale farmlands. Moreover, combining MPC algorithms with intelligent optimization algorithms enables precise decision-making in dynamic environments.
3. The investigation into the collaborative operation and multimachine scheduling algorithms is paramount in enhancing the application of agricultural robots. As agricultural practices become increasingly scalable and intelligent, the operational efficiency of a single agricultural robot in large-scale farmland scenarios becomes limited, making it challenging to meet the high-efficiency production demands of modern agriculture fully. The collaborative operation mechanism of agricultural robots facilitates data sharing, task distribution, and cooperative execution among multiple robots. By leveraging well-designed multimachine scheduling algorithms, the operation paths of robots can be optimized, enabling them to adapt to complex and dynamic farmland environments effectively.
4. The future path-planning system is anticipated to adopt a collaborative architecture integrating cloud computing, edge computing, and terminal devices. This architecture aims to achieve dynamic allocation of computing resources and hierarchical optimization of tasks. Specifically, the cloud will construct global maps and pretraining path models, while edge nodes will handle local path replanning and communication coordination. Terminal

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TABLE 38 | Summary table of mainstream path-planning algorithms.

Algorithm	Feasibility	Limitations	Solutions	Agricultural scene
APF	Simple and achievable with ease, and featuring high computational efficiency.	There might be issues of encountering local minima, which leads to the path not being globally optimal.	Incorporate other algorithms for path optimization to avoid local minima.	Obstacle avoidance of orchard-picking robots, dynamic obstacle avoidance in greenhouses, and detouring around temporary obstacles by weeding robots.
Dijkstra	It is applicable to diverse types of weighted graphs, encompassing directed and undirected ones.	It is not applicable to graphs featuring negative-weighted edges; the efficiency is relatively low in dense graphs.	The efficiency of the algorithm is optimized through the utilization of a priority queue; the Bellman–Ford algorithm is employed in graphs with negative weights.	Global path planning for greenhouse inspection robots, optimization of sowing paths in large fields, and inspection of fixed facilities.
A*	It is applicable to a wide range of graph and tree search spaces and is capable of efficiently identifying the shortest path.	The selection of the heuristic function exerts a remarkable influence on the performance of the algorithm.	Meticulously designing the heuristic function and integrating it with other optimization techniques can enhance the efficiency of the algorithm.	Path optimization for spraying in field ridges by plant protection drones and global path planning for combine harvesters.
DWA	It is applicable to partially unknown or dynamically evolving environments.	It is highly sensitive to dynamic environmental variations and necessitates frequent updates of path information.	By integrating local path-planning algorithms, rapid responses to dynamics can be achieved.	Dynamic obstacle avoidance for combine harvesters and terrain adjustment for autonomous tractors in muddy conditions.
ACO	It possesses strong robustness and global search capacity.	The setting of algorithm parameters is intricate, and the convergence rate is relatively slow; it is prone to getting trapped in a local optimum.	Employ an adaptive parameter adjustment strategy and integrate with other optimization algorithms.	Path optimization between orchard rows, multirobot collaborative operation, and complex farmland coverage path planning.
GA	It is applicable to large-scale and complex optimization issues.	The convergence rate is relatively slow, and it is prone to premature convergence; it is highly sensitive to parameter settings.	The elite retention strategy is employed to enhance the algorithm performance; it is combined with other optimization algorithms for hybrid optimization.	Global path planning for mountainous farmlands, optimization of harvesting paths in complex terrains, and multiconstrained task allocation.
PSO	It has few parameters, is easy to implement, and has a relatively fast convergence speed.	It is prone to being trapped in local optima, and the quality of the path is unstable.	Dynamic adjustment of inertia weight, hybrid optimization of the pheromone mechanism in ACO, and the introduction of chaotic initialization strategies.	Parameter optimization of plant protection drones and real-time adjustment and allocation of path planning for greenhouse robots.
RRT	Path-planning applicable to complex constraint conditions	The path may not be optimal; the search efficiency is relatively low in complex environments	Combine with other optimization algorithms for path optimization; adopt an incremental search strategy to improve search efficiency.	Unstructured farmland.

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TABLE 38 | (Continued)

Algorithm	Feasibility	Limitations	Solutions	Agricultural scene
PRM	It is applicable to path planning in complex environments and can effectively lower the dimension of the search space.	The quality of the path is dependent on the sampling density; the computational complexity is rather high in high-dimensional spaces.	Employ appropriate sampling strategies and graph search algorithms; combine with other optimization algorithms to enhance the quality of the path.	Path planning for fixed obstacles in vineyards and path design for warehouse-style agricultural facilities.
Q-learning	It exhibits outstanding dynamic environment learning capabilities and is applicable to partially observable environments.	Dimensional curse: incapable of handling continuous states or high-dimensional inputs.	Incorporating deep learning approaches, it forms the deep Q-network algorithm and employs the experience replay mechanism to accelerate the learning process.	Simple farmland grid map navigation, fixed facility inspection rounds, and task allocation for small robots in regular areas.
CNN	The visual navigation features high accuracy, supports perception in complex environments, and the real-time performance is gradually enhanced.	It relies on the quality of training data, has limited generalization ability, and requires a high amount of computing resources.	Employ data augmentation techniques to improve the robustness of the model; adopt transfer learning techniques to reduce the demand for training data.	Visual navigation in farmlands, fruit recognition and obstacle avoidance in orchards, and visual simultaneous localization and mapping in greenhouses.
RNN	It possesses a strong dynamic obstacle prediction ability and supports long-term and short-term dependency modeling.	The ability of long-distance dependency modeling is limited, the training stability is poor, the computational complexity is high, and the memory occupation is large.	Utilize a variant of the long short-term memory network to enhance the model's capability of capturing long-term dependencies.	Long-term farmland monitoring path planning, dynamic obstacle trajectory prediction and obstacle avoidance, and time-series environment-adaptive navigation.
DQN	It demonstrates a strong autonomous decision-making capacity in complex scenarios, supports multiobjective optimization, and is applicable to high-dimensional state spaces.	There are high demands for computational resources, poor training stability, the risk of overfitting, and low sample efficiency.	Employ the experience replay mechanism to accelerate the learning process; adopt techniques such as target networks and double Q-learning to stabilize the training process.	Fully autonomous adaptive navigation for farm robots, multimachine collaborative task allocation, and path planning under complex climatic conditions.

Abbreviations: ACO, ant colony optimization; APF, artificial potential field; CNN, convolutional neural network; DQN, deep Q-network; DWA, dynamic window approach; GA, genetic algorithm; PRM, probabilistic roadmap; PSO, particle swarm optimization; RNN, recurrent neural network; RRT, rapidly exploring random tree.

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devices will execute real-time obstacle avoidance and motion control. A distributed knowledge-sharing framework based on federated learning will also be established to enable experience sharing under data privacy protection. This design can overcome the computational power limitations of single machines, reduce energy consumption, and provide scalable technical support for large-scale collaborative operations in farmland environments.

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Data Availability Statement

The authors have nothing to report.

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